Interaction Obstructions, Resonant PT Breaking, and Doubled Jordan Symmetry in Fourth-Order Theories

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Abstract

We formulate a deformation problem for the canonical positive pseudo-Hermitian metric of the Pais–Uhlenbeck oscillator and for its Krein/ghost-parity alternative, and follow both into interacting fourth-order theories. Four results organize the paper. (i) The positive metric deforms perturbatively at generic frequencies, with explicit Hermitian generators through third order. (ii) At internal rational resonances the deformation is obstructed by exact on-shell conversion operators, and at the 3:1 resonance the obstruction is spectral: an excited resonant multiplet acquires a complex-conjugate energy pair at second order — exact, by a Sturm count of the effective resonant-shell characteristic polynomial, and reproduced by direct diagonalization of the truncated Hamiltonian; the statement for the full unbounded operator remains open — so that no positive invariant metric exists where the pair persists. (iii) In the perfect-square field theory the momentum continuum turns such isolated resonances into generic scattering shells: the branch-changing process $H + L \to L + L$ obstructs the analytic second-order deformation of the pointed-sector positive metric on an open subset of its shell, with an exact value at a rational kinematic point. (iv) In the two-field frame $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$ — the $O(1, 1)$ embedding of Bateman and Turok, whose exchange symmetry $U \leftrightarrow V$ they identified as ghost parity — the exchange maps two oppositely oriented null Jordan sectors into one another, its linearization is the bounded confluent limit of the regulated branch parity on the doubled space, and the obstruction of (iii) sits entirely in the exchange-odd component of the on-shell T, which is exactly ghost-parity pseudo-Hermitian. A self-contained charge-null lemma and regulated embedding further show that one-sided charge nullity is an enhanced boundary property, not a property of the split theory: the mapped vacuum obeys $S_{UU}/S_{VV} = (\delta/2g)^2 = \epsilon/g$ and becomes one-sided only on the confluent line, with the Bateman–Turok massless theory at $\epsilon = \mu^2 = 0$. Exact split-shell Krein pseudo-unitarity and boundary one-sidedness are therefore distinct; the two completions are inequivalent in the interacting theory. All finite-dimensional and modewise identities are machine-verified; derivations of the principal results are given in the appendices. An independent Science Forge computation extends the discrete resonance hierarchy to order six: the 5:3 and 7:1 loci vanish through orders two–five and obstruct at order six on exactly the predicted conversion monomials. Those two instances are imported here as externally reproduced exact evidence; the all-coprime hierarchy remains a conjecture. A separate exact `REDUCED-MODE` continuation of the scalar loop programme determines the final-state collinear logarithmic response. It finds that the real finite-part normalization changes by $3\lambda^6 \log c/(512\pi^4 s)$, whereas every continuous

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axis-compatible parent/daughter regulator gluing, including the physical pair threshold, has zero constant virtual response. This obstructs the ordinary independent-mass infrared prescription on that carrier; it is not a full NLO probability, failure of dressed-state or resummed constructions, or a gravitational result. An exact finite-carrier calculation changes the measured object. Correctly normalizing its dimensionless projector by the Born rate gives Gram coefficient $1/48$ per pair, amplitude $\sqrt{3}/12$, and hard/collinear blocks $-1/16$ and $+1/16$; multiplying by the Born rate cancels the physical $3/512$ response. The ordinary single-denominator cubic Fock generator is, however, exactly zero in the massless collinear limit. The unresolved problem is narrower: expanding the published composite map gives the exact order-λ two-annihilator kernel, and all of its apparent t, t^2 Jordan growth cancels on the two-field carrier. Its Krein Gram has cubic splitting-endpoint poles, however, so the distributional continuum projector and its dynamically derived $1/48$ coefficient are not yet constructed. An exact canonical-algebra countermodel further shows that CCR preservation, projector idempotence, trace preservation, and exchange symmetry alone retain three endpoint parameters. Thus $1/48$ is compatible but not selected without the deferred continuum transport or another independently justified physical endpoint condition. Retaining the parent–daughter energy deficit and the full radial daughter measure reduces those three flat-slice jets to one soft normalization on the declared three-dimensional chart, but the measured cross-Gram is still $-(1/2)\,dr/r$. Ordinary off-resonant composition is therefore logarithmically non-trace-class. Restoring the parent inverse metric, Bose factor, angular measure, and outgoing $2!/3!$ projector ratio does fix its unprojected response to $1/48$ per pair. However, its two logarithmic rows have generator charges $(+1, -1)$ and $(-1, +1)$ under the published fixed-vacuum oscillator grading; a prior one-sided nonpositive projection makes the residue zero. The physical neutral coefficient remains undecided. An exact vacuum-orbit completion with $Z = e^{\lambda\phi_0}$ makes both soft rows neutral and preserves $1/48$, but also turns the Appendix-C squeeze from apparent charge -2 into neutral $Z^2(b_\Upsilon^\dagger)^2$. Fixing $Z = 1$ restores the negative grading but destroys the charge derivation on that quotient. The missing full zero-mode module and trace in the deferred Eq. (19) proof are therefore still required. Moreover, the displayed Appendix-C squeezed vacuum fails a prior carrier test: in normalized finite-box modes its creation kernel is $1/(8|\mathbf{p}|^2)$, and its first two-particle positive-norm density is $\rho^2(\epsilon^{-1} - \Lambda^{-1})/(64\pi^2)$ for any constant member of the compatible fundamental-symmetry family. A six-mode lower bound shows that this density grows linearly with box size for every uniformly equivalent positive topology. The associated pair block is likewise non-Hilbert–Schmidt. Thus indefinite nullity does not produce an ordinary Fock–Krein vector; an extended or inequivalent non-Fock realization remains open and would need its domains and trace specified. The full exponential strengthens the condition to $\sup_p |\rho(p)/(4p^2)| < 1$. An explicit infrared weight satisfies this and has finite norm density, but its inverse metric is unbounded and its thermodynamic sector is inequivalent. The raw BT shear is also cross-Krein canonical rather than a positive-Hilbert Bogoliubov map, so existing extended positive-boson implementations do not supply the missing BT trace. A direct cross-Krein construction implements the covariant squeeze on paired cores and supplies a cyclic finite-rank trace. However, any finite normalized cyclic trace positive on the ghost-even orbit cone kills localized rank-one projectors, while the finite-rank trace has infinite identity weight. The transported vacuum projector also has exponentially growing trace norm. A semifinite finite-detector construction survives this obstruction. Its canonical trace gives localized orbit projectors weight one and the identity infinite weight; conditioning on a finite incoming projection gives nonnegative normalized weights for finite weakly ghost-symmetric output partitions. The normalized corner functional is explicitly noncyclic, and no normal thermodynamic state or Eq. (19) transport follows. Closing the order-λ scalar quadratic map over every leading Appendix-C inverse image removes the logarithmic carrier itself. The off-resonant symplectic phase of the opposite-sign preimage cancels its oscillatory inverse phase; the complete signed kernel is time independent, obeys the cross-CCR Ward identity, and has identically zero parent cross-Gram. Consequently the unsqueezed finite-mode nonlinear factor has neutral charge support through order λ, with $Q_1 = 0$ in that charge decomposition. This is not the full Eq. (19) statement because the public map is $R = SU$. The conditional $1/48$ and the associated finite-cell trace growth are not coefficients of this completed quadratic map. This does not determine a physical response because the S-matrix transition block is a distinct operator. The squeezed-vacuum and dynamical-zero-mode projector trace, continuum domain, and higher orders remain open within Eq. (19). On every finite paired core the covariant squeeze conjugates this Born operator and the detector together, so cyclicity preserves the zero trace; a nonzero bare squeezed-vacuum pair occupation is a fixed-detector comparison, not an additional Eq. (19) term. Trace similarity does not imply ghost evenness. For any homogeneous assignment $q(Z) = s$, the nonlinear and squeeze charges obey $q_K = 1 - s$ and $q_S = 2s - 2 = -2q_K$. If $s < 1$ the nonlinear tangent has a positive component, if $s > 1$ the free squeeze does, and at $s = 1$ the neutral squeezed $n = 1$ projector has exact non-null ghost-odd norm $-z^2(z^2 + 2)$. Thus no homogeneous regular assignment realizes the full Eq. (19) package on the finite regulator. The squeeze's non-trace-class thermodynamic limit prevents a thermodynamic pushforward statement, while object typing independently prevents promotion of the pushforward zero to an S-matrix zero. Nevertheless, finite paired-process probabilities have a non-normal thermodynamic cylinder limit: squeezed spectator corners have Krein trace one, so weak-ghost weights are volume independent even while their positive trace norm diverges. The zero quadratic coefficient survives this directed limit. Affiliation of an inclusive LSZ detector and the dynamical zero mode remain unproved. For an orthogonal inclusive detector, the leading block identity is exact: if $U = 1 + \lambda K + \lambda^2 L + \cdots$, its order-$\lambda^2$ transition weight contains only $(P_{\text{out}} K P_{\text{in}})^\dagger (P_{\text{out}} K P_{\text{in}})$. Its use requires the physical scattering kernel. The zero complete signed kernel instead belongs to the distinct Eq. (19) pushforward $R_t P R_t^\dagger$. The independently computed five-point S-matrix response remains $+1/512$ per pair and $+3/512$ over three pairs, or $1/48$ and $1/16$ after Born normalization. The axis-compatible physical virtual response is zero, leaving $+3/512$ uncancelled. The distinct R_t response is also zero but is not an S-matrix summand. A physical hard/dressed response $-3/512$, the complete NLO probability, and all-order Eq. (19) remain open. Physical-shell pseudo-unitarity fixes the coefficient conditionally. With $x^2 = \lambda^2 \log(c)/\pi^2$ and $S_{\text{phys}} = 1 + xA + x^2 B + \cdots$, one has $A^\dagger = -A$ and $B + B^\dagger + A^\dagger A = 0$. The measured real column has norm squared $1/16$, hence $2\,\mathrm{Re}\,B_{hh} = -1/16$ and the Born-rate hard response is exactly $-3/512$. This is phase independent and has an exact finite isometric witness, but remains conditional on a continuum incoming-plus-outgoing dressed S-matrix and its trace domain. The ordinary logarithmic carrier cannot supply that limit: disjoint shells approaching the endpoint have fixed real-column norm and mutual distance squared $1/8$. Their exact shell isometries likewise remain separated by $2\sin^2(x/4)$. A fixed abstract endpoint fibre pulls the family back to one isometry and preserves the conditional cancellation, but its local detector affiliation and dynamical origin remain open. At six points the output-side weak-ghost gate can be closed exactly. The twenty neutral three-Ω/three-Υ species assignments form ten complement pairs with Krein metric and ghost parity $\eta = \kappa = \left(\begin{smallmatrix} 0 & I_{10} \\ I_{10} & 0 \end{smallmatrix}\right)$. The complete coefficient (c, c) lies in the positive even ten-plane and has Krein norm $2\sum c_S^2$. Its 8×8 three-particle Choi matrix obeys $\kappa_3 A \kappa_3 = A$, so $A^\sharp = A^T$ and its generalized Born trace is the same sum of squares. Every fixed-shell nine-history range embeds isometrically in this public Fock plane with norm $9/8$. All ninety resolved histories do not: the simultaneous species map has rank ten and forgets an eighty-dimensional channel record. Thus the output history is now publicly affiliated, while a channel ancilla, the transported input projector, the virtual survival block and all-order Eq. (19) remain open. Preparing a state directly in that positive public sector bypasses the obstructed scalar-source pushforward. On its ghost-even three-particle four-plane the fixed-shell transition effect has exact spectrum $0, 1/16, (2-\sqrt{3})/8, (2+\sqrt{3})/8$. With $\zeta = \lambda^8 T\Delta\Xi/[128\pi^4\kappa^4 L_x L_y^2 L_z^2]$, the click/no-click effects ζG and $I - \zeta G$ are positive and normalized for $0 \le \zeta \le 16 - 8\sqrt{3}$. The source $(|\Upsilon^3\rangle + |\Omega^3\rangle)/\sqrt{2}$ has click rate $\lambda^8/[2048\pi^4\kappa^4 L_x L_y^2 L_z^2]$; normalization fixes the binary no-click complement, while genuine forward survival also depends on unobserved output blocks. This is a physical auxiliary-BT probability jet, not a transported perfect-square scalar source or an all-time result. The certified formal two-sidedness of R_t now pulls this source and both effects back to a normalized perfect-square scalar projector without changing the rate. The price is exact and unavoidable: every fixed-charge subspace of the odd three-particle cross-Krein carrier is null, so a positive source must break shift charge. Its projector has charge support $\{-6, 0, +6\}$ and scalar orbit support $\{Z^{-6}, 1, Z^6\}$. This gives a dressed physical-scalar probability jet, not the standard shift-invariant $P_\chi^{(\phi)}$ or general Eq. (19). On the finite covariant Gaussian core this dressed source is also explicit in the scalar Jordan oscillators. Its two three-mode branches carry Z^{-3} and Z^3; each is null and their cross CCR gives unit pairing, so the symmetric source has norm one. Because the complete six-point amplitude starts at λ^4, unknown $O(\lambda)$ corrections to the source first enter probability at λ^9 and cannot change the displayed λ^8 rate. The ordinary massless Fock thermodynamic source remains infrared-obstructed. For the tagged spectator process, the finite-time packet probability is complete through λ^6 in the declared normal-ordered $\overline{\mathrm{MS}}$ auxiliary scheme and transports to all nine recorded spectator labels. Along the exact fixed-energy family $s = 64\kappa^2/25$, $t = -32(1-c)\kappa^2/25$, $u = -32(1+c)\kappa^2/25$, its four resonant channels persist and the complete tree bracket obeys $W(c,T) \ge 13T/24 > 0$ for every $-1 < c < 1$ and $T > 0$. The six finite-time loop gaps are uniformly separated from zero on compact interior angle intervals, giving a bounded fibrewise q_6 coefficient and uniform small-coupling positivity. The angle remains a classical record at this fibrewise stage. For two orthogonal equal-weight hard modes, an off-diagonal positive detector erases that record without changing q_4 or q_6; its first known difference is an order-λ^8 angle variance. Exhausting the order-λ^8 ledger shows that this is the complete coherent-minus-recorded coefficient: the unknown higher cross is common to both effects. The complementary antisymmetric port removes that cross altogether. At $c = 0, 3/5$ and $\kappa T = 1$, exact finite-time tree and loop contrasts force a volume-uniform strictly positive lower bound on its absolute leading coefficient. The recorded and symmetric bright-port absolute q_8 coefficients, continuum coherence and both forward endpoints remain open. On the two-mode carrier, an explicit finite pointer Hamiltonian now selects the detector strength and relative phase; it is added apparatus physics, not a prediction of the public closed-system BT Hamiltonian. A local quadratic scalar density reproduces that Hamiltonian on the two selected modes, but no nonzero finite-derivative local density can isolate exactly those two zero-width angles in the continuum. The selected compression is therefore exact while its leakage into other angles must be retained. An order-20 projective Fejér filter with at most 38 transverse derivatives admits an antipodally even homogeneous lift to the complete invariant fixed-total-momentum two-body sphere. Two finite solid-angle packets capture all but less than 83/125000 of its exact phase-space norm, and exponentiating the residual-retaining star Hamiltonian gives bright-packet absorption efficiency above 124917/125000 at a half pulse. This is a fixed-P rotating-wave result. A Gaussian-switched version of the same local density has a square-integrable response on an open four-momentum set and retains more than $999117/10^6$ of its norm in the two packets and their momentum core. The target timelike pair sum is separated exactly from the non-timelike number-scattering transfer cone. An explicit off-shell finite-jet successor turns this pointwise gap into a Hilbert–Schmidt number-operator bound on $M_0/4 \le |k| \le 3M_0/4$; including the wrong-sign pair and adjoints, every undesired quadratic block is below $10^{-400000}$ relative to the desired pair block at first Dyson order. Unrestricted energies and complete time-ordered evolution remain open. Independently, the projected three-channel hard loop logarithm closes the Callan–Symanzik equation exactly with the Born rate and one-loop beta function. This yields the positive leading-log nonforward prediction $d\sigma_{\mathrm{hard}}/d\Omega = 24\pi^2/[25s\log^2(s/\Lambda^2)]$. It is a physical hard/window baseline, not a completed inclusive NLO probability; the collinear endpoint construction remains open.

AI authorship and computational accountability

GPT-5.6.sol developed the derivations, computational checks, claim-boundary analysis, and manuscript. Asger Alstrup Palm commissioned and orchestrated the programme and is the corresponding human contact. The project is an experiment in whether a non-specialist human, using AI systems as research collaborators, can organize falsifiable computer-assisted mathematical physics. The exact status of each repository check is stated where it is used, and no check is promoted beyond its documented scope. The manuscript remains subject to expert review.

Contents

1 Introduction

A companion series [12, 13, 14, 15] analyzed the *free* fourth-order theories — the Pais–Uhlenbeck (PU) oscillator, the fourth-order scalar field, and linearized quadratic gravity — and established a three-level structure: one complex spectral covariance, several real forms (involutions), and several completions. The positive pseudo-Hermitian completion of Bender and Mannheim [1, 2] and the Krein/ghost-parity completion of Bateman and Turok [4] are real forms of one free theory, distinguished at Jordan degenerations. This paper begins the interacting classification: which of these free structures can be continuously deformed when nonlinear processes are switched on?

The organizing mechanism, established here first for discrete oscillator resonances and then for continuum scattering shells, is

$$\text{metric deformation} \;\longrightarrow\; \text{on-shell cohomological obstruction} \;\longrightarrow\; \text{spectral complexification.}$$

Results. For the PU oscillator with the cubic vertex $V = -iy^3$ inherited from a real z^3 interaction: the deformation exists at first order for all $\omega_1 > \omega_2 > 0$ and at second and third order for generic frequencies (Sections 3–4); it is obstructed at second order at $\omega_1 = 3\omega_2$ by the exact on-shell conversion class $\propto a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3$, gauge-independently (Section 5); the obstruction is spectral — a complex-conjugate pair appears in the $E = 27\omega_2$ multiplet at $O(\zeta^2)$, certified exactly (Section 6); a third-order obstruction appears at $\omega_1 = \frac{3}{2}\omega_2$, and a transfer-lattice selection rule organizes the hierarchy (Section 7), whose first prediction is confirmed: 5:1 is unobstructed through order 3 and obstructed at order 4, exactly on the predicted monomial (Proposition 7.4); independent exact order-six calculations find the predicted first obstructions at 5:3 and 7:1 (Proposition 7.5) — the hierarchy is thereby sampled at five qualitatively distinct resonances through order six; the deformation diverges toward the Jordan boundary, with exact leading asymptotics at first order and measured geometric growth beyond (Section 8).

For the perfect-square field theory $S = -\frac{1}{2}\int(\Box\phi + \lambda(\partial\phi)^2)^2$, regulated by a mass splitting: the cubic vertex is exactly the Källén polynomial (Section 9); an even-ghost selection rule protects the positive metric at first order even above the ghost-decay threshold, while the sectorwise branch parity is obstructed there by the real decay (Section 10); at second order the branch-changing scattering $H + L \to L + L$ obstructs the analytic deformation of the pointed-sector positive metric on an open subset of its shell, with the exact value $401\sqrt{6}/39424$ at a rational kinematic point in fully specified conventions (Section 12); the exact two-field rewriting exhibits the extended theory's exchange symmetry, its two null Jordan sectors, and the interaction-generated Jordan dynamics (Section 11); and the regulated branch parity, singular on one confluent chain, converges on the doubled space to the linearization of that exchange (Section 13).

Relation to prior work. The two-field $O(1,1)$ embedding, the identification of the exchange $U \leftrightarrow V$ as ghost parity and internal charge conjugation, the quantum embedding relating the two-field and fourth-order Hamiltonians and S-matrices, and a Krein-space Born rule with positivity through weak ghost symmetry are all due to Bateman and Turok [4]; none of that structure is claimed here. Likewise standard are: complex eigenvalues of pseudo-Hermitian operators occurring in conjugate pairs with matching Jordan data [7], the forced indefiniteness of any compatible metric in their presence [8], antilinear (CPT-type) symmetry as the origin of real-or-conjugate-pair spectra [9], complex-ghost-pair unitarity of Lee–Wick type [10] (physically distinct: there the complex ghosts are excluded from asymptotic states), and positive invariant graph subspaces of block J-self-adjoint operators [11]. What this paper adds is the *deformation-obstruction theory of the positive pseudo-Hermitian metric* in the interacting theories: the obstruction complex itself, the exact 3:1 and 3:2 conversion classes and the transfer-lattice selection rule, the certified spectral pair and its broken-PT tongue, the continuum scattering-shell obstruction, the confluent bridge from the split branch parity to the doubled Jordan exchange, and the resulting separation statement: the Bateman–Turok Krein completion is *not* equivalent to a hidden positive pseudo-Hermitian completion, because the latter is obstructed by on-shell branch-changing interactions while the doubled Krein pairing remains exactly pseudo-unitary on the verified matrix elements (Proposition 12.4). The stronger one-sided-charge condition entering the Bateman–Turok Born rule is instead a boundary property (Proposition 14.2); its explicit Born-trace evaluation on a fixed split-mass shell does not test the charge-null mechanism, because the truncated shell carries no boost action. The final-state collinear logarithmic calculation is instead obstructed by the ordinary axis-compatible independent-mass prescription (Remark 14.3). A beyond-tree quotient trace therefore requires a new infrared/asymptotic-state architecture, not another fixed-shell evaluation.

Strength of the statements. Throughout we distinguish (a) *analytic metric obstruction* — nonexistence of an order-by-order deformation of the canonical metric in the specified representation and operator class — from (b) *absolute nonexistence of positive metrics.* For the oscillator at 3:1 the stronger statement (b) holds wherever the certified complex pair persists, since a positive metric would make the Hamiltonian similar to a self-adjoint operator; the qualifications (formal perturbation theory, unbounded cubic interaction, truncation) are stated where they apply. For the field theory only (a) is established here. "Generic" is likewise used in three distinct senses, always resolvable from context: *generic oscillator frequencies* (outside the finite resonance-candidate set at the stated order), *generic field obstruction* (nonzero on a nonempty open subset of an allowed scattering shell), and *kinematically open for every mass pair* (existence of the shell, which does not by itself imply nonvanishing of the obstruction at every point of it). The results have the following logical status: R_1 Jordan scaling — theorem; R_2, R_3 scaling — computational propositions; $R_n \sim \epsilon^{-3n/2}$ — conjecture; the 5:1 obstruction — computational proposition (confirmed at order 4, Proposition 7.4); the 5:3 and 7:1 order-six obstructions — clean-pinned externally reproduced computational proposition (Proposition 7.5); a full-field complex spectrum — open; the exact doubled classical involution — theorem (Bateman–Turok, restated); its quantum implementation — open; the κ-odd localization of the shell obstruction — computational proposition at the rational point (Proposition 12.4); one-sidedness of the regulated vacuum image — computational proposition, exact iff $\epsilon = 0$ in the stated charge frame (Proposition 14.2); the boundary Born-trace evaluation — open, with the fixed-shell route excluded rather than merely unattempted because the charge-null lemma's hypothesis is absent there (Section 14); the complete final-state collinear logarithmic response on the ordinary axis-compatible independent-mass regulator class — exact `REDUCED-MODE` obstruction (Remark 14.3); existence of a normalized finite coherent cross-multiplicity projector at the residual coefficient — exact `LOCAL-ALGEBRAIC` plus `REDUCED-MODE` classification (Remark 14.4); derivation of that transport from BT dynamics, a continuum dressed projector, full NLO quotient trace, and gravitational lift of that infrared statement — open.

Bookkeeping. The deformation order is counted by a formal parameter ζ: $H(\zeta) = H_0 + \zeta H_3 + \zeta^2 H_4$, defining $R_1, R_2, \ldots$ and the obstruction orders. The perfect-square coupling λ is a fixed parameter: in the two-field frame the quadratic Jordan structure and the vertex coefficients carry $g = \lambda^2$, while the nonlinear field map itself depends on λ. In the oscillator sections ζ multiplies the cubic vertex and coincides with the coupling.

Verification and reproducibility. Every finite-dimensional, modewise, and distributional identity in this paper is machine-verified at the root of the `area9innovation/weyl-gravity` repository accompanying the series (SymPy 1.14.0, NumPy 2.4.4, Python 3.14; cite the repository commit hash to fix the artifact). Appendix G maps checks to theorems and classifies each check as exact-symbolic, exact-rational (Sturm/charpoly), or high-precision numerical. Appendices A–F contain readable derivations of the principal results. Operator-domain caveats (unbounded interactions, formal series) are flagged where they occur. The order-six extension is pinned separately by `bridge/science_forge/certificates/SCIENCE_FORGE_PU_ORDER6_IMPORT.json` to Tango commit `ccee11d08de28d3e2681db209264887950666e87`. It records exact agreement between two Forge backends and an independent SymPy recomputation, while also recording that the upstream certificate was emitted by a dirty development toolchain. A second run built the compiler from a clean `git archive` of that commit and reproduced all 31 checks and all three SymPy golden comparisons; the clean replay receipt is part of the import. The scalar real–virtual continuation is pinned under `reverse_physics/certificates/` by certificate `REVERSE_PHYSICS_BT_REAL_VIRTUAL_AXIS_GLUING_V1.json` at standalone-repository commit `0daa9ed5507b983322b6e04cccf78907edf482b5`. Its producer and method-distinct invariant-Källén verifier use exact arithmetic and run sequentially under a 500,000 KB virtual-memory cap; the certificate preserves the `REDUCED-MODE` dependency boundary and explicitly leaves the full NLO quotient trace uncomputed. The finite coherent certificate `REVERSE_PHYSICS_BT_COLLINEAR_ PROJECTOR_TRANSPORT_V1.json` is retained but marked `SUPERSEDED_NORMALIZATION`: it used an absolute rate coefficient in a dimensionless projector. The dimensionless normalization is certified by `REVERSE_PHYSICS_BT_ASYMPTOTIC_GENERATOR_PREFLIGHT_V1.json`. Its exact $\mathbb{Q}(\sqrt{3})$ producer and method-distinct verifier reconstruct the Born normalization, projector equations, and cubic collinear scaling. The certificate labels the Jordan asymptotic generator and continuum projector `NOT_CONSTRUCTED`. The related certificate `REVERSE_PHYSICS_BT_RT_ JORDAN_KERNEL_V1.json` derives the resonant kernel while preserving those unresolved fields. It detects an internal oscillator-label inconsistency in Appendix C of Ref. [4], declares the minimal label repair used, derives the nonlinear coefficient directly from the published composite fields, and independently verifies exact secular cancellation. The same certificate fails closed at the nonintegrable endpoint Gram rather than promoting the formal finite-mode generator to a continuum projector. The hard result is pinned by `REVERSE_PHYSICS_BT_UV_HARD_SCATTERING_LAW_V1.json`. It imports the content-addressed

Born, beta-function, and projected-hard-log certificates, verifies their Callan–Symanzik identity by an independent rail, and keeps the full inclusive NLO probability `NOT_ESTABLISHED`.

Conventions. Oscillator conventions follow [12, 13]: normalized coordinates $V = (x, y, p, q)$, mode 1 = (x, p) at frequency ω_1, mode 2 = (y, q) at ω_2, positive normal form $h_0 = \frac{1}{2}[\omega_1\omega_2 p^2 + (\omega_1/\omega_2)x^2 + (\omega_2/\omega_1)q^2 + \omega_1\omega_2 y^2]$, rapidity $r = \log \frac{\omega_1+\omega_2}{\omega_1-\omega_2}$, $c = \cosh \frac{r}{2} = \omega_1/\sigma$, $s = \sinh \frac{r}{2} = \omega_2/\sigma$, $\sigma = \sqrt{\omega_1^2 - \omega_2^2}$. Operator computations use exact Weyl-symbol Moyal calculus (Appendix A).

2 The deformation complex

Definition 2.1 (Deformation problem and obstruction classes)**.** Let ρ_0 be the canonical free Dyson map, $h_0 = \rho_0 H_0 \rho_0^{-1} = h_0^\dagger$, and $v = \rho_0 V \rho_0^{-1}$ the transported vertex. Seek $\rho_\zeta = \exp[-\frac{1}{2}\sum_n \zeta^n R_n]\rho_0$ with $R_n = R_n^\dagger$ such that $h_\zeta = \rho_\zeta H_\zeta \rho_\zeta^{-1}$ is Hermitian order by order. The first two equations are

$$[h_0, R_1] = v^\dagger - v, \qquad [h_0, R_2] = (v_2^\dagger - v_2) + \tfrac{1}{2}[R_1,\, v + v^\dagger], \tag{1}$$

where v_2 is the transported $O(\zeta^2)$ vertex when present (derivation: Appendix A). For *discrete* spectrum, $h_0 = \sum_E E\, P_E$, the equation $[h_0, R] = S$ has a Hermitian solution iff $P_E S P_E = 0$ for every E; the obstruction class is $\mathfrak{o}_+ = \Pi_{\ker \mathrm{ad}_{h_0}} S = \sum_E P_E S P_E$, and the solution is $R = \sum_{E \neq E'} P_E S P_{E'}/(E - E')$ plus an arbitrary Hermitian element of the commutant. The parallel Krein problem deforms the ghost parity: $[H_0, K_1] = [\kappa_0, V]$, $\{\kappa_0, K_1\} = 0$, with $\mathfrak{o}_K = \Pi_{\ker \mathrm{ad}_{H_0}}[\kappa_0, V]$.

Remark 2.2 (Discrete versus continuous spectrum). The kernel projection above is defined for point spectrum. All field-theoretic computations in this paper are performed at *finite volume* (discrete momenta), where ad_{h_0} has point spectrum and Definition 2.1 applies verbatim; continuum statements are then statements about the finite-volume family, with shell conditions realized by tuning masses to lattice momenta. A distributional (rigged-Fock or spectral-measure) formulation of the continuum obstruction is deferred.

Anti-Hermitian (unitary) generator components are free and do not affect the Hermiticity conditions; commutant additions to R_n are the metric ambiguity. *Gauge independence* of an obstruction class means invariance under both, at all lower orders.

3 The cubic PU oscillator at first order

The real interaction z^3 continues, under the PU rotation $z = iy$, to $V = -iy^3$; the transported vertex is $v = -i(cy + isp)^3$, and $v^\dagger - v = 2ic\,(c^2y^3 - 3s^2yp^2)$.

Theorem 3.1 (First-order deformation)**.** *For every $\omega_1 > \omega_2 > 0$ the first-order equation is solved by the Hermitian Weyl-ordered cubic*

$$R_1 = \mathcal{W}\left[\frac{2c^3}{\omega_1\omega_2}qy^2 + \frac{4c^3}{3\omega_1^3\omega_2}q^3 - \frac{6cs^2(2\omega_1^2 - \omega_2^2)}{\omega_1\omega_2(4\omega_1^2 - \omega_2^2)}p^2q + \frac{12cs^2\omega_1}{\omega_2(4\omega_1^2 - \omega_2^2)}pxy - \frac{12cs^2\omega_1}{\omega_2^3(4\omega_1^2 - \omega_2^2)}qx^2\right], \tag{2}$$

with no singular denominator in the physical region; the equivalent Hermitian interaction is $h_1 = \frac{1}{2}(v + v^\dagger) = s(3c^2y^2p - s^2p^3)$. (Verification: substitute (2) into $[h_0, \mathcal{W}[f]] = i\mathcal{W}[\{h_0, f\}]$, exact for quadratic h_0.)

4 Second order: generic existence

Theorem 4.1 (Generic second-order deformation)**.** *For incommensurate frequencies the second-order obstruction $\Pi_{\ker \mathrm{ad}_{h_0}} \frac{1}{2}[R_1, v+v^\dagger]$ vanishes; the solved R_2 is Hermitian, and the assembled $h_\zeta = e^{-X/2}\star(h_0+\zeta v)\star e^{X/2}$, $X = \zeta R_1 + \zeta^2 R_2$, is Hermitian through $O(\zeta^2)$, verified end to end in exact Moyal calculus (including the $\frac{1}{8}[[h_0, R_1], R_1]$ term and the Λ^3 contribution). R_2 carries the resonance denominators $(\omega_1 - \omega_2)$ and $(\omega_1 - 3\omega_2)$; no others can vanish for $\omega_1 > \omega_2 > 0$. (Monomial system: Appendix A.)*

5 The exact 3:1 obstruction

Theorem 5.1 (Second-order obstruction at 3:1)**.** *At $\omega_1 = 3\omega_2$ the first-order deformation still exists, but the second-order class is nonzero and exact:*

$$\mathfrak{o}_+^{(2)} = \frac{27\sqrt{3}}{320\,\omega_2^4}\left(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3\right), \tag{3}$$

the anti-Hermitian on-shell conversion of one mode-1 quantum into three mode-2 quanta. The class is gauge independent under the full relevant first-order freedom: additions to R_1 of Hermitian commutant elements including the quadratic invariants N_1, N_2, N_1N_2 and the resonant quartic operators $a_1 a_2^{\dagger 3} + a_1^\dagger a_2^3$ *and $i(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3)$, and anti-Hermitian first-order generators:* $\Pi_{\ker \mathrm{ad}_{h_0}}[G, v+v^\dagger] = \Pi_{\ker \mathrm{ad}_{h_0}}[v - v^\dagger, G] = 0$ *for all such G. (Second-order kernel additions to R_2 commute with h_0 and cannot affect the second-order class; they enter first at third order.) Derivation: Appendix B.*

There is thus no analytic second-order deformation of the canonical positive metric at 3:1, within the Weyl-algebra class, continuously connected to the free metric.

6 Resonant PT breaking

Proposition 6.1 (The antilinear symmetry)**.** *Let $\mathcal{A} = \Pi \circ \Theta$, where Θ is complex conjugation in the (x, y)-Schrödinger representation ($x, y \mapsto x, y$, $p, q \mapsto -p, -q$, $i \mapsto -i$) and Π the total parity $(x, y, p, q) \mapsto -(x, y, p, q)$. (The symbol Θ is reserved for this conjugation; it is distinct from the Krein parity generators K_n, the effective spectral matrix $K^{(2)}$, and the momentum cutoff $K_{\max}$.) Then $\mathcal{A}$ is antiunitary, $\mathcal{A}^2 = 1$, and*

$$[\mathcal{A}, h_0] = 0, \qquad [\mathcal{A}, v] = 0, \qquad \textit{hence} \qquad [\mathcal{A}, h_\zeta] = 0 \quad (\zeta \in \mathbb{R}): \tag{4}$$

h_0 is even and Θ-real; v is odd under both Π and Θ separately ($\Theta v \Theta = -v$, $\Pi v \Pi = -v$), hence invariant under their composition (machine-verified). "PT breaking" below means the appearance of eigenvalues not fixed by this antilinear symmetry, i.e. complex-conjugate pairs.

Theorem 6.2 (Certified complex pair in the $E = 27\omega_2$ multiplet)**.** *At $\omega_1 = 3\omega_2$ ($\omega_2 = 1$), the second-order effective matrix on the ten-dimensional shell $\{|j, 27-3j\rangle\}_{j=0}^{9}$ is real tridiagonal with antisymmetric off-diagonal $K_{j,j+1} = -\frac{27\sqrt{3}}{640}\sqrt{(j+1)n(n-1)(n-2)}$, $n = 27 - 3j$, and explicit rational diagonal (Appendix C). Because only the products $K_{j,j+1}K_{j+1,j}$ enter, its* *characteristic polynomial has* rational *coefficients; an exact Sturm count shows it has exactly eight real roots, hence exactly one complex-conjugate pair, with certified thirty-digit enclosure*

$$\kappa_\pm = 2.448807382199383966\ldots \pm 0.224586593808125108\ldots\, i. \tag{5}$$

Exact diagonalization of the truncated $h_0 + \zeta v$ *at* $\zeta = 0.02$ *(two cutoffs) exhibits the pair with* $\operatorname{Im} E = \pm 0.2246\,\zeta^2$ *to* 0.5%*. The lowest resonant doublet* $\{|1,0\rangle, |0,3\rangle\}$ *remains real.*

Corollary 6.3. *Wherever the complex pair persists in the interacting spectrum, no positive-definite invariant metric exists — analytic in* ζ *or not. The hierarchy of claims is: (1) the complex pair in the second-order effective resonant-shell matrix is* exact *(Sturm certificate); (2) its persistence in the interacting truncations is supported by cutoff-stable direct diagonalization; (3) an operator-theoretic statement for the full unbounded cubic Hamiltonian remains open.*

Proposition 6.4 (Broken-PT tongues of the effective multiplet)**.** *Within the second-order effective multiplet, detuning* $\delta_r = \omega_1 - 3\omega_2$ *adds* $j\,\delta_r$ *to the shell diagonal, and the spectrum is complex for* $|\delta_r| < \delta_c(\zeta) = 38.4151\ldots \times \zeta^2 + o(\zeta^2)$*. That an order-n resonance produces an* $O(\zeta^n)$ *tongue is the corresponding general perturbative expectation, not a proved proposition.*

7 Third order and the selection rules

Theorem 7.1 (Transfer-lattice selection rule)**.** *Grade monomials by energy transfer,* $[h_0, M] = [(\Delta n_1)\omega_1 + (\Delta n_2)\omega_2]M$*. Order-n objects have transfers in the n-fold sumset of the vertex transfer set, total degree* $\le n+2$*, and degree parity* $\equiv n \bmod 2$ *(proof: Appendix D). The order-n obstruction is supported on the finite candidate sets:* $\{2{:}1\}$ *at order 1,* $\{3{:}1\}$ *at order 2,* $\{3{:}2, 2{:}1, 4{:}1\}$ *at order 3, with* 5:1 *first reachable at order 4. The lattice gives candidates; coefficients decide.*

Theorem 7.2 (Third order)**.** *At generic frequencies the third-order obstruction vanishes and* R_3 *is Hermitian with odd total degrees* ≤ 5*. At* $\omega_1 = \frac{3}{2}\omega_2$ *the class is nonzero; at* $(\omega_1, \omega_2) = (3,2)$ *its exact value is*

$$\mathfrak{o}_+^{(3)} = -\frac{117\sqrt{30}}{1120}\, i\,\big(a_1^2 a_2^{\dagger 3} + a_1^{\dagger 2} a_2^3\big), \tag{6}$$

and the coefficient scales exactly as $\omega_2^{-13/2}$ *(verified by the ratio* $2^{-13/2}$ *between* $(\omega_1, \omega_2) = (6,4)$ *and* $(3,2)$*), so the general form carries the dimensional factor* $\omega_2^{-13/2}$*. The class is gauge independent in the sense of Theorem 5.1. The candidates* 2:1 *(orders 1 and 3) and* 4:1 *(order 3) have vanishing coefficients.*

Conjecture 7.3 (Resonance hierarchy)**.** *The coprime ratio* $\omega_1/\omega_2 = p{:}q$ *first obstructs at order* $p + q - 2$ *when the mode-2 transfer p is odd. (Verified:* 3:1 *at order 2,* 3:2 *at order 3, and* 5:1 *at order 4 — Proposition 7.4; externally reproduced exact instances:* 5:3 *and* 7:1 *at order 6 — Proposition 7.5. The mechanism excluding even mode-2 transfers and the all-coprime statement remain open.) The resonance set is an expanding hierarchy, not (so far) a dense set.*

Computational Proposition 7.4 (The 5:1 obstruction at fourth order)**.** *At* $(\omega_1, \omega_2) = (5,1)$ *the order-2 and order-3 kernel projections vanish (the selection rule makes* 5:1 *unreachable* *below order 4) and R_2, R_3 exist and are Hermitian. The fourth-order class is nonzero, exactly*

$$\mathfrak{o}_+^{(4)} = -\frac{203125\sqrt{5}}{2341011456}\big(a_1 a_2^{\dagger 5} - a_1^\dagger a_2^5\big) = -\frac{13 \cdot 5^{13/2}}{2^{16}\, 3^6\, 7^2}\big(a_1 a_2^{\dagger 5} - a_1^\dagger a_2^5\big), \tag{7}$$

the on-shell $1 \leftrightarrow 5$ conversion on the predicted monomial, gauge independent in the sense of Theorem 5.1. The coefficient scales exactly as ω_2^{-9} (ratio 2^{-9} between $(10,2)$ and $(5,1)$), extending the dimensional series $\mathfrak{o}_+^{(n)} \propto \omega_2^{-(5n-2)/2}$ for $n = 2,3,4$; and $R_4 = O(\epsilon^{-6})$ toward the Jordan boundary, the fourth data point of $R_n = O(\epsilon^{-3n/2})$ (Section 8). The order-n sources are generated programmatically from the adjoint series of Appendix A, and the same code path re-derives the order-2 and order-3 classes exactly before running order 4. (Checks FO1–FO9, `verify_51_order4.py`.)

Computational Proposition 7.5 (The 5:3 and 7:1 obstructions at sixth order). *At $(\omega_1, \omega_2) = (5,3)$ and $(7,1)$, the exact kernel projections at orders two through five vanish and the corresponding $R_2, \ldots, R_5$ are Hermitian. The first nonzero classes occur at order six and are*

$$\mathfrak{o}_{+,5:3}^{(6)} = \frac{3863828151875\sqrt{15}}{6463101113204736}\big(a_1^3 a_2^{\dagger 5} - a_1^{\dagger 3} a_2^5\big), \tag{8}$$

$$\mathfrak{o}_{+,7:1}^{(6)} = \frac{28633766567\sqrt{7}}{2656254925209600000}\big(a_1 a_2^{\dagger 7} - a_1^\dagger a_2^7\big). \tag{9}$$

They are the predicted $3 \leftrightarrow 5$ and $1 \leftrightarrow 7$ on-shell conversion kernels, with the real coefficient and antisymmetric combination required by even-order parity. Rescaling $(5,3)$ to $(10,6)$ multiplies the result by exactly 2^{-14}, matching the sampled dimensional law $\mathfrak{o}_+^{(n)} \propto \omega_2^{-(5n-2)/2}$ at $n = 6$.

The evidence consists of a 31-check exact Forge battery on two code generators, an independent end-to-end Hermiticity assembly through order five, and a from-scratch SymPy recomputation agreeing term by term. The bounded import certificate pins the source commit and artifacts. This proposition does not prove the hierarchy for every coprime ratio, convergence, or nonremovability under every further admissible canonical transformation. The upstream dirty-toolchain marker is retained for provenance and independently discharged by the clean pinned replay recorded with the import.

8 The Jordan divergence hierarchy

Theorem 8.1 (Exact first-order Jordan asymptotics). *With $\omega_{1,2} = \omega \pm \epsilon$, the explicit R_1 of (2) has the exact leading Laurent behaviour*

$$R_1 = \epsilon^{-3/2}\,\mathcal{W}\left[\frac{xyp}{2\sqrt{\omega}} + \frac{y^2 q}{4\sqrt{\omega}} - \frac{p^2 q}{4\sqrt{\omega}} - \frac{x^2 q}{2\omega^{5/2}} + \frac{q^3}{6\omega^{5/2}}\right] + O(\epsilon^{-1/2}), \tag{10}$$

so $R_1 = \Theta(\epsilon^{-3/2})$, while the free generator diverges only logarithmically.

Computational Proposition 8.2 (Higher orders). *The solved R_2 and R_3 scale as ϵ^{-3} and $\epsilon^{-9/2}$ (measured decade exponents -3.00 and -4.49 over $\epsilon = 10^{-2} \to 10^{-3}$), with no small-denominator enhancement from $\omega_1 - \omega_2$ at second order. These are high-precision numerical statements at fixed orders, not proved big-O estimates.*

Conjecture 8.3. *$R_n = \Theta(\epsilon^{-3n/2})$, so the deformation's radius of convergence in ζ collapses as $\zeta_c \sim \epsilon^{3/2}$ toward the Jordan boundary.*

9 The perfect-square vertex

The interacting theory is $S = -\frac{1}{2}\int(\Box\phi + \lambda(\partial\phi)^2)^2$ [4], with cubic vertex $V_3 \propto \Box\phi(\partial\phi)^2$ and quartic $V_4 \propto \frac{1}{2}(\partial\phi)^4$.

Proposition 9.1 (Källén form)**.** *On $p_1 + p_2 + p_3 = 0$ the symmetrized cubic vertex is exactly*

$$\mathcal{V}_3 = p_1^2(p_2\cdot p_3) + p_2^2(p_1\cdot p_3) + p_3^2(p_1\cdot p_2) = \tfrac{1}{2}\lambda_{\mathrm{K}}(p_1^2, p_2^2, p_3^2). \tag{11}$$

The polynomial vertex factor therefore has a second-order zero at the massless confluent point*: $\lambda_{\mathrm{K}}(0,0,0) = 0$ and $\nabla\lambda_{\mathrm{K}}(0,0,0) = 0$, while the Hessian is nonzero. On the regulated decay channel, $\lambda_{\mathrm{K}}(m_1^2, m_2^2, m_2^2) = m_1^2(m_1 - 2m_2)(m_1 + 2m_2)$: the vertex vanishes exactly at threshold and turns on continuously above it. (A full external-state selection rule for generalized Jordan legs would additionally require the confluent limit of leg normalizations, shell measures, and mode factors; we do not claim it here.)*

10 First order in the regulated field theory

Regulate by $P_\delta = (\Box + m_1^2)(\Box + m_2^2)$, $\delta = m_1^2 - m_2^2 > 0$, at finite volume (Remark 2.2). In the mode expansion every leg is on its branch shell, so on the kernel the vertex coefficient is $\frac{1}{2}\lambda_{\mathrm{K}}$ of the branch masses. The transported field mode, derived from the free Dyson transport and verified against the spectral Wightman function of [14] (Appendix E), is

$$\phi'(k) = \frac{i}{\sqrt{2\delta}}\left[\frac{a_2 + a_2^\dagger}{\sqrt{\omega_2}} + \frac{a_1 - a_1^\dagger}{\sqrt{\omega_1}}\right], \qquad \sigma = \sqrt{\delta}\ (k\text{-independent}). \tag{12}$$

Theorem 10.1 (First-order protection; sectorwise parity obstruction)**.** *(i)* Even-ghost selection rule*: the transported reality factors make $v^\dagger - v$ vanish on every odd-ghost-count monomial (Appendix E). Even-ghost shells are kinematically closed for all $m_1 > m_2 > 0$. Hence $\mathfrak{o}_+^{(1)} = 0$ identically — below and above the decay threshold. (ii) The sectorwise branch parity $\kappa_0 = (-1)^{N_{\mathrm{ghost}}}$ has $\mathfrak{o}_K^{(1)} = 0$ below threshold and, above it, a nonzero class on the open $1 \to 22$ shell, equal there to $\lambda_{\mathrm{K}}(m_1^2, m_2^2, m_2^2)$ times the leg normalizations.*

Computational Proposition 10.2 (Split-basis scaling)**.** *At massive confluence both R_1 and K_1 diverge as $\delta^{-3/2}$; on massless Jordan paths $m_2^2 = \alpha\delta$ the positive R_1 scales as $\delta^{-1/2}$ with path-independent power, while the Krein K_1 is $\delta^{-1/2}$ generically and $\delta^{-3/2}$ on the exceptional path $\alpha = 2/7$. These are statements about coefficients in the split mode basis, which itself degenerates at $\delta = 0$; a basis-independent formulation on confluent states is deferred.*

11 The exact two-field rewriting

The two-field form below is, up to normalization and sign conventions, the $O(1,1)$ embedding of Bateman and Turok [4], who introduced the pair (Ω, Υ), identified the full $O(1,1)$ symmetry including the exchange as ghost parity and internal charge conjugation, and constructed the quantum embedding relating the two theories. The theorem restates their embedding in our conventions; the results specific to this section and the next two are the chart-transition description of the exchange, the null-Jordan-sector and interaction-generated Jordan statements, the location of the sectorwise obstruction inside this structure, and the confluent-parity bridge.

Theorem 11.1 (Two-field form and exchange symmetry [4, adapted])**.** *Introducing the auxiliary field ψ and the change of variables $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$ (unit pointwise Jacobian),*

$$\mathcal{L} = -\partial_\mu U\, \partial^\mu V + \frac{\lambda^2}{2}U^2V^2 = -\tfrac{1}{2}(\partial X)^2 + \tfrac{1}{2}(\partial Y)^2 + \frac{\lambda^2}{8}(X^2 - Y^2)^2, \qquad X, Y = \tfrac{U \pm V}{\sqrt{2}}. \quad (13)$$

The exchange $U \leftrightarrow V$ is an exact symmetry of the extended two-field theory. *In the original variables it is the transition map between the charts $\phi_A = \frac{1}{\lambda}\log U$ and $\phi_B = \frac{1}{\lambda}\log V$,*

$$(\phi, \psi) \;\longmapsto\; \left(\tfrac{1}{\lambda}\log\tfrac{\psi}{\lambda} - \phi,\ \psi\right), \quad (14)$$

defined on the chart overlap ($U \neq 0 \neq V$), singular at the perturbative vacuum $\psi = 0$, and requiring a branch choice in the complexified theory. The original (ϕ, ψ) chart covers only $U \neq 0$ and does not contain the mirror background: the exchange is a global symmetry of the extension, not an everywhere-defined involution of the original field space, and it is invisible at any finite perturbative order around the original vacuum.

Bateman and Turok warn, and we adopt the caveat, that the two theories are inequivalent beyond perturbation theory: the fourth-order path integral corresponds to $U > 0$, while the two-field model integrates over all U and has an indefinite kinetic form [4]. Statements about the mirror background $(0, 1)$ are therefore statements about the extended theory; whether both null branches are physical sectors of one completion is open.

Proposition 11.2 (Null sectors and interaction-generated Jordan dynamics)**.** *The constant zero-action stationary backgrounds form the two null branches $UV = 0$ (whether they are vacua in a given completion is a separate question, since the kinetic form is indefinite). The perturbative background $(U, V) = (1, 0)$ is a nonzero null configuration; the exchange maps it to $(0, 1)$. Expanding about $(1, 0)$: $\mathcal{L}^{(2)} = -\partial u\partial v + \frac{g}{2}v^2$, $\mathcal{L}^{(3)} = g\, uv^2$, $\mathcal{L}^{(4)} = \frac{g}{2}u^2v^2$ ($g = \lambda^2$), with equations $\Box v = 0$, $\Box u = gv$: the exact $\Box^2$ Jordan pair, the off-diagonal supplied by the interaction coupling; about $(0, 1)$ the mirror expansion has the oppositely oriented chain. A chain-preserving involution commuting with a single Jordan block is $\pm I$ (companion lemma), so exchange between oppositely oriented sectors is the only possible realization of a parity-like symmetry. In this frame the interactions are pure potentials: $H_{\rm int} = -L_{\rm int}$ exactly, with no Ostrogradsky–Legendre corrections.*

12 The sectorwise second-order obstruction

Regulate the sector-A theory by $\mathcal{L}_0 = -\partial u\partial v - \mu^2 uv + \frac{g}{2}v^2 + \frac{\epsilon}{2}u^2$, with masses $m_\pm^2 = \mu^2 \pm \sqrt{\epsilon g}$ and branch eigenvectors $v = \rho_b u$, $\rho_\pm = \mp\delta/(2g)$, $\delta = m_+^2 - m_-^2$ (the regulator necessarily breaks the exchange symmetry). All computations below set $g = 1$. The positive-frame quantization and the complete normalization conventions — finite volume with unit-normalized discrete modes $[a, a^\dagger] = 1$, unit-norm external states, lattice momenta, $m_L = 4$ — are collected in Appendix E. The second-order on-shell source element is computed from the exact matrix-element form

$$\langle \text{out}|\, \mathcal{S}_+^{(2)}\, |\text{in}\rangle = \langle v_2^\dagger - v_2\rangle + \sum_n \frac{\langle \text{out}|v_1^\dagger|n\rangle\langle n|v_1^\dagger|\text{in}\rangle - \langle \text{out}|v_1|n\rangle\langle n|v_1|\text{in}\rangle}{E - E_n}, \quad (15)$$

derived in Appendix E from $(v^\dagger - v)(v + v^\dagger) + (v + v^\dagger)(v^\dagger - v) = 2(v^\dagger v^\dagger - vv)$.

Theorem 12.1 (Sectorwise obstruction on an open shell subset)**.** *Consider the branch-changing scattering $H+L \to L+L$, kinematically open at suitable momenta for every $m_H > m_L$. At the rational center-of-mass point $m_H = \frac{3}{2}m_L$ (both incoming particles at rest; outgoing momenta $\pm\frac{3}{4}m_L$; in lattice units $m_L = 4$, $m_H = 6$, momenta ± 3, all shell conditions exact), the source element* (15) *has the exact value*

$$\boxed{\mathcal{M}_{\text{obs}} = \frac{401\sqrt{6}}{39424\, g^2} \xrightarrow{g=1} \frac{401\sqrt{6}}{39424}} \tag{16}$$

(the $1/g^2$ dependence at fixed masses is verified exactly: each vertex carries $g\rho^2 \propto \delta^2/g$), with the quartic contact piece exactly zero. The computation is truncation-complete (Lemma 12.2) and the element is analytic in the shell data on the nonsingular part of the shell; hence the second-order obstruction is nonzero on a nonempty open subset of the shell containing this point. *Independent tuned kinematic points give* 0.5233, 0.3109, 0.3970 *(numerical), consistent with nonvanishing on a large region; a characterization of the zero set is left open, and no claim is made for every mass pair.*

Lemma 12.2 (Truncation completeness)**.** *For external momenta $\{k_a, k_b\} \to \{k_c, k_d\}$ pairwise distinct, every intermediate Fock state contributing to* (15) *has all momenta in the finite set generated by the externals and their pairwise sums and differences ($s = k_a + k_b$, $t = k_a - k_c$, $u = k_a - k_d$): in each contributing chain the surviving spectator particles must coincide with external particles, which fixes every internal momentum; free-momentum loops arise only in forward/disconnected kinematics, excluded here. (Proof: Appendix E; corroborated by exact cutoff independence between $K_{\max} = 4$ and 6.)*

Remark 12.3 (Oscillator versus field theory)*.*

PU oscillator	field theory
discrete energy spectrum	continuous momentum spectrum
isolated rational resonance	generic scattering shell
3:1 first appears at order 2	$H + L \to L + L$ appears at order 2
finite resonant multiplet	continuum of degenerate in/out states

The continuum turns the discrete resonance obstruction into a scattering-shell obstruction. The exact exchange symmetry does not cure it: it maps the obstructed sector-A construction to the mirror sector-B construction.

Computational Proposition 12.4 (Krein consistency of the obstructed element)**.** *Let $G = (-1)^{N_{\text{ghost}}}$ be ghost-branch number parity in the positive-frame Fock space, and let ι be the mode identification $(b,k)_B \simeq (b,k)_A$ of the mirror sector. At the rational kinematic point of Theorem 12.1, exactly: (i) $H_B = \iota H_A \iota^{-1}$ (the mirror sector is the same theory in the mirrored frame) and $H_A^\dagger = G H_A G$ term by term, so the mirror-adjoint relation $H_B = W H_A^\dagger W^\dagger$ holds with $W = \iota \circ G$ — i.e. the "doubled pairing" $\eta_{\text{dbl}} = \text{offdiag}(W^\dagger, W)$ is precisely the two-sector unfolding of the ghost-parity Krein metric of [4], not additional structure. (ii) On the degenerate shell $\{|H(0)L(0)\rangle, |L(3)L(-3)\rangle\}$ (exhaustively the whole reachable shell) the second-order on-shell T satisfies $GTG = T^\dagger$ exactly, while Hilbert Hermiticity fails: the parity-even (diagonal) block is real, and the entire obstruction is the parity-odd block, $T(\text{out}, \text{in}) = -401\sqrt{6}/78848 = -\mathcal{M}_{\text{obs}}/2$. (iii) Consequently the finite-time interaction-picture S obeys $S_B^\dagger W S_A = W$ identically — Krein pseudo-unitarity is exact on the same matrix elements on* *which every positive pointed metric is obstructed. A positive H-invariant graph subspace of the doubled space exists if and only if a positive pointed metric exists (standard J-self-adjoint graph argument [11], instantiated and machine-verified), so doubling preserves pseudo-unitarity but cannot evade the obstruction. (Checks DQ1–DQ9, `verify_doubled_theory.py`.)*

The obstruction is therefore carried exactly by the κ-odd component of the on-shell T — the component that the weak-ghost-symmetry positivity mechanism of [4] does not use. Whether it maps into their null orthogonal (C) component under their asymptotic embedding is the natural next question (Section 14).

13 The confluent parity theorem

Theorem 13.1 (Confluent limit of the regulated parity)**.** *Let $P_\delta b_\pm = \pm b_\pm$ be the regulated branch parity and $c = (b_+ + b_-)/2$, $d = (b_+ - b_-)/\delta$ the confluent Jordan basis, so $P_\delta c = \frac{\delta}{2} d$ and $P_\delta d = \frac{2}{\delta} c$; in the standard column convention on the ordered basis (c, d),*

$$P_\delta = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}, \qquad P_\delta^2 = 1, \tag{17}$$

with no bounded limit as $\delta \to 0$: the regulated parity cannot converge as a chain-preserving parity inside one Jordan sector. On the doubled space with the oppositely oriented *confluent identification in the mirror sector ($c_B = (b_+ - b_-)/2$, $d_B = (b_+ + b_-)/\delta$), the cross parity $b^A_\pm \leftrightarrow \pm b^B_\pm$ equals, exactly and δ-independently, the sector exchange $(c_A, d_A) \leftrightarrow (c_B, d_B)$. The regulated branch parity thus converges, after doubling by the oppositely oriented Jordan sector, to* the linearization of the exact sector-exchange involution around the two mirror backgrounds. *(Identifying the full nonlinear quantum operator requires a separate construction, not supplied here.)*

14 Discussion

The mechanism. Away from resonances the positive pseudo-Hermitian metric deforms locally, with explicit generators. The failures are structural: on-shell branch-changing processes produce gauge-independent cohomological obstructions — isolated rational loci for the oscillator, generic scattering shells in the field theory; at the oscillator resonances the obstruction is certified spectral; and independent of resonances, the deformation loses uniformity toward Jordan degeneration. The sectorwise branch parity is broken by real ghost decay above threshold; what survives at the massless perfect-square boundary is the exact sector-exchanging involution of the extended two-field theory, whose linearization is the bounded confluent limit of the regulated parity on the doubled space.

Symmetry enhancement at the boundary. The massless perfect-square point is not the split theory with parameters set to zero: the Källén vertex degenerates there, the doubled $O(1, 1)$ structure emerges, and the Jordan dynamics is generated by the interaction about its own null backgrounds — a boundary-emergent structure mirroring the Weyl enhancement at the conformal boundary of the gravity companion [15].

Separation of the two completions. Proposition 12.4 sharpens the free-theory picture of the companion series: in the interacting theory the two real forms are not merely distinguished at Jordan degenerations — they *separate*. The doubled Krein pairing is exactly preserved at the split rational shell ($GTG = T^\dagger$ on shell, finite-time pseudo-unitarity), while every analytic positive pseudo-Hermitian completion of the pointed sector is obstructed on an open shell subset. This already excludes identifying that Krein pairing with an ordinary positive pseudo-Hermitian completion. It does *not* establish the stronger Bateman–Turok weak-ghost-symmetry positivity mechanism at split mass: Proposition 14.2 shows that the regulated vacuum image contains both charge signs there. The sharpest remaining form of the reconciliation is therefore a boundary transport question: under their asymptotic embedding, physical operators decompose into a ghost-symmetric part and a null orthogonal part, and only the former contributes to their generalized Born rule [4]; since the obstruction is κ-odd (Proposition 12.4), the conjecture is that it is carried into the null component — obstructing every positive metric while remaining invisible to the Krein probabilities. Verifying this on the computed $H + L \to L + L$ element is the concrete next calculation; the lemma and the proposition below supply its verified ingredients.

The charge-null mechanism. Throughout this paragraph $\dagger$ denotes the Krein adjoint and τ the Krein trace. The following finite-particle lemma makes the null-relocation mechanism self-contained (it is the statement used in [4], whose general proof is deferred to their companion paper; for boost weights it is a three-line grading argument).

Lemma 14.1 (Charge null)**.** *Let a Krein space carry an $SO^+(1,1)$ charge grading whose pairing couples charge q only to charge $-q$, and suppose the Krein adjoint preserves charge — as it does for boost weights, where c and $c^\dagger$ of one field carry the* same *charge. Then $\tau(A_q) = 0$ for every $q \neq 0$. Consequently an operator $C = \sum_{q<0} C_q$ containing only strictly negative charges satisfies $\tau(C^\dagger C) = 0$ and $\tau(B^\dagger C) = 0$ for every neutral B: C is null and orthogonal to the neutral sector, and the generalized Born trace of a weakly ghost-symmetric process depends on its neutral part alone.*

Proof. Boost invariance of the trace gives $\tau(A_q) = \tau(\alpha_\sigma A_q) = e^{q\sigma}\tau(A_q)$ for all σ, hence $\tau(A_q) = 0$ for $q \neq 0$; equivalently, on a charge-graded Fock space a grading-raising operator is traceless. Charge preservation under the adjoint gives $q(C_q^\dagger C_r) = q + r < 0$ and $q(B^\dagger C_r) = r < 0$ term by term. $\square$

Computational Proposition 14.2 (Regulated embedding and its charge structure)**.** *At the rational kinematic point, per momentum sector, the regulated split theory admits a canonical Bogoliubov map onto the cross-paired charge basis of a neutral reference theory $-\partial u\,\partial v - \tilde{\mu}^2 uv$ (operators c_U, c_V with the only nonvanishing commutators $[c_U, c_V^\dagger] = [c_V, c_U^\dagger]$; charges $q(c_U^{(\dagger)}) = +1$, $q(c_V^{(\dagger)}) = -1$, the analogue of Bateman–Turok's b_Ω, b_Υ). The pointed vacuum maps to a Gaussian state whose squeezing components obey, exactly and for every reference dispersion,*

$$\frac{S_{UU}}{S_{VV}} = \left(\frac{\delta}{2g}\right)^2 = \frac{\epsilon}{g}, \qquad \delta^2 = 4\epsilon g, \tag{18}$$

i.e. the charge $+2$ component is sourced by the regulator $\frac{\epsilon}{2}u^2$ and the charge -2 component by the interaction-generated $\frac{g}{2}v^2$. The embedding is strictly one-sided — no positive charges *in the vacuum image, the hypothesis of Lemma 14.1 for the mapped theory — if and only if $\epsilon = 0$: the $O(1,1)$-symmetric confluent line. The massless Bateman–Turok theory is the point $\epsilon = \mu^2 = 0$ on that line. At confluence the matched-reference limit is $S_{VV} \to -g/(4w^2)$, or $-1/(4w^2)$ in the $g = 1$ normalization used in the computation; at $\mu^2 = 0$, where $w = |\mathbf{k}|$, this is precisely the coefficient structure of their Eqs. (C5)–(C6). At split masses the exact ratio above rules out removing the $+2$ component by changing the reference dispersion. Under the residual charge-preserving Bogoliubov freedom, numerical solution of the adaptation equations runs to a degenerate (confluent-type) frame rather than finding a bounded adapted frame. (Checks ON1–ON4, `verify_obstruction_null.py`.)*

The proposition is the charge-frame image of the first-order result of Section 10: real ghost decay above threshold breaks the sectorwise branch parity, and in the charge frame the same ϵ sources the charge $+2$ contamination that violates weak ghost symmetry. The relocation of the κ-odd obstruction into a null negative-charge component is therefore exact on the one-sided confluent line, with the Bateman–Turok identification at its massless point, and holds at split masses with charge contamination of relative size ϵ/g (not assumed small away from the boundary). The remaining, precisely-posed step — the obstruction-to-null theorem — is the boundary Born-trace evaluation: the neutral component's blindness to the obstruction coefficient and its limit along a specified on-shell family with $(\epsilon, \mu^2) \to (0, 0)$. One exact rational family is $m_L = 4s$, $m_H = 6s$, $|\mathbf{k}_{\text{out}}| = 3s$, $\mu^2 = 26s^2$, $\epsilon g = 100s^4$, followed by $s \to 0$; the process operator must be transported along this family rather than held fixed while only the embedding is varied. Lemma 14.1 makes that step self-contained.

That the transport is essential, rather than a technical scruple, can be quantified. Evaluating the generalized Born trace of [4] directly on the two-dimensional shell of Proposition 12.4, with the process operator held fixed at the split-mass rational point, gives $\tau(T^\dagger T) = \|T_+\|^2 - \|T_-\|^2$ — the κ-graded components being orthogonal in the shell trace — with

$$\|T_-\|^2 = 2\Big(\frac{401\sqrt{6}}{78848}\Big)^2 = \frac{482403}{1554251776} \neq 0. \tag{19}$$

Lemma 14.1 sends exactly this quantity to zero. A truncated shell carries no boost action, so the lemma's hypothesis is absent and its conclusion fails by precisely the obstruction squared; the fixed-shell trace is nonetheless positive, but only because $\|T_+\|^2$ exceeds $\|T_-\|^2$ by a factor of about 26, which is a statement about the relative size of two matrix elements and not about nullity. A fixed-shell evaluation therefore cannot decide the obstruction-to-null theorem in either direction. We record it because the positive sign is easy to mistake for confirmation. (Machine-checked in `REVERSE_PHYSICS_BT_BORN_TRACE_V1`; the exact value of $\|T_+\|^2$ and the positivity witness are given there, and the same certificate repairs a floating-point contamination of the diagonal of T in `verify_doubled_theory.py` which affected no published quantity, every such quantity being an off-diagonal element or a reality statement.)

One structural simplification is worth recording alongside it: on a finite shell the weak ghost symmetry of [4] — existence of a decomposition $A = B + C$ with B ghost symmetric and C null and orthogonal to B — is *equivalent* to the single inequality $\|A_-\| \leq \|A_+\|$, since the κ-eigenspaces are orthogonal in the shell trace and $\tau(A^\dagger A) = \|A_+\|^2 - \|A_-\|^2$. There is nothing to search for. This does not extend to the charge-graded trace, where Lemma 14.1 rather than a norm comparison carries the positivity.

Complementarity of the two premises. The separation above is easily read as a scoreboard, and it is not one. Each completion rests on a premise that fails somewhere the other's holds, and the two failures occur on *different* boundaries. The positive pseudo-Hermitian completion requires the dynamics to be diagonalizable with real spectrum; that fails precisely on the coincident-pole locus, where the similarity transformation is singular and only the Krein real form continues as a nondegenerate completion [15]. The Krein completion requires the internal charge to be one-sided; Proposition 14.2 shows this holds exactly iff $\epsilon = 0$, and at split masses both signs are present. Neither premise is generic. They are boundary conditions on different boundaries, so neither completion is a specialization of the other and neither is discardable in favour of it.

Two consequences follow, and the second is the one that bears on conformal gravity. First, the free-level agreement of [15] is not a weaker form of the interacting separation: it says that any dispute between the two completions conducted at free-field level concerns the involution and not the predictions, since both induce the same functional on the gauge-invariant algebra. Second, at loop order neither completion reaches the Jordan locus — which is where a pure $1/k^4$ propagator sits. The positive pseudo-Hermitian route withdraws the standard cutting rules there by its own account ([3], Sec. VI; the argument is set out in [15]), while [4] establish positivity at tree level and name collinear infrared divergence of the massless theory as the obstacle to higher orders. The two limitations are unrelated in origin and coincide in location.

That scalar limitation admits an exact statement on one complete declared carrier.

Remark 14.3 (Axis-compatible collinear normalization obstruction). At order λ^6, on the complete final-state collinear logarithmic normalization carrier, differential in the nonsingular outer two-body solid angle, the three identical real-emission pair boundaries respond to a common daughter mass-ratio normalization $r \mapsto cr$ as

$$\Delta_c \frac{d\sigma_{\text{real}}}{d\Omega} = \frac{3\lambda^6}{512\pi^4 s} \log c. \tag{20}$$

Let the recombined virtual parent regulator be $G(x, y) = xg(y/x)$, with g continuous at zero and $0 < |g(0)| < \infty$, while the spectator regulators are held fixed. Then the complete virtual external-mass logarithmic response is zero at constant order. Hence the real and virtual normalization responses do not cancel on this axis-compatible class. The statement has dependency tag `REDUCED-MODE`; it is not a complete NLO probability or a theorem against distributional, dressed-state, enlarged-degenerate-state, or resummed constructions.

Certificate summary. For $x_i = \delta a_i$ and pair invariant $t = \delta\tau$, the exact three-spectator square-free jet of the complete 25-graph tree gives

$$[a_2 a_3 a_4]C^2 = \frac{3(a_0 - a_1)^2\big((a_0 - a_1)^2 - 2\tau(a_0 + a_1)\big)}{8\tau^3}, \qquad A_5 = \frac{M_5}{8\lambda^3} = \delta^2 C + O(\delta^3). \tag{21}$$

The splitting fraction and outer scattering ratio cancel identically, so the inner solid angle contributes 4π. The exact threshold finite part shifts by $-(3/8)\log c$. Restoring the five-delta-prime sign $(-1)^5$, the $1/(2!3!)$ projector weight, factorized three-body phase space, and $|M_5|^2 = 64\lambda^6 A_5^2$ gives $\lambda^6 \log c/(512\pi^4 s)$ per unordered pair and therefore (20). The certified virtual boundary rate is

$$\frac{d\sigma_{\text{virt,boundary}}}{d\Omega} = \frac{3\lambda^6}{128\pi^4 s} \sum_{i=1}^{4} \log \frac{-\mu^2}{X_i}. \tag{22}$$

For an axis-compatible parent, $G(x, cy)/G(x, y) = g(cr)/g(r) \to 1$ as $r = y/x \to 0$; its logarithm therefore has zero constant response. The hard Mandelstam-ratio logarithm also has zero daughter-ratio response, while continuous cut-free terms and local analytic counterterms cannot supply one. Both the dot-product graph producer and a method-distinct invariant-Källén verifier establish these identities exactly in the axis-gluing certificate cited above.

Remark 14.4 (Finite coherent collinear projector transport). Let $B = 3\lambda^4/(32\pi^2 s)$ be the Born rate and set $\eta = \lambda^2 \log c/\pi^2$. The absolute real response $\lambda^6 \log c/(512\pi^4 s)$ per pair is therefore $B\eta/48$, not a dimensionless $1/512$ projector coefficient. Place one hard channel and the three unordered collinear-pair channels in the carrier $(h, r_{12}, r_{13}, r_{23})$ and let

$$P_0 = \mathrm{diag}(1, 0, 0, 0), \qquad K = \begin{pmatrix} 0 & -a & -a & -a \\ a & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ a & 0 & 0 & 0 \end{pmatrix}, \qquad a = \frac{\sqrt{3}}{12}.$$

Then $K^T = -K$, and the exact formal transport $P(\epsilon) = e^{\epsilon K} P_0 e^{-\epsilon K}$ obeys $P(\epsilon)^2 = P(\epsilon)$ and $\mathrm{tr}\, P(\epsilon) = 1$ through order two. Its order-two diagonal response is

$$(P_2)_{hh} = -3a^2 = -\frac{1}{16}, \qquad \sum_{r=1}^{3} (P_2)_{rr} = 3a^2 = \frac{1}{16}.$$

More generally, if the order-one hard–collinear block is A, the equation $P^2 = P$ uniquely forces $(P_2)_{hh} = -AA^T$ and $(P_2)_{cc} = A^T A$; the displayed negative term is therefore normalization, not an adjustable weight. Multiplication by $B\eta$ gives the physical hard response $-3\lambda^6 \log c/(512\pi^4 s)$ and cancels all three real pairs.

There is nevertheless an exact dynamical obstruction to the ordinary dressing ansatz. For massless collinear daughters the published cubic vertex has $\mathcal{M}_3 = -i\lambda t^2$, whereas $\Delta E = t/(2E) + O(t^2)$. Hence the ordinary Dyson/Fock kernel $\mathcal{M}_3/\Delta E$ vanishes as $t \to 0$ and its Gram cannot be $1/48$. Direct expansion of the published composite map sharpens this statement. As printed, its Appendix C mode expansion cannot imply both associated oscillator formulas; exchanging the ordinary and growing labels in that mode expansion repairs both. With that declared repair the complete resonant two-annihilator order-λ coefficient follows from $R^\dagger \Omega R = \lambda^{-1} e^{\lambda\phi}$ and $R^\dagger \Upsilon R = \lambda^{-1} e^{-2\lambda\phi} \Box e^{\lambda\phi}$. Although its intermediate a-basis entries contain t and t^2, every such term cancels after inversion onto the (Ω, Υ) carrier. A formal finite-wave-packet anti-Krein cubic generator therefore exists without fitting its norm.

This does not yet give the displayed $1/48$. Contracting the ordered daughter slots of the derived kernel with the off-diagonal BT one-particle form, before Bose and phase-space factors, produces a cross Gram proportional to

$$-\frac{E^2(e_1^2 + e_2^2 - e_1 e_2)}{2e_1^3 e_2^3}, \qquad E = e_1 + e_2,$$

which has cubic poles at both splitting endpoints. The continuum distributional extension, oscillatory creation and vacuum-squeeze sectors, background charge bookkeeping, complete incoming projector, and physical NLO probability remain unconstructed. The bare broken-vacuum cubic has charge -1, so neutrality also cannot be inferred before that missing bookkeeping is supplied.

The endpoint extension has an exact classification. The dimensionless cross shape is exactly $-\frac{1}{2}[z^{-3}+(1-z)^{-3}]$. Reflection-even scaling-degree-three extensions retain three independent endpoint distributions. In particular, the triple-plus prescription evaluates to zero on the inclusive constant test, while the symmetric cutoff finite part evaluates to $1/2$; adding $c_0(\delta_0+\delta_1)$ shifts that value by $2c_0$. Thus $c_0=1/96$ would reproduce $1/48$, but only by fitting it. The omitted oscillatory and squeezed sectors must dynamically fix the endpoint constants before the finite projector becomes a prediction.

On the published fixed-vacuum oscillator grading, charge selection excludes that last proposed source: the oscillatory $a_1^\dagger$ term maps to $b_\Upsilon^\dagger$ with charge -1, while every term in Q_t has charge -2; the BT dagger preserves these charges and the certified inclusive kernel preserves the strictly negative trace radical. This statement is conditional on that grading and does not transfer to the covariant broken-vacuum carrier constructed below. The apparent coisometric range gap is, however, absent on the formal perturbative branch. The published free cross-CCR gives $[A_\Upsilon, A_\Omega^\dagger]=(2E)^3/(4E^2)=2E$, while the homomorphism gives the same commutator as $2E\Pi$, $\Pi=R_t^\dagger R_t$; hence $\Pi_0=1$. Expanding $\Pi^2=\Pi$ then forces every positive-order coefficient of $\Pi-1$ to vanish recursively. Formal inversion of R_t is therefore cleared, and the support projection cannot supply the missing coefficient. The exact certificate `REVERSE_PHYSICS_BT_CANONICAL_ENDPOINT_AMBIGUITY_V1` then tests the remaining algebraic conditions directly. On a carrier with one hard and three reflection-even endpoint directions, every $K(u)=\left(\begin{smallmatrix}0 & -u^T\\ u & 0_3\end{smallmatrix}\right)$ is skew, and $e^{\epsilon K}P_0e^{-\epsilon K}$ preserves the CCR, idempotence, and trace for arbitrary u. Its order-ϵ^2 hard and endpoint traces are $-u^Tu$ and $+u^Tu$. Hence the listed canonical identities admit a three-parameter countermodel to uniqueness. The choice $u=(\sqrt{3}/12,0,0)$ is compatible with $1/48$, but selecting it would be a fit until the deferred continuum map or an independent physical endpoint condition fixes u. This insufficiency witness does not claim that every u is realized by the BT dynamics.

The full off-resonant composition is determined exactly. Keep $d=e_1+e_2-E_{\text{parent}}$ through the symplectic extractor and the repaired inverse map. All explicit t, t^2 terms again cancel, $d=0$ recovers the resonant table entry by entry, and additional channels proportional to d survive. The two spatial momentum delta functions diagonalize the parent block of $K^\sharp K$ and leave the daughter measure $d^3p_1/(2\pi)^3=r^2dr\,d\Omega/(2\pi)^3$. On the soft blow-up $e_1=r$, $e_2=1$, $d=\alpha r$, the ordered cross-Gram begins $-1/(2r^3)$ independently of α, so the measured leading term is $-(1/2)dr/r$. A common cutoff rescaling $\epsilon\mapsto c\epsilon$ shifts the finite part by $(1/2)\log c$. Consequently the three local constants of the exactly collinear slice collapse to one soft normalization on this full chart, but ordinary composition remains logarithmically non-trace-class and has a universal leading response. Raising the parent index gives $-1/4$, the Bose factor gives $-1/8$, and the angular measure gives $-\lambda^2dr/(16\pi^2r)$. A cutoff rescaling therefore produces $+\lambda^2\log c/(16\pi^2)$; the outgoing-projector ratio $2!/3!=1/3$ fixes $+1/48$ per unordered pair and $+1/16$ over three pairs. Projector idempotence supplies the opposite hard block $-1/16$, or $-3/512$ after multiplication by the Born coefficient.

This normalization is not yet the physical neutral answer. With the published fixed-vacuum charges $q(\Omega)=+1$, $q(\Upsilon)=-1$ and the BT dagger preserving charge, the two logarithmic contractions have generator charges $(+1,-1)$ and $(-1,+1)$, each with residue $-1/4$. Consequently no logarithmic row lies wholly in the one-sided nonpositive image; imposing that projection before the Gram makes the residue exactly zero. A completed broken-vacuum pushforward might instead use background shifts of both signs to place the pair in its neutral component, but neither the required zero-mode operator nor $R_tP_2R_t^\dagger$ is available in the Letter. Thus the exact alternatives are: the neutralized 1/48, zero after one-sided projection, or positive-charge leakage outside the present radical argument. This exact `LOCAL-ALGEBRAIC`, `REDUCED-MODE` classification is recorded by `REVERSE_PHYSICS_BT_SOFT_CHARGE_ RESOLVED_FLOW_V1`, downstream of `REVERSE_PHYSICS_BT_FULL_OFF_ RESONANT_PROJECTOR_V1`. Its finite-cutoff formal flow $K_t = e^{-idt}D - e^{idt}D^\sharp$ is anti-Krein and has $H_{\rm as}^{(1)} = d(e^{-idt}D + e^{idt}D^\sharp)$; sharp and smooth cutoffs have the same leading logarithmic response but retain a finite matching constant. It neither establishes the physical neutral coefficient nor a complete probability.

The global broken-vacuum orbit admits an exact operator realization without a c-number spurion. Set $\phi = \phi_0 + \varphi$ and $Z = e^{\lambda\phi_0}$. Then Eq. (16) factorizes as $\Omega = Z\widehat{\Omega}$ and $\Upsilon = Z^{-1}\widehat{\Upsilon}$ on $\mathbb{Q}[Z, Z^{-1}]$, with $\delta Z^n = nZ^n$. Covariance uniquely dresses a quadratic coefficient by $Z^{q_p - q_{d_1} - q_{d_2}}$. All eight number-lowering rows are consequently neutral, and the two logarithmic orbit powers cancel in their Gram, preserving the conditional 1/48 response. However, the repaired Appendix-C pullback implies $A_1 = Zb_\Upsilon$; its squeeze is therefore $Z^2(b_\Upsilon^\dagger)^2$, of total charge zero rather than -2. Conversely the fixed-vacuum ideal $I = (Z - 1)$ is not stable under the charge derivation, since $\delta(Z - 1) = Z \equiv 1 \pmod{I}$. Thus fixed-vacuum negative grading and covariant zero-mode completion do not commute. The certificate `REVERSE_PHYSICS_BT_ZERO_MODE_ EQ19_TRILEMMA_V1` records this exact obstruction. It does not replace the missing full dynamical zero-mode module, invariant generalized-Born trace, or nonlinear pushforward, and therefore does not prove Eq. (19), the physical 1/48, or its negation.

There is also a topology obstruction before that missing trace can be used on the ordinary Fock carrier. The certificate `REVERSE_PHYSICS_BT_SQUEEZED_ VACUUM_IMPLEMENTABILITY_V1` normalizes the Appendix-C creation generator in a periodic box and classifies the compatible charge-exchanging fundamental symmetries. If the positive metric weights the Υ direction by $\rho(\mathbf{p})$, uniform equivalence requires $0 < m \le \rho \le M$. Direct Wick contraction then gives

$$\frac{1}{V}\|Q_+|0\rangle\|_\kappa^2 = \frac{\rho^2}{64\pi^2}\left(\epsilon^{-1} - \Lambda^{-1}\right)$$

for constant ρ, and the six lowest box modes give a lower-bound density $3m^2L/(256\pi^4)$. The pair-creation block has twice this divergent norm. The squeeze is therefore Krein-null but not a vector, nor ordinarily Bogoliubov-implementable, in the massless positive Fock topology. A weight $\rho(p) \sim p^\alpha$ could soften the integral only for $\alpha > 1/2$, when ρ^{-1} is unbounded and the topology is inequivalent. This exact `LOCAL-ALGEBRAIC`, `REDUCED-MODE` obstruction neither refutes the algebraic map nor excludes a rigorously defined extended/non-Fock representation of Eq. (19).

The full exponential sharpens this carrier test. For an unordered momentum pair its exact normalized amplitude is $z(p) = \rho(p)/(4p^2)$, so a vector requires both $\sup_p |z(p)| < 1$ and square summability. Any uniformly equivalent $\rho \ge m > 0$ violates the first condition once $L \ge 4\pi/\sqrt{m}$. An explicit inequivalent repair is

$$\rho_\mu(p) = \frac{4\gamma\mu^2 p^2}{p^2 + \mu^2}, \qquad z_\mu(p) = \frac{\gamma\mu^2}{p^2 + \mu^2}, \qquad 0 < \gamma < 1,$$

whose unordered square-sum density is $\gamma^2\mu^3/(16\pi)$. Its inverse metric is unbounded at the origin and its thermodynamic vacuum leaves the ordinary Fock sector. Moreover, the raw BT shear has positive-Hilbert coefficients $u = 1, v = z$ and hence $u^2 - v^2 = 1 - z^2 \neq 1$: it is cross-Krein canonical, not a positive-Hilbert Bogoliubov transformation. Generic extended positive-boson implementation theorems therefore do not apply verbatim and do not provide the missing cyclic BT trace. The certificate `REVERSE_PHYSICS_BT_EXTENDED_SQUEEZE_ CARRIER_V1` records this constructive but inequivalent vacuum carrier and the remaining trace obstruction.

The cross-Krein construction can nevertheless be completed at finite regulator. Complete the orbit on $\ell^2(\mathbb{Z})$ with $[e_m, e_n] = \delta_{m+n,0}$ and $J_0 e_n = e_{-n}$. Then Z is the bilateral shift and $Z^\dagger = Z$. On the weighted oscillator carrier,

$$Q = \sum_{\{p,-p\}} \left(z_p Z^2 A_p^* A_{-p}^* - \bar{z}_p Z^2 D_p D_{-p} \right)$$

obeys $Q^\dagger = -Q$ and defines a densely defined closable squeeze on paired polynomial/Gaussian cores. The finite-rank trace $\mathrm{Tr}_{\mathrm{fin}}\, \Theta_{x,y} = [y, x]$ is cyclic and invariant under this transport.

This finite construction has an exact extension obstruction. For $E_n = Z^n E_0 Z^{-n}$, cyclicity gives $\tau(E_n) = c$. If $\tau(1) = 1$ and τ is positive on the J_0-even projection cone, then $P_N = \sum_{n=-N}^{N} E_n$ gives $(2N+1)c \le 1$ for every N, hence $c = 0$. The finite-rank trace instead has $\mathrm{Tr}_{\mathrm{fin}}(E_0) = 1$ but infinite identity weight. There is also no normal trace-norm limit: for the explicit weight above,

$$\frac{1}{V} \log \|P_V\|_1 \longrightarrow \ell(\gamma, \mu) = \frac{\mu^3}{12\pi} \left[(1+\gamma)^{3/2} + (1-\gamma)^{3/2} - 2 \right] > 0.$$

The certificate `REVERSE_PHYSICS_BT_CROSS_KREIN_TRACE_ LIMIT_V1` therefore isolates the open problem as an explicit semifinite, relative, or non-normal Born weight and the full nonlinear Eq. (19) transport. It does not establish the physical 1/48.

The semifinite branch has an exact finite-detector realization. On $B(\ell^2(\mathbb{Z}))$, the canonical faithful normal semifinite trace satisfies $\mathrm{Tr}_\infty(E_n) = 1$ and $\mathrm{Tr}_\infty(1) = +\infty$. Let P_{in} be a finite-rank J-even Krein projection, let S be cross-Krein isometric on its range, and let P_i be a finite exhaustive output partition there. If every $A_i = P_i S P_{\mathrm{in}} = B_i + C_i$ has the weak ghost decomposition, then

$$p_i = \frac{\mathrm{Tr}_{\mathrm{fin}}(A_i^\dagger A_i)}{\mathrm{Tr}_{\mathrm{fin}}(P_{\mathrm{in}})} \ge 0, \qquad \sum_i p_i = 1.$$

The proof uses weak-ghost trace orthogonality for positivity and $S^\dagger S = 1$ for normalization; it never uses a finite trace of the identity. This does not contradict the preceding no-go because the normalized corner functional is not cyclic. The exact matrix-unit witness $P = E_{00}$, $X = E_{01}$, $Y = E_{10}$ gives $\omega_P(XY) = 1$ and $\omega_P(YX) = 0$. The certificate `REVERSE_PHYSICS_BT_SEMIFINITE_ RELATIVE_BORN_WEIGHT_V1` records the construction with dependency tags `LOCAL-ALGEBRAIC` and `REDUCED-MODE`. It does not prove that the unbounded squeeze preserves the required trace ideal, construct a non-normal thermodynamic state, reproduce Eq. (19), or establish the physical 1/48.

The available zero-mode-completed logarithmic detector sector admits a sharper domain test. For N orthogonal logarithmic cells, the conditional amplitude $a = \sqrt{3}/12$ defines the skew finite-carrier generator $Kh = a \sum_i d_i$, $Kd_i = -ah$. The transported rank-one projection has

$$P_1 = a \sum_i \left(|d_i\rangle\langle h| + |h\rangle\langle d_i| \right), \qquad P_2 = a^2 \sum_{i,j} |d_i\rangle\langle d_j| - N a^2 |h\rangle\langle h|,$$

and satisfies projector idempotence through order λ^2. The unique orbit dressing makes this certified sector neutral, with no strictly negative radical. Each finite-N coefficient remains finite rank on the weighted squeeze core. Nevertheless,

$$\operatorname{Tr} P_1 = \operatorname{Tr} P_2 = 0, \qquad \|P_1\|_1^2 = \frac{N}{12}, \qquad \|P_2\|_1 = \frac{N}{24}.$$

The signed hard/soft box trace $-N/48 + N/48$ cancels while its absolute trace weight grows. Hence the finite detector lies in the semifinite paired ideal, but the logarithmic soft-cutoff limit is not uniformly trace class in this sector.
This exact distinction is recorded by `REVERSE_PHYSICS_BT_FINITE_DETECTOR_PUSHFORWARD_V1`. It leaves open an operator-level cancellation from omitted pushforward sectors, a hard matching construction, or a local non-normal renormalized weight, and does not prove Eq. (19) or the physical 1/48.

The complete signed quadratic composition supplies that operator-level cancellation at order λ. A target $b_\Upsilon(s,p)$ has, at leading order, the three inverse images $a_1(s,p)$, the secular part of $a_2(s,p)$, and the oscillatory part of $a_2(-s,-p)$. Symplectic extraction multiplies the last image by the inverse of its Appendix-C oscillatory phase. Summing all sign sectors gives

$$\delta b_\Omega = \frac{s_1 e_1 + s_2 e_2}{2e_1 e_2} b_\Omega^{(s_1)} b_\Omega^{(s_2)},$$

$$\delta b_\Upsilon = -\frac{s_2}{2e_1} b_\Omega^{(s_1)} b_\Upsilon^{(s_2)} - \frac{s_1}{2e_2} b_\Upsilon^{(s_1)} b_\Omega^{(s_2)},$$

with every other row zero. The daughter Krein contraction has $G_{\Omega\Upsilon} = G_{\Upsilon\Omega} = 0$; because the parent inverse metric is off diagonal, the quadratic parent trace has no dr/r term. The mixed-sign rows close the order-λ cross-CCR Ward identity, and the unique Z dressing makes the associated finite-mode cubic generator neutral. Thus the unsqueezed factor obeys $UP_0U^{-1} = P_0 + \lambda[K, P_0] + O(\lambda^2)$ with wholly neutral charge support and $Q_1 = 0$. The certificate `REVERSE_PHYSICS_BT_FULL_SIGNED_ QUADRATIC_CLOSURE_V1` records 128 exact preimage contributions and 48 exact Ward fixtures with dependency tags `LOCAL-ALGEBRAIC` and `REDUCED-MODE`. The logarithmic normalization is a conditional theorem for the resonant truncation; it does not survive the complete public quadratic map. No physical coefficient follows from this projector-pushforward kernel: the S-matrix transition block is a distinct object. The squeezed-vacuum and dynamical-zero-mode projector trace, continuum domain, and higher composite orders remain open within Eq. (19). The Appendix-C squeeze does not supply another additive coefficient on this finite paired core. Writing the finite-regulator map as $R(\lambda) = S(1 + \lambda K + O(\lambda^2))$, cross-Krein isometry gives

$$(SKS^{-1})^\dagger (SKS^{-1}) = S(K^\dagger K)S^{-1}.$$

Transporting the detector by the same similarity and using the finite-rank cyclic trace leaves the completed quadratic coefficient at zero. The exact 3/16 bare one-pair occupation of the normalized $z = 1/2$ squeezed vacuum holds a b-Fock detector fixed and is not an Eq. (19) pushforward probability. This distinction is certified by `REVERSE_PHYSICS_BT_SQUEEZED_ DETECTOR_SIMILARITY_V1`. Its `LOCAL-ALGEBRAIC`, `REDUCED-MODE` zero is a finite-regulator pushforward result. The non-trace-class thermodynamic limit, dynamical zero mode, and higher orders remain open for Eq. (19), and the distinct S-matrix kernel must be evaluated for a physical response. The finite paired-process algebra does admit a thermodynamic conditional functional without trace-norm convergence. Tensoring by squeezed spectator projections of Krein trace one leaves all local weak-ghost Born weights unchanged; an exact fixture gives $9/25, 16/25, 0$ at every spectator volume while the positive trace norm grows as $(4/3)^N$. Hence the pointwise zero quadratic coefficient has a regulator-independent cylinder limit. Certificate `REVERSE_PHYSICS_BT_CYLINDER_BORN_LIMIT_V1` does not identify this pair-cylinder algebra with an inclusive LSZ or spacetime-local detector. The orthogonal-block lemma does not identify its input kernel. The complete signed kernel above is the $R_t P R_t^\dagger$ projector pushforward, whereas the physical real process is $P_{\text{out}}(S-1)P_{\text{in}}$. Certificate `REVERSE_PHYSICS_BT_INCLUSIVE_NLO_OBJECT_LEDGER_V1` keeps the exact zero pushforward trace but assigns the five-point process its independently computed response: $+1/512$ per pair, $+3/512$ in total, and $1/48$ per pair after Born normalization. The axis-compatible physical virtual response vanishes, leaving $+3/512$ uncancelled. The public R_t response also vanishes but is a distinct Eq. (19) object, not a physical summand; an independently normalized hard/dressed term $-3/512$ is missing. The complete NLO probability and all-order Eq. (19) remain open. The missing coefficient is nevertheless universal under a correctly typed physical-shell pseudo-unitarity assumption. For $S_{\text{phys}}(x) = 1 + xA + x^2 B + O(x^3)$,

$$A^\dagger = -A, \qquad B + B^\dagger + A^\dagger A = 0.$$

The certified real column gives $\|Ah\|^2 = 1/16$, so $\operatorname{Re} B_{hh} = -1/32$ and the hard survival response is $-1/16$, or $-3/512$ after multiplication by the Born coefficient.

Certificate `REVERSE_PHYSICS_ BT_PHYSICAL_SHELL_ PSEUDOUNITARY_ COMPLETION_V1` supplies an exact finite $\mathbb{Q}(\sqrt{3})$ isometric witness. Phases and skew second-order freedom do not affect the result. It does not construct the continuum dressed S-matrix, its degenerate trace domain, or beyond-tree positivity. The ordinary continuum limit is nevertheless decidable. In logarithmic coordinate $y = -\log r$, choose normalized disjoint shell vectors $u_{n,i}$ moving to the endpoint. Then

$$\|A_n h - A_m h\|^2 = \frac{1}{8} \quad (n \neq m), \qquad \|e^{xA_n} h - e^{xA_m} h\|^2 = 2\sin^2(x/4).$$

Thus no strong Møller limit exists on the ordinary logarithmic L^2 carrier. The weak limit is a contraction with order-x^2 defect $-1/16$. Certificate `REVERSE_PHYSICS_BT_LOG_SHELL_MOLLER_LIMIT_V1` constructs instead a fixed endpoint fibre with embeddings $J_n e_i = u_{n,i}$; the pulled-back shell isometry and its $-1/16 + 1/16 = 0$ probability ledger are independent of n. This leading-log dressed bundle is not affiliated with a local LSZ/AQFT detector and is not derived from the full BT Hamiltonian. It does have a canonical asymptotic resolution affiliation. For $q_R(y) = q(y - R)$ with unit endpoint jump, the positive detector cell $d_{R,a} = q_{R+a} - q_R$ satisfies

$$\int_{\mathbb{R}} d_{R,a}(y)\, dy = a, \qquad u_{R,a} = \sqrt{d_{R,a}/a}, \qquad u_{R+b,a} = T_b u_{R,a}.$$

Hence $J_{R,a} e_i = u_{R,a}$ is the cutoff-dilation cocycle of the asymptotic momentum-resolution algebra, rather than a fitted endpoint identification. Certificate `REVERSE_PHYSICS_BT_DETECTOR_RESOLUTION_DILATION_V1` checks a sharp step and a C^1 cubic smoothstep exactly. Both give the same generalized-Born response: $+1/48$ per pair and $+1/16$ in total, while pseudo-unitarity gives the hard response $-1/16$. Thus the physical NLO leading-log resolution response vanishes on the declared final-pair cylinder for every admissible monotone profile. The resolution translation is generated by Abel-regularized formal soft time flow. Indeed,

$$-i \int_0^\infty e^{-\epsilon t} H_{\text{as}}^{(1)}(t)\, dt : \qquad A_\epsilon(d) = \frac{-id}{\epsilon + id}, \qquad |A_\epsilon(d)|^2 = \frac{d^2}{\epsilon^2 + d^2}.$$

On $d = \alpha r$ this is exactly $q_R(y) = (1 + e^{2(y-R)})^{-1}$ with $R = \log(\alpha/\epsilon)$, so inverse Abel time dilates detector resolution. The ordinary coherent columns nevertheless remain non-Cauchy: at fixed $c > 1$ their limiting squared distance is $\log(c)(c-1)/(c+1)$, not the detector response $\log c$. The canonical orthogonal-increment purification

$$\Xi_{R,a}(s, y) = \mathbf{1}_{[R,R+a]}(s)\sqrt{\operatorname{sech}^2(y - s)/(2a)}$$

has unit norm and marginal $(q_{R+a} - q_R)/a$; disjoint resolution intervals are orthogonal. Certificate `REVERSE_PHYSICS_ BT_ABEL_NAIMARK_ ASYMPTOTIC_ DILATION_V1` records the exact intertwiner and obstruction. Its auxiliary s label is a resolution/noise coordinate, not spacetime. This `LOCAL-ALGEBRAIC`, `REDUCED-MODE` result identifies the common detector automorphism, not the public R_t lowering operator with the physical splitting operator; a spacetime-local detector, complete sectors, the finite NLO probability, and Eq. (19) remain outside the claim. The remaining operator comparison is now exact on the external mass-jet cylinder. If

$$L = -\frac{(a_0 - a_1)^2}{4\tau}, \qquad Q = \frac{2\tau(a_0 + a_1) - (a_0 - a_1)^2}{4\tau^2},$$

the complete five-point tree has relevant components (Q, L), while the recombined four-point parent tree has components $(1/2, 1/2)$. In the one-leg delta-prime cross basis this fixes

$$T = \operatorname{diag}(2Q, 2L), \qquad -T^\sharp T = \rho I_2, \qquad \rho = \frac{(a_0 - a_1)^2[2\tau(a_0 + a_1) - (a_0 - a_1)^2]}{4\tau^3} > 0$$

above unequal-mass threshold. The common tree phase cancels, and the exact phase-space ratio turns the integrated ρ response into $1/48$ per pair. By contrast, the completed public quadratic R_t map has covariant parent Gram $\operatorname{diag}(0, 2)$ for equal daughter signs; raising with the cross metric gives $\left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$, a nonzero rank-one nilpotent. Rank and Jordan type exclude every metric-compatible basis, phase, nonzero scalar, zero-mode similarity, or Abel multiplier from identifying the two operators. Certificate `REVERSE_PHYSICS_BT_PHYSICAL_ COLLINEAR_OPERATOR_ FACTORIZATION_V1` records the method-distinct graph derivations. It factors the physical leading map on this reduced jet cylinder and proves an obstruction to the public-D identification; it does not construct the full physical Møller/S operator or prove Eq. (19). The exact pointwise map nevertheless has a regulated direct-integral realization. On $a_0 = 1$, $a_1 = r$, $\tau = u$, set

$$d\mu_r(u) = \frac{\sqrt{\lambda(u, 1, r)}}{u}\, du, \qquad (V_r h)(u) = T(r, u)h.$$

Then $V_r^\sharp V_r = I(r) I_2$, where

$$\begin{aligned} I(r) &= -\frac{2}{3} H(r) \\ &= \frac{5r^3 - 6r^2 \log r - 3r^2 - 6r \log r + 3r - 5}{24(r - 1)} \\ &= \frac{5}{24} + r\left(\frac{1}{4}\log r + \frac{1}{12}\right) + O(r^2 \log r). \end{aligned}$$

Thus a finite-regulator pseudo-unitary block exists and the massless Gram limit is finite, but no strongly C^1 column into a fixed Krein carrier with bounded fundamental symmetry can realize the delta-prime derivative: strong differentiability of V_r would make $V_r^\sharp V_r$ differentiable, whereas $I'(0+) = -\infty$. The divided germ $D(r) = [I(r) - I(0)]/r$ still has the exact relative cocycle

$$\lim_{r\to 0}[D(cr) - D(r)] = \frac{1}{4}\log c,$$

or $\log(c)/48$ per pair after physical normalization. On $R = -\log r$ this affine law is linearized by $N = \left(\begin{smallmatrix} 0 & -1/4 \\ 0 & 0 \end{smallmatrix}\right)$ on $\mathrm{span}\{1, s\} \subset \mathcal{S}'(\mathbb{R}_s)$. The constant and coordinate distributions are not L^2 vectors, so the Abel–Naimark translation supplies a rigged relative-resolution cocycle but not an endpoint state. Certificate `REVERSE_PHYSICS_BT_RIGGED_ RESOLUTION_JORDAN_ MOLLER_V1` records the exact threshold integration and the independent rapidity rail. The rigged coordinate is resolution data, not spacetime; a generalized-Born functional on this rigging, a physical Hamiltonian affiliation, the complete Møller/S operator, and Eq. (19) remain open. There is a canonical positive realization of the relative functional on the resolution-local CCR net. The physical response endomorphism is γI_2, $\gamma = 1/48$, so its minimal Kolmogorov factor retains both jet species:

$$k_s(y) = \sqrt{\gamma p_s(y)}I_2, \qquad \mathrm{tr}_{\mathrm{sp}}(X) = \tfrac{1}{2}\,\mathrm{Tr}_2(X).$$

For a bounded resolution interval I and three unordered pairs, set

$$F_I(i; s, y) = \mathbf{1}_I(s)k_s(y), \qquad \|F_I\|^2 = 3\gamma|I| = \frac{|I|}{16}.$$

The local Weyl displacement $W(F_I)$ is unitary, compatible under interval inclusion, translation covariant, and factors over disjoint intervals. Its coherent Born state gives the exact all-count law

$$P_n(a) = e^{-a/16}\frac{(a/16)^n}{n!}, \qquad \sum_{n\geq 0} P_n(a) = 1,$$

with three independent pair means $a/48$ and hard no-emission probability $e^{-a/16}$. Thus the certified $-a/16 + a/16$ response is the tangent of a positive normalized local process, not merely a signed truncation. The global amplitude has $\|F_{[0,L]}\|^2 = L/16$, so no global Fock vector or inner Weyl implementer exists; the coherent state is locally normal and non-Fock at infinity. Certificate `REVERSE_PHYSICS_BT_RESOLUTION_ LOCAL_COHERENT_BORN_ PROCESS_V1` records this resolution-local leading-log Møller automorphism. The Poisson law is uniquely forced only under the declared stationary independent- increment coherent completion. The complete six-point tree now tests and rejects that assumption in the nested strongly ordered double-collinear sector. An exact 220-tree subset recursion gives the same two-scale external-mass kernel at three unrelated exact hard fixtures. After both Källén threshold reductions, the mixed nonanalytic coefficient is 15/32. Restoring the $2 \to 4$ amplitude and phase-space factors, the twelve labeled nested histories, and the ordered simplex gives

$$P_2(a) = \frac{5a^2}{512}, \qquad P_2^{\mathrm{Poisson}}(a) = \frac{a^2}{512}, \qquad \kappa_2(a) = \frac{a^2}{64}.$$

Thus the nonlinear tree dynamics is super-Poisson by a factor of five at leading ordered double-logarithmic order. Certificate `REVERSE_PHYSICS_BT_SIX_POINT_ STRONGLY_ORDERED_TREE_V1` records the complete graph count, an explicit-tree independent rail, the invariant-cutoff finite part, and every factorial. It leaves the relative 1/48 weight and rank-two one-emission GNS factor intact, but the coherent state is not the state dynamically selected at two emissions. The complete seven-point tree resolves the next ordered coefficient. Summing all 2485 trees and taking three physical-invariant-cutoff threshold finite parts gives the third count coefficient and factorial cumulant

$$P_3(a) = \frac{9a^3}{8192}, \qquad M_3(a) = \frac{27a^3}{4096}, \qquad \kappa_3(a) = \frac{7a^3}{2048}.$$

The gamma–Cox law fixed by the first two moments instead predicts $P_3(a) = 15a^3/8192$ and is therefore excluded. Scalar Cox positivity itself survives: the exact Stieltjes bound is satisfied, and the unique law on at most two intensity atoms which matches the three moments has

$$x_\pm = \frac{11 \pm \sqrt{113}}{64}, \qquad p_- = \frac{\sqrt{113}+7}{2\sqrt{113}}, \qquad p_+ = \frac{\sqrt{113}-7}{2\sqrt{113}}.$$

The corresponding convex mixture of local Weyl-displaced states is positive, normalized, locally normal, inclusion compatible, and translation covariant. Certificate `REVERSE_PHYSICS_BT_SEVEN_POINT_ COX_SELECTION_V1` records the full tree sum, independent exact rail, moment test, and controlled two-atom dilation. The dilation is a minimal-support completion through three moments, not a dynamically derived all-order law or a single Møller automorphism. The complete eight-point preflight shows why a fourth scalar moment cannot yet be appended by the same bare-limit recipe. Summing all 34300 trees and forming the complete spectator projector before any hierarchy limit gives ordered valuation $(0, 0, -1)$. At identical soft data, changing only one hard adjacent invariant from 33 to 34 changes the leading residue by the exact nonzero amount

$$\Delta R_8 = \frac{257}{1568}.$$

A separate rational-series implementation with the invariant triangle cubic vertex reproduces both finite-hierarchy projected values. Certificate
`REVERSE_PHYSICS_BT_EIGHT_POINT_HARD_PROFILE_OBSTRUCTION_V1`
therefore withdraws the provisional hard-independent coefficient 629 and leaves the two-atom Cox prediction undecided. It does not prove that a physical fourth moment is absent: the outer fixed-invariant Källén reduction or a hard-profile quotient could still remove the dependence. Retaining the outer invariant $u = \tau_4$ resolves the first possibility. The two complete profiles have leading difference

$$e_3(F_{33} - F_{34})\big|_{e_3=0} = \frac{771}{1568u}.$$

The outer measure sends $1, u^{-1}, u^{-2}, u^{-3}$ to J_1, J_2, J_3, J_4, whose fixed-invariant $r \log r$ coefficients are all one. Hence the difference is a pure J_2 row and survives the threshold functional with coefficient $771/(1568e_3)$; evaluation at the earlier $u = 3$ gives exactly $257/(1568e_3)$. With $r = e_3\tau_3/a_4 = 5e_3$, the actual leading outer logarithms differ by 3855/1568. Certificate
`REVERSE_PHYSICS_BT_EIGHT_POINT_OUTER_THRESHOLD_PROFILE_V1`
records the complete profiles and an independent pole-derivative reconstruction of the four moment coefficients. The remaining three threshold reductions may still collapse or retain this profile. Restoring τ_3 symbolically closes the next one. The complete profile difference is

$$\frac{3(27213\tau_3^2 - 13462\tau_3 + 29485)}{156800\tau_3^2\tau_4}.$$

After the outer J_2 functional, the rows $1, \tau_3^{-1}, \tau_3^{-2}$ map at $a_3 = 2$ to $a_3 J_1, J_2, a_3^{-1} J_3$. Their unit nonanalytic coefficients give the strictly nonzero second-threshold difference

$$\frac{334239}{313600}.$$

Certificate
`REVERSE_PHYSICS_BT_EIGHT_POINT_MIDDLE_THRESHOLD_PROFILE_V1`
records the two symbolic profiles and a separate exact $\tau_3 = 4$ replay. The next reduction can evaluate the first two unit-J_n functionals at $\tau_4 = a_4 = 1$ and $\tau_3 = a_3 = 2$ while retaining τ_2 symbolically. The resulting complete hard-fixture difference is

$$\frac{3(2090\tau_2^2 + 1146\tau_2 + 981)}{12800\tau_2^2}.$$

It replays the preceding coefficient at $\tau_2 = 7$. At $a_2 = 3$, its J_1, J_2, J_3 rows leave the third-threshold difference

$$\frac{23229}{6400} \neq 0.$$

Certificate
`REVERSE_PHYSICS_BT_EIGHT_POINT_TAU2_THRESHOLD_PROFILE_V1`
therefore leaves only the independent-mass τ_1 reduction. Its value is a profile-functional obstruction, not a normalized fourth moment: the nested mass-ratio factors and final inner residue remain unassembled. That final scalar test can be performed without choosing a common mass ray. Set $a_0 = 1$, $r = a_1/a_0$, and $u = \tau_1/a_0$. The complete profiles differ by

$$\frac{3\{9r^2 - 18r(u+1) + 34u^2 - 18u + 9\}}{128u^2}.$$

The invariant-cutoff rationalization $u = 1 + m^2 + m(z + z^{-1})$, $r = m^2$, gives the two $r \log r$ coefficients $-6699/128$ and $-7149/128$. Their independently derived difference is

$$\frac{225}{64} \neq 0.$$

Certificate
`REVERSE_PHYSICS_BT_EIGHT_POINT_INNER_THRESHOLD_OBSTRUCTION_V1`
therefore obstructs a hard-independent scalar fourth jump on the declared complete fixtures after all four threshold functionals. It does not obstruct a channel/profile-valued successor, normalize a fourth probability, or decide the Cox law. The surviving direction is also distinct from the forced cross-Krein complement. Both fixtures have $(a_0, a_1, \tau_1) = (1, 4, 10)$ and therefore the same $\rho = 819/4000$, the same missing Gram $\begin{pmatrix} 0 & -\rho \\ -\rho & -2 \end{pmatrix}$, and the same endpoint coefficient $c_1 = 14819/8000$, while their fourth responses differ. Certificate
`REVERSE_PHYSICS_BT_COMPLEMENT_HARD_PROFILE_SEPARATION_V1`
therefore requires an additional hard-profile base coordinate beyond ρ; it does not construct the corresponding dynamical bundle. The induced two-point profile effect is in fact strictly negative:

$$K_4 = \mathrm{diag}(-6699/128, -7149/128).$$

The eight external delta-prime sign is positive, as are the inherited threshold scale, lower rate chain, phase-space, counting, simplex, squared phase, and Born normalizations. Hence every faithful positive profile trace of K_4 is negative, and no ordinary completely positive or Hudson–Parthasarathy jump can extend the finite positive column diagonally. Certificate `REVERSE_PHYSICS_BT_EIGHT_POINT_PROFILE_POSITIVITY_OBSTRUCTION_V1` also gives the minimal indefinite alternative. One copy of $G_{\mathrm{miss}}(\rho)$ over each hard idempotent supplies two negative directions; mapping into their canonical norm--2 lines gives an exact $B^\sharp B = K_4$ and a Krein-skew block generator. This is not a positive fourth probability or a charge-compatible BT derivation: it makes an Eq. (19)-type trace or a derived off-diagonal channel recombination necessary on this carrier. The lift is not the purely negative-charge remainder in Eq. (19). The certified leg $C = \left(\begin{smallmatrix} -\rho & -1 \\ 0 & 1 \end{smallmatrix}\right)$ obeys $C^T J C = G_{\mathrm{miss}}$ and transports the actual target charge action. The null charge basis of G_{miss} is $n_+ = (1,0)^T$, $n_- = (1,-\rho)^T$, and its transported charge generator is

$$H_G = C^{-1}\,\mathrm{diag}(1,-1)C = \begin{pmatrix} 1 & 2/\rho \\ 0 & -1 \end{pmatrix}, \qquad H_G^2 = I, \qquad H_G^T G_{\mathrm{miss}} + G_{\mathrm{miss}} H_G = 0.$$

For the norm--2 vector f_2, both $\Pi_+ f_2$ and $\Pi_- f_2$ are null and their two cross pairings are -1. Hence the complete profile block obeys

$$B_+^\sharp B_+ = B_-^\sharp B_- = 0, \qquad B_+^\sharp B_- + B_-^\sharp B_+ = K_4.$$

Certificate `REVERSE_PHYSICS_BT_EIGHT_POINT_KREIN_CHARGE_LOCALIZATION_V1` therefore localizes the nonzero effect in total-charge-zero cross interference. A one-sided negative Q term has zero pullback on this fibre; a neutral higher-composite or additional dynamical trace is required, but not constructed. The minimal neutral bosonic carrier can nevertheless be fixed exactly. For two hard-profile modes, the charge-zero degree-$2k$ symmetric-Fock sector has inertia

$$\left(\frac{(k+1)(k+2)}{2}, \frac{k(k+1)}{2}\right).$$

Degree two has only one negative direction and cannot carry rank-two K_4; degree four has inertia $(6,3)$. In its occupation basis $20, 11, 02$ for each charge sign, the two vectors

$$u_1 = \frac{e_{20,11} - e_{11,20}}{\sqrt{2}\,\rho^4}, \qquad u_2 = \frac{e_{11,02} - e_{02,11}}{\sqrt{2}\,\rho^4}$$

are symmetric-bosonic, total-charge zero, orthogonal, and have Gram $-2I_2$. Multiplication by the same amplitudes $\sqrt{6699}/16, \sqrt{7149}/16$ gives a neutral $B_4^\sharp B_4 = K_4$ and a charge-compatible Krein-skew generator. Certificate `REVERSE_PHYSICS_BT_NEUTRAL_BOSONIC_COMPOSITE_LIFT_V1` proves degree-four minimality and also exposes the next gate. Occupation swap is the induced ghost parity, and both u_i are odd. With the positive profile source ghost even, this block is ghost odd rather than the ghost-even neutral P term asserted in Eq. (19). In fact any ghost-even forward block D obeys $D^\sharp D = D^T(G_4\kappa)D \succeq 0$ and cannot equal negative K_4. A ghost-odd source partner, a larger complete projector, or an additional dynamical trace remains unconstructed. The minimal odd-source extension is itself exactly solvable. Give the added two-dimensional source metric $-I_2$, retain the selected composite metric $-2I_2$, and put

$$T = \mathrm{diag}(\sqrt{6699}/16, \sqrt{7149}/16), \quad L = \begin{pmatrix} I_2 \\ T \end{pmatrix}, \quad \eta = \mathrm{diag}(-I_2, -2I_2).$$

Then $T^T(-2I_2)T = K_4$, and the Krein-orthogonal graph projector

$$P = L(L^T\eta L)^{-1}L^T\eta$$

has rank two and obeys $P^2 = P = P^\sharp$, $[P,H] = [P,\kappa] = 0$, and $\mathrm{tr}(P^\sharp P) = 2$. Certificate
`REVERSE_PHYSICS_BT_NEUTRAL_GRAPH_PROJECTOR_EXTENSION_V1`
therefore constructs a finite neutral ghost-even projector candidate, but only on the added odd sector. Direct affiliation to the original even positive source is exactly obstructed: parity gives $-F = F$ and hence $F = 0$, while every graph pullback is $-A^TMA \preceq 0$ for $M = \mathrm{diag}(6827/128, 7277/128) > 0$. The finite trace is a rank weight, not the BT continuum trace or a fourth probability; a larger signature-balanced R_t projector remains unconstructed. The odd source itself need not be external to the public carrier. On the two-profile public J-Fock space, the neutral degree-four occupation sector has inertia $(6,3)$, and

$$w_1 = \frac{e_{20,11} - e_{11,20}}{2}, \qquad w_2 = \frac{e_{11,02} - e_{02,11}}{2}$$

have Gram $-I_2$, total charge zero, and odd ghost parity. For the certified complement charge basis $S = ((1,0)^T, (1,-\rho)^T)$, the missing leg satisfies $CS = -\rho I_2$. Consequently

$$\mathrm{Sym}^4(C) = \rho^4 I_9, \qquad \mathrm{Sym}^4(C)u_i = \sqrt{2}\,w_i, \qquad \left(\frac{I_2}{\sqrt{2}}\right)^T(-2I_2)\left(\frac{I_2}{\sqrt{2}}\right) = -I_2.$$

Certificate
`REVERSE_PHYSICS_BT_PUBLIC_FOCK_ ODD_SOURCE_AFFILIATION_V1`
therefore realizes the graph's odd source as an explicit public neutral Fock subspace and gives its exact ghost-even metric affiliation to the composite. This removes the carrier-existence obstruction, but not the dynamical one: the induced isometry is not the graph slope T. Moreover T belongs to the physical eight-point response ledger, whereas $R_tP_\chi R_t^\dagger$ is a projector pushforward. Identifying this particular graph with the Eq. (19) projector would require an explicit intertwiner; fitting T as an R_t coefficient is not authorized. Equation (19), its continuum trace, and a physical fourth probability remain unproved. The charge-support part of Eq. (19) can instead be settled formally without that identification. Retain the orbit operator $Z = e^{\lambda\phi_0}$ and put $\phi = \phi_0 + \varphi$. The exact pullback is

$$\alpha(\Omega) = \lambda^{-1}Ze^{\lambda\varphi}, \qquad \alpha(\Upsilon) = Z^{-1}e^{-\lambda\varphi}[\Box\varphi + \lambda(\partial\varphi)^2].$$

Every coupling coefficient has orbit power equal to its target boost charge, so $\delta_\phi\alpha = \alpha\delta_{1,1}$ to all formal orders. The two free Hamiltonians used in R_t are charge neutral, so the same intertwining holds after time translation. Perturbative two-sidedness transfers this identity to $\beta = \alpha^{-1}$. Hence for any finite nonzero-mode projector with $\delta_\phi P = 0$,

$$A = \beta(P), \qquad A^2 = A = A^\dagger, \qquad \delta_{1,1}A = 0.$$

Certificate
`REVERSE_PHYSICS_BT_COVARIANT_FORMAL_ EQ19_CHARGE_SUPPORT_V1`
therefore proves the formal charge decomposition with $P_{\mathrm{neutral}} = A$ and $Q_{\mathrm{negative}} = 0$ to every order. This does not prove ghost evenness or time independence, and it does not descend to the fixed-vacuum quotient: for $I = (Z-1)$, $\delta_\phi(Z-1) = Z \equiv 1 \pmod{I}$. The asymptotic

limits, continuum projector and generalized-Born trace remain unconstructed. The parity calculation for the unsqueezed nonlinear factor is independent of its charge support. On the exact resonant block with daughter energies $1, 2$ and parent energy 3, the complete signed kernel defines a Krein-skew species generator K_+ with $[q, K_+] = K_+$. Its unique covariant dressing is $K_{\mathrm{pub}} = Z^{-1}K_+$. If P_0 is the standard projector onto the two one-particle species, then

$$P_1 = Z^{-1}[K_+, P_0]$$

is a nonzero rank-four, neutral and stationary projector tangent. Ghost parity reverses the orbit and gives $\kappa P_1 \kappa = Z\kappa[K_+, P_0]\kappa$. Since the Laurent powers Z^{-1} and Z are independent, P_1 is not ghost even. Moreover the canonical even/odd split obeys, for the finite relative orbit trace,

$$\tau_0(B_1^\sharp C_1) = 0, \qquad \tau_0(B_1^\sharp B_1) = \frac{7}{288}, \qquad \tau_0(C_1^\sharp C_1) = -\frac{7}{288}.$$

Thus this complete $U(\lambda)$ tangent is not ghost even and its odd part is not null. It is not by itself the order-λ coefficient of the full projector: the public factorization $R = SU$ conjugates it by the Appendix-C squeeze. Certificate
`REVERSE_PHYSICS_BT_COVARIANT_GHOST_ PARITY_BRANCH_OBSTRUCTION_V1`
also gives the support-minimal finite repair

$$K_{\mathrm{even}} = \frac{1}{2}(Z^{-1}K_+ + Z\kappa K_+\kappa), \qquad P_{\mathrm{even}} = e^{\lambda K_{\mathrm{even}}} P_0 e^{-\lambda K_{\mathrm{even}}}.$$

This projector is neutral, stationary and ghost even to all formal orders, but the conjugate Z branch is not supplied by the public nonlinear factor. The repair therefore requires a new source-affiliation theorem; it is not a physical probability or an asymptotic completion. There is an exact obstruction to such an affiliation if it is required to be regular on the same off-shell perturbative local-symbol chart. Let ϵ be the zero-jet augmentation. The two Eq. (16) symbols have

$$O = \lambda^{-1}Ze^{\lambda\varphi}, \qquad O^{-1} = \lambda Z^{-1}e^{-\lambda\varphi}, \qquad \epsilon(O) = \lambda^{-1}Z,$$

whereas

$$Y = Z^{-1}e^{-\lambda\varphi}[\Box\varphi + \lambda(\partial\varphi)^2], qquad \epsilon(Y) = 0.$$

Thus O is a unit and Y is not. Since a unital automorphism preserves units, no regular source ghost parity can exchange them, even up to an invertible normalization. Certificate
`REVERSE_PHYSICS_BT_PERTURBATIVE_GHOST_ PARITY_UNIT_OBSTRUCTION_V1`
also records the sharp boundary: this is a local-symbol theorem, not a zero-oscillator augmentation of the CCR algebra. It rules out a filtration-continuous quantum automorphism with that regular classical symbol, but not a singular or unbounded correspondence. Localizing at the bracketed field strength excludes the zero-jet vacuum chart; doubling the sheet retains two vacua but changes the source theory.

The complete map admits a stronger charge–squeeze exhaustion that does not treat the preceding tangent as the full coefficient. Give the orbit factor an arbitrary homogeneous real charge $q(Z) = s$. The nonlinear and squeeze charges are locked:

$$q_K = 1 - s, \qquad q_S = 2s - 2 = -2q_K.$$

For $s < 1$, the full order-λ squeezed tangent has charges $q_K, -q_K, -3q_K$; its unique positive component has rank four. This includes the fixed-vacuum oscillator choice $s = 0$. For $s > 1$, the free squeezed projector already has positive components $q_S, 2q_S$, of ranks four and two after tensoring with the public two-species $n = 1$ projector.

Only $s = 1$ avoids positive charge. Then the whole projector is neutral, so directness forces $Q_{<0} = 0$. On one disjoint unordered momentum pair, with basis $(|0\rangle, |\Omega\Omega\rangle, |\Upsilon\Upsilon\rangle)$, its exact Krein-skew squeeze generator and transported vacuum projector at $t = 0$ are

$$Q = \begin{pmatrix} 0 & -z & 0 \\ 0 & 0 & 0 \\ z & 0 & 0 \end{pmatrix}, \qquad SPS^{-1} = \begin{pmatrix} 1 & zZ^2 & 0 \\ 0 & 0 & 0 \\ zZ^2 & z^2Z^4 & 0 \end{pmatrix}, \qquad Q^3 = 0.$$

Tensoring with the public $n = 1$ projector gives canonical even and odd parts with

$$\tau_0(B^\sharp C) = 0, \qquad \tau_0(C^\sharp C) = -z^2(z^2 + 2).$$

This is nonzero for every real nonzero Appendix-C amplitude; the normalized finite-box fixture $z = 1/4$ gives $-33/256$. Thus the only neutral assignment fails the required ghost evenness already in the squeezed free projector. Certificate
`REVERSE_PHYSICS_BT_EQ19_SPURION_ SQUEEZE_DICHOTOMY_NO_GO_V1`
therefore refutes every homogeneous regular realization of the public full map on this finite regulator. The earlier certificate `REVERSE_PHYSICS_BT_REGULAR_COVARIANT_ EQ19_NO_GO_V1` retains its exact rank-four unsqueezed-factor calculation, but its tangent is not used as the complete squeezed coefficient. This is not a universal refutation of Eq. (19): nonhomogeneous, localized, doubled, singular, non-Fock and nonperturbative completions remain open. Resolving the labels gives a different, channel-faithful lift. The ordered histories are rooted combs with dimensions 3, 12, and 60; canonical one-leaf insertion has respectively 3, 4, and 5 children. Dividing the selected-history coefficients successively fixes the positive per-extension rate squares

$$q_0 = \frac{1}{48}, \qquad q_1 = \frac{5}{64}, \qquad q_2 = \frac{27}{400}.$$

Their total exit rates 1/16, 5/16, and 27/80 reproduce the three tree probabilities as $\Lambda_0 a$, $\Lambda_0\Lambda_1 a^2/2$, and $\Lambda_0\Lambda_1\Lambda_2 a^3/6$. Jump operators $L_{h\to c} = \sqrt{q_k}|c\rangle\langle h| \otimes I_2$ give a finite completely positive trace-preserving branching instrument. Its first three children are exactly the physical pair channels, each with Gram $(1/48)I_2$. Certificate `REVERSE_PHYSICS_BT_CHANNEL_ RESOLVED_BRANCHING_ INSTRUMENT_V1` records the independent comb enumeration, channel Grams, and normalized semigroup. The higher I_2 lift and absorbing three-emission closure are positive constructions, not derived amplitude phases or an all-order BT Hamiltonian. Resolving the complete six-point amplitude before its scalar spectator trace gives an exact obstruction to affiliating the second of those identity-species jumps. With $S_1 = a_3 + a_4 + a_5$ and $S_2 = a_3a_4 + a_3a_5 + a_4a_5$, write the recombined outer parent as $H_0 = L_0S_1 + Q_0S_2$ and $H_1 = L_1S_1 + Q_1S_2$. On the spectator quotient relevant to $[a_3a_4a_5]C_6^2$, the strong amplitude has the unique factorization $C_{6,\mathrm{rel}} = uH_0 + vH_1$, but

$$u = 2Q_{\mathrm{inner}}, \qquad v = \frac{a_2}{2} \neq 2L_{\mathrm{inner}}.$$

Thus the second parent coefficient already depends on the outer daughter history. More decisively, the singleton/complement pairing gives the raised profile-normalized endomorphism

$$N = \begin{pmatrix} ua_2/2 & a_2(\tau_2 + a_2)/(\tau_2 - 2a_2) \\ -u^2a_2(2\tau_2 - a_2)/(\tau_2 - 2a_2) & ua_2/2 \end{pmatrix},$$

whose characteristic discriminant is

$$-\frac{4u^2a_2^2(\tau_2+a_2)(2\tau_2-a_2)}{(\tau_2-2a_2)^2}<0$$

for $\tau_2 > a_2 > 0$, away from the surface where the parent cross normalization itself vanishes. Hence no parent-fibre basis change identifies this block with a positive scalar multiple of I_2. Certificate `REVERSE_PHYSICS_BT_SIX_POINT_ PARENT_JET_INTERFERENCE_V1` records the amplitude components and an explicit enumeration of all 220 trees. It retains the scalar 5/3072 history weight and the abstract finite GKSL completion, but refutes its amplitude affiliation above the first jump on the declared two-species carrier.

The grading-faithful enlargement resolves this scoped obstruction. Retain the constant and linear parent jet and singleton and pair spectator profile as the four-component tensor carrier with

$$\eta = J\otimes 3J, \qquad D=\operatorname{diag}(u,u,v,v), \qquad R=\begin{pmatrix} I_2 & I_2\end{pmatrix}.$$

The complete physical pullback

$$A=\eta^{-1}D^TR^T(3J)RD=\begin{pmatrix} uv & 0 & v^2 & 0\\ 0 & uv & 0 & v^2\\ u^2 & 0 & uv & 0\\ 0 & u^2 & 0 & uv\end{pmatrix}$$

satisfies $A^2=2uvA$. Hence $P=A/(2uv)$ is a regular Krein-selfadjoint projector above threshold. Its kernel is generated by $(v,0,-u,0)$ and $(0,v,0,-u)$, is nondegenerate, and is annihilated exactly by RD; its orthogonal image is generated by the corresponding plus-sign vectors and has raised endomorphism $2uvI_2>0$. For every four-component profile vector X,

$$(RDX)^T(3J)(RDX)=2uv(PX)^T\eta(PX),$$

so the quotient reproduces the complete six-point scalar amplitude rather than only its final trace. The same construction on the pure-profile five-point parent gives hard norm 3/2, projected norm 3/4, and signed ratio $-4LQ=\rho$. Therefore the first physical jump embeds consistently, and

$$q_1=\frac{5/3072}{1/48}=\frac{5}{64}$$

is amplitude-affiliated on the canonical quotient fibre. The certificate
`REVERSE_PHYSICS_BT_SIX_POINT_PROFILE_QUOTIENT_COMPLETION_V1`
records the arbitrary-vector identity, exact kernel/image Grams, prefix replay, and independent fixtures.

The complete seven-point amplitude closes that next pre-trace gate on the same carrier. For the three spectator jets set $S_1=a_4+a_5+a_6$ and $S_2=a_4a_5+a_4a_6+a_5a_6$. Its leading amplitude has one scalar, three equal singleton, and three equal complementary-pair components. Defining

$$A=(a_0-a_1)^2-2\tau_1(a_0+a_1)+2\tau_1^2, \qquad C=a_2\{a_2A+2\tau_2(-A+3\tau_1^2)\}+2\tau_2^2(A+\tau_1^2),$$

the two relevant coefficients factor uniquely through the recombined six-point constant/linear parent profiles as

$$C_{7,\mathrm{rel}}=F_1S_1+F_2S_2=uH_0+vH_1, \qquad u=-\frac{A}{2\tau_1^2},$$

$$v = \frac{C\tau_3^2 - A\tau_2^2(a_3^2 - 2a_3\tau_3 + 2\tau_3^2)}{4\tau_1^2\tau_2^2(\tau_3 + a_3)}.$$

The parent profile determinant is $3a_3^3(a_3+\tau_3)/(16\tau_3^4)$, so this factorization is unique. On the nested physical domain, $0 < A < 2\tau_1^2$. Writing $r = a_2/\tau_2$, $s = a_3/\tau_3$, and $g(x) = (x-1)^2 + 1$, the numerator controlling v is proportional to $A\{g(r) - g(s)\} + 2\tau_1^2(3r+1) > -A + 2\tau_1^2 > 0$. Thus $u < 0 < v$. The seven-external-leg delta-prime sign reverses the raw pullback, and its nonzero physical quotient eigenvalue is therefore $-2uv > 0$. With the same Krein-selfadjoint projector P, the exact identity

$$-(RDX)^T(3J)(RDX) = (-2uv)(PX)^T\eta(PX)$$

holds for every profile vector X; the negative raw image orientation is Hilbertized by the profile fundamental symmetry $-J$. An independent rail enumerates all 2485 trees at separate exact kinematics and matches all seven pre-trace components. Hence

$$q_2 = \frac{9/81920}{5/3072} = \frac{27}{400}$$

is amplitude-affiliated on the signed quotient. The certificate
`REVERSE_PHYSICS_BT_SEVEN_POINT_PROFILE_QUOTIENT_AFFILIATION_V1`
records the component tensor, threshold sign proof, quotient identity, and independent tree replay. All three available jumps are therefore affiliated with the five-, six-, and seven-point amplitudes.

They also admit one common reversible finite coupling jet. If B_k is the rooted-comb child–parent incidence, then $B_k^T B_k = (k+3)I$. The unique positive insertion-covariant edge weights for $K = F - F^\dagger$, $F_k = \alpha_k B_k \otimes I_2$, are

$$\alpha_k^2 = (k+1)q_k, \qquad (\alpha_0, \alpha_1, \alpha_2) = \left(\frac{\sqrt{3}}{12}, \frac{\sqrt{10}}{8}, \frac{9}{20}\right).$$

Thus $U_x = e^{xK}$ reproduces, with $a = x^2$, the aggregate leading column

$$P_1(a) = \frac{a}{16}, \qquad P_2(a) = \frac{5a^2}{512}, \qquad P_3(a) = \frac{9a^3}{8192}$$

through the complete available tree order. The 152-dimensional generator is Krein-skew on the Hilbertized quotient, has rank 52, and includes the nonzero reverse three-to-two-emission block; the level-three boundary is not made absorbing. Its incoming radial block has characteristic polynomial $z^4 + (17/10)z^2 + 81/1280$ and two positive frequency squares, so the finite witness is oscillatory rather than unstable. Certificate
`REVERSE_PHYSICS_BT_THREE_JUMP_ KREIN_MOLLER_JET_V1`
records this finite Møller Taylor column and its independent comb replay. It also exposes the remaining barrier exactly. The first transition has $\|P_\perp U_{\sqrt{a}} e_0\|^2 = a/16 + O(a^2)$, whereas every bounded strongly differentiable family $V(a) = I + aG + o(a)$ on a fixed finite carrier has off-diagonal probability $O(a^2)$. Hence the jet is not an additive-resolution Hamiltonian or semigroup. The required quantum-stochastic continuation can nevertheless be constructed on the same finite quotient. Put one Boson noise channel on each of the 75 rooted-comb insertion edges and

$$J_e = \sqrt{q_k}|c\rangle\langle h| \otimes I_2, \qquad D = \frac{1}{2}\sum_e J_e^\dagger J_e.$$

The Hudson–Parthasarathy equation [5]

$$dU_a = \left(\sum_e J_e \, dA_e^\dagger - \sum_e J_e^\dagger dA_e - D \, da \right) U_a$$

obeys both exact Itô structure identities and defines a strongly continuous unitary additive-resolution cocycle on $\Gamma_s(L^2(\mathbb{R}_+, da) \otimes \mathbb{C}^{75})$. Its vacuum reduction is exactly the certified branching semigroup. For a selected k-emission history, the ordered time simplex gives leading amplitude

$$a^{k/2} \sqrt{\frac{\prod_{j<k} q_j}{k!}} = x^k \frac{\prod_{j<k} \alpha_j}{k!}, \qquad x = \sqrt{a},$$

so its first three Fock sectors reproduce the finite Møller jet exactly. The 75 edge channels are minimal for the pinned channel-resolved completely positive map: their vectorized Kraus Gram is diagonal and has rank 75. Certificate
`REVERSE_PHYSICS_BT_QUANTUM_ STOCHASTIC_MOLLER_DILATION_V1`
records an independent reconstruction of both unitarity identities, Kraus rank, population resolvents, and all three amplitude coefficients. This breaks the finite additive-resolution barrier by white-noise scaling, not by restoring ordinary strong differentiability. The noise coordinate remains auxiliary resolution history, not a spacetime dimension. A physical fourth jump, continuum Møller/LSZ/S operator, complete probability, and Eq. (19) remain open, as do the non-strongly-ordered many-body probabilities. The first of those continuum-affiliation gates can be sharpened. For the physical direct-integral column V_r, $V_r^\sharp V_r = I(r) I_2$ with nonconstant I; in particular

$$I(1/16) - I(1/4) = \frac{11}{80} \log 2 - \frac{57}{2048} > \frac{419}{10240}.$$

Hence the raw columns are not related by isometric resolution translations. After the explicit polar normalization $E_R = V_{e^{-R}} / \sqrt{I(e^{-R})}$, however, $E_R^\sharp E_R = I_2$ and $C_b(R) = E_{R+b} E_R^\sharp$ transports their positive ranges. With $p_s(y) = \frac{1}{2} \operatorname{sech}^2(y - s)$,

$$(Af)_i(s, y) = \sqrt{p_s(y)} E_y f_i(s)$$

is an isometry from the three-pair, two-species HP one-particle carrier onto the corresponding Abel-regularized physical range and exactly intertwines resolution shifts. Multiplication by $\sqrt{q_0}$ gives $a/48$ per pair and $a/16$ in total on an interval of length a. The target must retain both the system and noise: a single pinned pair-noise mark has rank at most one, while the physical pair Gram has rank two. Certificate
`REVERSE_PHYSICS_BT_ABEL_FOCK_ PHYSICAL_INTERTWINER_V1`
records the independent polar, translation, rate, and rank checks. It physically affiliates the first three edge marks. At six points the positive quotient eigenvalue supplies the next physical measure rather than merely a rate. With $r = a_1/a_0$, $w = \tau_1/a_0$, $\Delta_r = \sqrt{\lambda(w, 1, r)}$, and

$$q = \frac{2w(1+r) - (1-r)^2}{2w^2}, \qquad \lambda_6 = a_2 q,$$

define

$$d\sigma_r = \frac{\lambda_6}{a_2(1+r)} d\mu_r = \frac{q \Delta_r}{(1+r)w} \, dw.$$

This density is positive above threshold, invariant under daughter exchange, and tends to $d\log w$ at infinity. An exact rationalized primitive proves that σ_r maps the threshold ray bijectively onto $\mathbb{R}_+$. If $E_6 = N_+/\sqrt{6qv}$ is the normalized image of the four-component parent/profile quotient, then

$$(B_r f)(w) = \sqrt{\frac{\lambda_6}{a_2(1+r)}} E_6(r,w) f(\sigma_r(w))$$

is unitary onto its physical direct integral. Composing it with the first Abel column gives an isometry from the ordered two-noise carrier and affiliates all twelve second-level edge marks. The exact rates obey $q_0 q_1 = (1/48)(5/64) = 5/3072$; the ordered simplex and twelve histories give $5a^2/512$. Certificate
`REVERSE_PHYSICS_BT_SIX_POINT_NESTED_ CONTINUUM_INTERTWINER_V1`
records the quotient, primitive, exchange, dense-domain, shift, and channel checks. Together the two continuum constructions physically affiliate marks 0 through 14. The seven-point signed quotient supplies the last available finite layer. Put

$$\alpha = A/\tau_1^2, \qquad s = a_3/\tau_3, \qquad w = \tau_2/a_2,$$

so that $1 \le \alpha < 2$, $0 < s < 1$, and $w > 1$, and define

$$H_0 = 2 + \alpha s(2-s), \qquad H(w) = H_0 + \frac{6-2\alpha}{w} + \frac{\alpha}{w^2}.$$

The signed eigenvalue and its asymptotic scale are

$$\lambda_7 = -2uv = \frac{\tau_3 \alpha H(w)}{4(1+s)}, \qquad S_7 = \frac{\tau_3 \alpha H_0}{4(1+s)}.$$

With the massless-hierarchy middle Källén measure $d\nu_0 = (w-1)dw/w$, the normalized coordinate

$$d\chi_{\alpha,s} = \frac{\lambda_7}{S_7} d\nu_0 = \frac{H(w)}{H_0} \frac{w-1}{w}\, dw$$

is positive and has unit linear asymptote. Its elementary primitive proves that it maps $[1,\infty)$ bijectively onto $\mathbb{R}_+$. Normalizing the signed image by $E_7 = N_+/\sqrt{-6uv}$ therefore gives the third conditional direct-integral unitary. Its composition with the two-emission column affiliates marks 15 through 74; the rate identity $(1/48)(5/64)(27/400) = 9/81920$, the ordered simplex, and sixty histories give $9/8192$. Certificate
`REVERSE_PHYSICS_BT_SEVEN_POINT_NESTED_ CONTINUUM_INTERTWINER_V1`
records the signed range, primitive, finite-hierarchy, shift, and channel checks. Thus all 75 marks in the available three-emission hierarchy have physical continuum columns. No fourth jump or all-order inductive operator is supplied. Their direct sum nevertheless completes the physical vacuum column of the finite stochastic model. If I_Ω is vacuum inclusion, U_a is the 75-channel Hudson–Parthasarathy unitary, and $\mathcal{A}_{\le 3} = I_{\mathrm{vac}} \oplus \mathcal{A}_1 \oplus \mathcal{A}_2 \oplus \mathcal{A}_3$, then

$$\mathcal{M}_a = \mathcal{A}_{\le 3} U_a I_\Omega, \qquad \mathcal{M}_a^* \mathcal{M}_a = I_2.$$

The exact ordered-history kernels include the survival factors with amplitude drifts

$$(d_0, d_1, d_2, d_3) = \left(\frac{1}{32}, \frac{5}{32}, \frac{27}{160}, 0\right).$$

Their $(1,3,12,60)$ history norms are nonnegative and sum to one for every $a \geq 0$, while their first allowed coefficients remain $1/16, 5/512, 9/8192$. Hence the hard response $-1/16$ and the physical real response $+1/16$ occur on the same finite physical column. Multiplication by the Born coefficient $3/32$ gives $-3/512 + 3/512 = 0$. The zero public R_t projector-pushforward response is not used as an S-matrix summand.

The public-to-physical mismatch also has a minimal exact completion. With $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, choose the public representative $D = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, so that its raised Gram is the rank-one nilpotent $\left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$. For the physical $T = \mathrm{diag}(2Q, 2L)$, one has $T^T JT = -\rho J$, $\rho = -4LQ > 0$. Retaining D as an orthogonal compression forces the missing pullback

$$C^T JC = -\rho J - D^T JD = \begin{pmatrix} 0 & -\rho \\ -\rho & -2 \end{pmatrix}, \qquad C = \begin{pmatrix} -\rho & -1 \\ 0 & 1 \end{pmatrix}.$$

For $F = (D, C)^T$, $\eta = J \oplus J$, and $W = FT^{-1}$, one has $F^T \eta F = T^T JT$, $W^T \eta W = J$, and $\pi_{\text{public}} WT = D$. Thus the public leg is a compression of a common Krein dilation, not the physical operator itself. The missing Gram has determinant $-\rho^2$ and raised trace -2ρ; it needs a two-dimensional cross-Krein complement and cannot be supplied by either a positive-noise fibre or a trace-null Eq. (19) remainder alone. Certificate
`REVERSE_PHYSICS_BT_FINITE_PHYSICAL_ MOLLER_COLUMN_V1`
records the exact simplex, population, compression, inertia, trace, and typing checks. The complement's algebraic form is fixed but is not derived from BT zero-mode or higher-composite dynamics. The result is a vacuum column of the pinned finite resolution model, not a two-sided spacetime S operator, a complete NLO theorem, or an all-order proof of Eq. (19).

At the level of abstract operator theory, that column does admit a complete two-sided unitary extension, and the exact freedom in doing so is useful. Let $I : H_{\text{hard}} \to K_{\text{phys}}$ be the incoming hard inclusion and put $P_{\text{in}} = II^*$, $P_{\text{out}} = \mathcal{M}_a \mathcal{M}_a^*$. Every unitary with $SI = \mathcal{M}_a$ is

$$S_W = \mathcal{M}_a I^* + W(1 - P_{\text{in}}), \qquad W^* W = 1 - P_{\text{in}}, \quad WW^* = 1 - P_{\text{out}}.$$

Conversely these identities make S_W unitary. On an exact four-outcome, two-species compression both defects have rank six; a Householder completion and its precomposition by a nontrivial rational defect rotation give two different exact unitaries with the same column. On the actual nested continuum the defect is countably infinite already in the one-emission L^2 range. Certificate
`REVERSE_PHYSICS_BT_PHYSICAL_MOLLER_ DEFECT_COMPLETION_V1`
therefore removes abstract unitarity as the obstruction but proves that the vacuum amplitudes leave an entire incoming-continuum unitary undetermined. Deriving that defect action from the BT asymptotic Hamiltonian, with crossing, trace and spacetime affiliation, remains the physical problem; choosing one mathematically is not a BT S-matrix construction.

The already constructed HP cocycle does not by itself select this missing action on the certified physical carrier. Let L count rooted-comb level and N_F count edge-noise quanta. Its structure coefficients preserve $Q_{\text{HP}} = N_F - L$, since creation raises both counts and reverse annihilation lowers both. However, the physical maps $\mathcal{A}_k$ cover only $0 < t_1 < \cdots < t_k < a$. On using such vectors as incoming states, the next edge time can occupy any of the $k+1$ positions among the ancestral times, so the canonical chamber is not two-sided invariant. The minimal no-spectator reducing closure has $k!$ chambers per rooted history. Thus the multiplicities $(1,3,12,60)$ become $(1,3,24,360)$: 312 crossed history sheets, or 624 species-resolved copies, lack physical intertwiners. Certificate

REVERSE_PHYSICS_BT_CROSSED_ORDER_ CHAMBER_COMPLETION_V1
checks a nonzero exact reversed-chamber witness at $q_1 = 5/64$. The first missing physical block is therefore a crossed two-time six-point intertwiner, not another abstract completion and not the vacuum fourth jump.

The one-branch analytic crossing of that block is now exact and fails the fixed-sharp isometry test. Put $w = -x < 0$ on the massless spacelike sheet. The outgoing coefficient becomes

$$q(r, -x) = -q_\times(r, x), \qquad q_\times = \frac{2x(1 + r) + (1 - r)^2}{2x^2} > 0.$$

The complete four-component quotient continues, but its nonzero raised eigenvalue is $-a_2 q_\times$ and its certified profile-swap Hilbertization is $-3a_2 q_\times I_2$. The six external delta-prime factors retain even sign. Thus no isometry from a positive HP reversed chamber can preserve this one-branch sharp, uniformly for all twelve histories. The certificate
REVERSE_PHYSICS_BT_CROSSED_ SIX_POINT_KALLEN_ OBSTRUCTION_V1
also constructs the conditional alternative. Reversing the branch sharp makes the image positive, while

$$d\sigma_\times = \frac{q_\times \sqrt{x^2 + 2(1 + r)x + (1 - r)^2}}{(1 + r)x}\, dx$$

has infinite length at both endpoints and maps the spacelike ray onto $\mathbb{R}$. It is therefore a bilateral minimal dilation of the reversed HP half-line, not another vacuum column. Neither the public R_t data nor Eq. (19) derives the required sharp reversal. A complete crossed $3 \to 3$ detector block, retaining both crossed orientations and their pre-trace interference, must supply it or prove the full crossed-channel obstruction.

The ordinary two-orientation recombination does not supply it on the certified factorized quotient. Momentum crossing $p \mapsto -p$ fixes p^2 and hence acts trivially on the constant/linear virtuality jet. Both tree orientations therefore carry the same negative species block up to phase. Any positive orientation density M gives

$$M \otimes (-6q_\times v I_2) \preceq 0,$$

so detector interference has no positive direction. More generally, the species-uniform collapse $R_t = [I_2, tI_2]$ has nonzero raised eigenvalue $-2q_\times tv$. A real unit-relative-sign repair is consequently unique:

$$R_- = [I_2, -I_2].$$

It selects the formerly complementary quotient and gives $+6q_\times v I_2$ under the fixed Hilbertization. Certificate
REVERSE_PHYSICS_BT_CROSSED_DETECTOR_ ORIENTATION_NO_GO_V1
shows that the inducing operation is the internal dual-number parity $\epsilon \mapsto -\epsilon$, with $S_\epsilon^T J S_\epsilon = -J$. It is anti-Krein and is not ordinary scalar crossing. Its BT dynamical affiliation, or a nonfactorizing crossed pre-trace term that replaces it, remains the exact missing datum.

The incoming Wightman dual does not furnish that parity as a universal one-particle rule. Introduce the spectral families $W_\mu^\pm = \theta(\pm p^0)\delta(p^2 - \mu)$. Momentum reflection exchanges W_μ^+ and W_μ^- and commutes with $-\partial_\mu$, so its action on the simple/dipole mass jet is I_2, not S_ϵ. There is no oriented Jacobian sign in the distributional pushforward. The amplitude check is decisive: at $\tau = -x$, the certified five-to-four operator becomes

$$T_\times = \mathrm{diag}(-q_\times, \ell_\times), \qquad q_\times, \ell_\times > 0,$$

and obeys $T_\times^\sharp T_\times = -\rho_\times I_2$. The fifth external delta-prime sign makes its physical Gram $+\rho_\times I_2$. A universal S_ϵ would instead make that Gram $-\rho_\times I_2$ and spoil the already healthy first crossed splitting. Certificate
`REVERSE_PHYSICS_BT_CROSSED_WIGHTMAN_ DUAL_NO_GO_V1`
therefore confines any surviving repair to a profile-selective, higher-composite or doubled operation, or to a nonfactorizing crossed $3 \to 3$ pre-trace term. It proves no full LSZ crossing theorem and leaves all twelve reversed physical intertwiners unaffiliated.

Regular profile selection on the same carrier also fails. With $\eta = J \otimes 3J$, every Krein isometry changes the crossed raised pullback only by similarity and therefore preserves its negative nonzero eigenvalue. Every anti-isometry reverses that eigenvalue, but it also reverses the norm of the certified five-point prefix

$$h = (0, \tfrac{1}{2}, \tfrac{1}{2}, 0)^T, \qquad h^T \eta h = \frac{3}{2}.$$

The canonical $S_{\text{parent}} = \operatorname{diag}(I_2, -I_2)$ indeed changes R_+ to R_-, but sends this prefix norm to $-3/2$. An exhaustive sixteen-sign census finds no metric-compatible repair preserving the prefix, and continuity forbids an identity-to-anti-Krein switch at the no-spectator boundary on a fixed nondegenerate carrier. Certificate
`REVERSE_PHYSICS_BT_CROSSED_PROFILE_ SELECTIVE_PARITY_OBSTRUCTION_V1`
also verifies that the public neutral degree-four Fock ghost parity is a Krein isometry, even on its selected ghost-odd negative plane. Hence inherited higher-composite parity cannot supply the anti-Krein sign. A singular chart, new doubled/off-diagonal amplitude, or the nonfactorizing crossed $3 \to 3$ pre-trace block remains open.

That pre-trace loophole closes on the complete correlated square-free leading jet even before the strong hierarchy limit. The full 220-tree sum at finite e has one common row for singleton masks $1, 2, 4$, one common row for pair masks $3, 5, 6$, and zero cubic mask. The outer profile matrix is invertible, with determinant $-3a_2^2/(8\tau_2^2)$, and reconstructs unique coefficients $u(e), v(e)$ with identically zero nonfactorizing residual. After $\tau_1 = -x$ and $(a_0, a_1) = (1, r)$, their signs are fixed throughout the positive domain by

$$-6a_2^2 x^2 u_\times = 3a_2^2\big[(r-1)^2 + 2x(r+1)\big] + 2e^2x^2(r^2+r+1) > 0$$

and

$$6a_2 x v_\times = 3a_2^2 x + 3a_2 e\big[(r-1)^2 + x(r+1)\big] + 2e^2 x(r^2+r+1) > 0.$$

Consequently the fixed-sharp quotient retains its negative rank-two eigenvalue $2u_\times v_\times$ for every finite positive hierarchy ratio. Certificate
`REVERSE_PHYSICS_BT_CROSSED_SIX_POINT_ NONFACTORIZING_PRETRACE_NO_GO_V1`
therefore rules out a hierarchy-limit artifact or hidden pre-trace repair on this cylinder. It does not cover the complete non-correlated six-body phase space, subleading external-mass orders, or doubled/off-diagonal source data; one of those genuine enlargements is now required.

The first enlargement changes the conclusion at the level of the complete local Born density. Take three exact future-null incoming momenta with total momentum $(16/5, 0, 0, 0)$ and rotate the outgoing triple in their plane by

$$c(t) = \frac{1-t^2}{1+t^2}, \qquad s(t) = \frac{2t}{1+t^2}.$$

The resulting all-incoming six-tuple is massless and conserves momentum for every real t. On this continuous physical family the complete 220-tree amplitude in the 64-slot six-mass jet begins at degree three. Its twenty middle-degree coefficients obey ten exact complement identities $c_S(t) = c_{S^c}(t)$. This self-duality is absent from each of the three tree topology sectors separately and appears only after their perfect-square relative coefficients are summed. Hence

$$[x_0 \cdots x_5]|\mathcal{M}_6|^2 = 2 \sum_{S<S^c} c_S(t)^2 > 0$$

away from the internal propagator poles. Strictness follows because the ten reduced numerator polynomials have gcd one; every pole has even multiplicity. Moreover $\mathcal{M}_6$ starts at mass degree three, so derivatives of a regular local phase-space weight cannot mix into this top coefficient. Certificate
`REVERSE_PHYSICS_BT_SIX_POINT_PLANAR_ PHYSICAL_BORN_DENSITY_V1`
therefore proves that the negative correlated quotient is not the complete physical local density. It establishes a positive nonforward planar continuum, not positivity over nonplanar six-body phase space, a regulated integral, normalized probability, or Eq. (19).

Coplanarity is not essential to that positive mechanism. Compose the in-plane rational rotation with an out-of-plane rotation about an independent spatial axis whose stereographic parameter is $u = t/2$. The six momenta remain exactly null and conserved, while the outgoing triple has generically nonzero z components and spans a different plane. On this exact nonplanar continuum, the full amplitude again has 42 mass-jet terms beginning at degree three, all ten identities $c_S = c_{S^c}$ hold, and complement antisymmetry cancels only after all three topology sectors are combined. The local kernel is again $2\sum c_S^2$; its ten numerator polynomials have gcd one and its ten pole factors have even multiplicity. Certificate
`REVERSE_PHYSICS_BT_SIX_POINT_NONPLANAR_ DIAGONAL_PHYSICAL_BORN_DENSITY_V1`
therefore proves strict local positivity away from poles without planar kinematics. It remains a one-parameter diagonal, not the full two-parameter nonplanar phase space or its regulated integral.

The diagonal restriction can now be removed exactly. Let the two stereographic parameters in $R_x(u)R_z(t)$ be algebraically independent. A sparse-FLINT computation of the complete 220-tree mass jet over $\mathbb{Q}(t,u)$ finds no term below degree three and proves all ten identities $c_S(t,u) = c_{S^c}(t,u)$. Therefore

$$[x_0 \cdots x_5]|\mathcal{M}_6|^2 = 2 \sum_{S<S^c} c_S(t,u)^2 \geq 0$$

at every regular real pair, with strict inequality on a dense open regular subset. Certificate
`REVERSE_PHYSICS_BT_SIX_POINT_TWO_ANGLE_ PHYSICAL_BORN_DENSITY_V1`
records the exact rational-function equalities and six independent explicit 220-tree checks. A bivariate elimination was stopped at the memory boundary, so isolated or lower-dimensional common zeros are not excluded. Moreover the two angles rigidly rotate a fixed outgoing energy triangle: they do not cover the full five-dimensional final-state phase space. Pole regulation, integration, normalization, and Eq. (19) remain open.

Both restrictions can in fact be removed without constructing a five-variable rational amplitude. For the ten unordered three–three channels set $s_A = (\sum_{i\in A} p_i)^2$. The complete middle coefficient has the universal form

$$c_S = c_{S^c} = \frac{1}{4} \sum_{A \neq S} \frac{1}{s_A}.$$

The auxiliary $O(1,1)$ derivation makes the formula transparent: S labels three external Ω and three external Υ legs, and a pair of $\Omega^2\Upsilon^2$ vertices joined by the cross propagator exists in every unordered $3|3$ channel except the one that places all three like species on one side. The resulting ten-channel incidence matrix is $J-I$, with $\det(J-I)=-9$. Hence the ten c_S have no common zero at any regular finite kinematic point, and

$$[x_0\cdots x_5]|\mathcal{M}_6|^2 = 2\sum_{S<S^c} c_S^2 > 0$$

throughout the complete regular massless physical $3\to 3$ phase space. An exact chart with two outgoing-shape and three orientation parameters has Jacobian rank five; a nonzero minor is 864/3125. Six full-chart rational fixtures independently reproduce the formula by explicit enumeration of all 220 trees. Certificate
REVERSE_PHYSICS_BT_SIX_POINT_FULL_PHASE_ SPACE_BORN_POSITIVITY_V1
therefore establishes the complete local tree sign theorem, not a regulated or integrated probability. The ten channel-pole hypersurfaces, normalization, loops, Eq. (19), and any gravitational lift remain open.

The ordinary exclusive integral is in fact obstructed. Writing $y_A = 1/s_A$ reduces the exact density to

$$D = \left(\sum_A y_A\right)^2 + \frac{1}{8}\sum_A y_A^2,$$

so an isolated channel pole has leading term $9/(8s_B^2)$. At the exact positive-energy rank-five chart point $(a,b,t,u,v) = (2,-2,3/5,1/2,1/3)$, s_{11} is the unique zero and $\partial_t s_{11} = -1152/425$. Thus

$$D = \frac{180625}{1179648(t-3/5)^2} + O((t-3/5)^{-1}),$$

and every ordinary neighborhood integral diverges positively. Certificate
REVERSE_PHYSICS_BT_SIX_POINT_PHASE_ SPACE_POLE_OBSTRUCTION_V1
shows that principal value supplies no sign cancellation: a detector or inclusive real–virtual prescription is mandatory and remains unconstructed.

The pole also admits no positive exclusive distributional completion. Indeed, positivity would make the extension a locally finite Radon measure, whereas agreement with the leading density on $\epsilon < |s_B| < L$ forces mass $\frac{9}{4}(\epsilon^{-1} - L^{-1}) \to \infty$. Delta-supported counterterms do not change this punctured mass. Scaling-degree-preserving linear extensions exist with ambiguity $c_0\delta + c_1\delta'$, reduced to $c_0\delta$ by reflection symmetry of the leading pole, but every finite-part extension is nonpositive. The symmetric Feynman-modulus regulator displays the unresolved history as $9\pi\delta(s_B)/(8\epsilon)$. Certificate
REVERSE_PHYSICS_BT_SIX_POINT_POSITIVE_ DISTRIBUTION_COMPLETION_NO_GO_V1
also type-checks the proposed complements: the available five-point NLO term is order λ^6 on an external collinear boundary, rather than the order-λ^8 internal $3|3$ density, while the finite Møller result fixes only one input column and its two-sided defect partial unitary is not selected by the public BT data. The next physical object is therefore a BT-derived finite-time or wave-packet Møller action that counts the on-shell sequential four-point history separately. No finite inclusive probability is yet constructed.

The leading history carrier can nevertheless be constructed exactly. In the ten-channel basis the residue map is $R = (J-I)/4$, so

$$D = y^T\left(J+\frac{1}{8}I\right)y = \frac{9}{8}\sum_A \frac{1}{s_A^2} + 2\sum_{A<B}\frac{1}{s_A s_B}.$$

For each channel the residue has nine entries 1/4 and one zero, giving Born norm 9/8; the complete Gram has spectrum 81/8 and 1/8 with multiplicity nine. The diagonal term therefore contains every positive double pole as a separately labelled sequential outcome. The zero-diagonal interference block is $J - I$, with inertia $(1, 9)$, and is not itself a positive outcome Gram. For $F_T(\omega) = \int_0^T e^{i\omega t}dt$ and $s = 2E\omega + O(\omega^2)$, the exact finite-time normalization gives $|F_T|^2/(4E^2) \to (\pi T/E)\delta(s)$, or $9\pi T\delta(s)/(8E)$ for the BT leading history. Certificate `REVERSE_PHYSICS_BT_SIX_POINT_SEQUENTIAL_ HISTORY_CARRIER_V1` constructs this reduced carrier and coefficient. It does not derive the finite-time block from the BT Hamiltonian, prescribe the signed interference, or provide the matching survival term.

The channel labels themselves admit a normalized finite completion once the final species assignment S and intermediate channel A are kept as distinct types. The auxiliary quartic vertex allows the ninety histories (S, A) with $S \neq A$. The lift $B_{(S,A),B'} = \delta_{AB'}/4$ obeys $B^T B = 9I/16$, so $W_{\text{hist}} = 4B/3$ is an isometry, while coherent collapse $C_{S',(S,A)} = \delta_{S'S}$ gives $CB = (J - I)/4$. The resolved and coherent positive weights pull back to $9I/8$ and $J + I/8$; their difference, rather than a negative outcome, is the signed interference $J - I$. Moreover

$$P_\mu = \frac{(1-\mu)I + \mu C^T C}{9}, \qquad 0 \le \mu \le 1,$$

is a genuine history-sector effect with positive complement. The skew block $K = \begin{pmatrix} 0 & -W_{\text{hist}}^T \\ W_{\text{hist}} & 0 \end{pmatrix}$ obeys $K^3 = -K$; its exact rotation, followed by $\{P_\mu, I - P_\mu\}$, gives survival, detected and unresolved effects whose sum is I_{10}. Certificate `REVERSE_PHYSICS_BT_SIX_POINT_HISTORY_ INCIDENCE_ISOMETRY_V1` therefore fixes the finite equal-edge channel-label part of the Møller defect action. The rotation angle, detector coherence, finite-time energy kernel and momentum-wave-packet defect embedding remain unaffiliated with BT dynamics.

The transverse shell coordinate now supplies the missing finite-time kernel locally. At the exact rank-five pole above, the channel-11 intermediate momentum is $q = (1, 9/17, -4/85, -72/85)$, so $q^2 = 0$ and $E = 1$. Exact coarea reduction of the labeled three-body phase measure gives

$$d\Phi_3 = \frac{54}{2125(2\pi)^5}\, da\, db\, dt\, du\, dv = \frac{3}{320(2\pi)^5}\, da\, db\, ds\, du\, dv.$$

For a finite observation interval, $\alpha_{T,E}(s) = F_T(s/(2E))/(2E)$ is square integrable and obeys $\int |\alpha_{T,E}|^2 ds = \pi T/E$. Tensoring it with the fixed-channel history vector of norm squared 9/8 gives $Q_T = 9\pi T/(8E)$. The exact local column $M_g = (\sqrt{1 - g^2 Q_T}, gA_{T,E})^T$ is isometric when $0 \le g^2 Q_T \le 1$, and is the source column of a rank-one unitary rotation. At the fixture, the frozen labeled phase measure contributes $27T/(81920\pi^4)$ before the common BT multiplier, incoming flux, tangential detector normalization, and generalized-Born convention. Certificate `REVERSE_PHYSICS_BT_SIX_POINT_FINITE_TIME_ SHELL_COLUMN_V1` therefore constructs a finite local sequential probability column. It does not calibrate g, glue the ten channel tubes, prescribe their connected interference, construct a global Møller operator, or prove Eq. (19).

The Hamiltonian part of that calibration is fixed. Restoring the public factors $V_3 = -2i\lambda F_3$, $V_4 = -4i\lambda^2 F_4$ and $P = -i/K^4$, the V_4^2, $V_3^2 V_4$ and V_3^4 six-point topology classes carry the common factor $16i\lambda^4$ with relative signs $+, -, +$. The reduced density therefore acquires $256\lambda^8$. The finite shell norm is $288\pi\lambda^8 T/E$, and the exact fixture gives the labeled local coefficient

$$\frac{27\lambda^8 T}{320\pi^4}$$

per unit tangential chart volume. Certificate `REVERSE_PHYSICS_BT_SIX_POINT_SHELL_TREE_NORMALIZATION_V1` also fixes why this is not yet a probability. The coefficient has mass dimension -2; the public two-particle example supplies the compensating transverse-area normalization, but the Letter gives no corresponding three-particle incoming characteristic cell or $3 \to 3$ flux convention. A dimension-two incoming projector weight and a tangential detector function must therefore be declared and derived through the full generalized-Born trace. The tree coupling is no longer missing, but a universal scalar g still is not defined.

For a declared finite-volume detector, that last local normalization can be completed without postulating a universal three-body flux. Scale the exact three-beam fixture by κ, fix eight spatial momentum components, and replace the p_{0x} constraint by $E_0 + E_1 + E_2 = 16\kappa/5$. The on-shell constraint Jacobian is one. In the finite-volume conventions of Appendix B of Bateman–Turok, one ordered cell therefore has weight

$$\frac{5}{48\kappa^3 L_0 L_x^2 L_y^3 L_z^3}.$$

The external $\delta_4(0) = L_0 L_x L_y L_z$ leaves

$$N_{\text{in}} = \frac{5}{48\kappa^3 L_x L_y^2 L_z^2},$$

which has the required mass dimension two. The permutation-symmetric incoming cell and physical outgoing cell each contain six disjoint label orbits, cancelling the two 3! factors in the generalized-Born trace. Hence the local shell rate per unit dimensionless tangential volume is

$$\Gamma_\Xi = \frac{9\lambda^8}{1024\pi^4\kappa^4 L_x L_y^2 L_z^2}, \qquad q_\Xi(T) = \Gamma_\Xi T\,\Delta\Xi + O(1).$$

Certificate `REVERSE_PHYSICS_BT_THREE_PARTICLE_ CHARACTERISTIC_CELL_V1` proves this coefficient and its factorial audit. It also proves that the normalization is detector data: replacing p_{1x} rather than p_{0x} by the energy characteristic changes the Jacobian to $3/5$ and the point-cell weight by $5/3$. Thus a dimensionless probability coefficient exists for the declared cell, but no detector-independent $3 \to 3$ cross section, compact wave-packet limit, ten-channel gluing, or global scattering operator has been inferred.

The finite-time kernel in this coefficient is now dynamically affiliated at the cut-probability level. In the interaction-picture generalized Born trace, a spectral intermediate state of mismatch ω propagates for relative durations τ and τ' on the amplitude and conjugate, hence

$$\int_0^T d\tau \int_0^T d\tau'\, e^{i\omega(\tau-\tau')} = |F_T(\omega)|^2 = \int_{-T}^{T} (T - |\sigma|) e^{i\omega\sigma} d\sigma.$$

The incoming energy characteristic cancels the independent external $\delta_1(0) = L_0$; the internal relative duration T remains. This derives the sinc-squared shell and its $\pi T/E$ norm from the public spectral Hamiltonian/Born structure and reproduces Γ_Ξ without fitting a time regulator. Certificate `REVERSE_PHYSICS_BT_FINITE_TIME_HAMILTONIAN_ CUT_AFFILIATION_V1` also isolates why this does not yet derive survival from BT evolution. Pseudo-unitarity conserves a signed Krein trace, but a J-unitary boost with $J = \mathrm{diag}(1, -1)$ has complementary weights $\cosh^2 r$ and $-\sinh^2 r$. The positive $\cos^2\theta/\sin^2\theta$ history rotation is therefore still an operational detector dilation until its carrier is embedded as a weakly ghost-symmetric BT subquotient or obtained from Eq. (19).

That output-carrier embedding can now be made inside the public neutral six-leg Fock sector. Its twenty three-Ω/three-Υ assignments come in ten complement pairs. In the basis of representatives followed by complements, both the cross-Krein metric and ghost parity are

$$\eta = \kappa = \begin{pmatrix} 0 & I_{10} \\ I_{10} & 0 \end{pmatrix}, \qquad U_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} I_{10} \\ I_{10} \end{pmatrix}, \qquad U_+^T \eta U_+ = I_{10}.$$

The certified complement identity $c_S = c_{S^c}$ therefore puts the complete coefficient $a = (c, c) = \sqrt{2}U_+ c$ entirely in this positive ghost-even plane, with

$$a^T \eta a = 2\sum_S c_S^2.$$

Equivalently, arrange the twenty neutral coefficients as the 8×8 three-in/three-out species matrix A and let $\kappa_3|x\rangle = |7-x\rangle$. Then $\kappa_3 A \kappa_3 = A$, hence $A^\sharp = A^T$ and $\mathrm{tr}(A^\sharp A) = 2\sum_S c_S^2$. For each fixed intermediate channel B, the nine normalized histories $S \neq B$ embed isometrically as $U_+ e_S$; their residue maps to coefficient $1/4$ on both S, S^c and has norm $9/8$, exactly matching the Hamiltonian-cut shell norm. Certificate `REVERSE_PHYSICS_BT_SIX_POINT_GHOST_EVEN_ HISTORY_EMBEDDING_V1` therefore closes the fixed-shell output-history affiliation gate.

It also identifies the remaining record problem without ambiguity. The map that simultaneously has this action on all ninety resolved histories has rank ten and an eighty-dimensional kernel: public species data coherently forgets which intermediate channel occurred. An explicit positive channel-label ancilla is required to retain that detector record. More importantly, output ghost symmetry does not compute $R_t P_\chi^{(\phi)} R_t^\dagger$. The transported physical input projector, vacuum/zero-mode trace, virtual survival block, global shell gluing and all-order Eq. (19) remain open.

There is nevertheless a direct physical route inside the public auxiliary theory. Restrict the three-particle source and output to the positive even frame

$$u_x = \frac{|x\rangle + |7-x\rangle}{\sqrt{2}}, \qquad x = 0, 1, 2, 3.$$

The fixed channel-$B = 1$ Choi residue becomes

$$R_+ = \frac{1}{4}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix},$$

so the click effect $G = R_+^T R_+$ has spectrum

$$0, \quad \frac{1}{16}, \quad \frac{2-\sqrt{3}}{8}, \quad \frac{2+\sqrt{3}}{8}.$$

It is therefore positive on an actual public BT source sector. Dividing the full rate by its full residue norm 9/8 gives

$$\gamma_0 = \frac{\lambda^8}{128\pi^4\kappa^4 L_x L_y^2 L_z^2}, \qquad \zeta = \gamma_0 T \Delta\Xi.$$

The two effects

$$E_{\text{click}} = \zeta G, \qquad E_{\text{no}} = I_4 - \zeta G$$

are positive and sum to I_4 for $0 \le \zeta \le 16 - 8\sqrt{3}$. For $u_0 = (|\Upsilon\Upsilon\Upsilon\rangle + |\Omega\Omega\Omega\rangle)/\sqrt{2}$, they give

$$q_{\text{click}} = \frac{\zeta}{16}, \qquad q_{\text{no}} = 1 - \frac{\zeta}{16}, \qquad \Gamma_{u_0,\Xi} = \frac{\lambda^8}{2048\pi^4\kappa^4 L_x L_y^2 L_z^2}.$$

The BT cut fixes the click coefficient, and the binary operational complement fixes the no-click effect. A minimal skew completion has Hermitian forward coefficient $-G/2$, but this is genuine BT survival only if the selected output is exhaustive. Certificate
`REVERSE_PHYSICS_BT_ SIX_POINT_POSITIVE_ SECTOR_PHYSICAL_ DETECTOR_EFFECT_V1`
thus supplies a normalized leading physical probability jet for a source prepared directly in the public auxiliary theory. It deliberately bypasses, and does not prove, the scalar projector pushforward: $P_u \neq R_t P_\chi^{(\phi)} R_t^\dagger$. Smooth finite-duration terms, higher-order resummation, global shell gluing and all-order Eq. (19) remain open.

The distinction is exact. If A_C is the selected click block and B_{out} contains positive transitions outside the detector, then

$$\mathrm{Herm}(M_8)_{\text{source}} = -\frac{1}{2}(A_C^* A_C + B_{\text{out}}^* B_{\text{out}}).$$

This is the $C_4^* C_4$ cut component; other perturbative products and cut-free order-λ^8 terms remain separate. True survival loses both terms, whereas outside leakage restores $B_{\text{out}}^* B_{\text{out}}$ inside the no-click event. Thus the detector still has $E_{\text{no}} = I - A_C^* A_C$, but its local click data do not determine the forward virtual coefficient. Two exact rational pseudo-unitary completions with the same click block and different leakage blocks prove this nonidentifiability. Certificate
`REVERSE_PHYSICS_BT_COMPACT_DETECTOR_ SURVIVAL_LEAKAGE_FACTORIZATION_V1`
therefore removes the virtual graph from the binary detector-probability gate while retaining it for genuine survival, complete evolution, and an S-matrix. The minimal Julia dilation is the zero-leakage completion, not a uniquely derived BT evolution.

The connected order-λ^4 output type can now be closed. Since the public cubic and quartic vertices carry coupling degrees one and two, every connected graph obeys

$$d_\lambda = E + 2L - 2.$$

At $d_\lambda = 4$ the types are $(E, L) = (6, 0), (4, 1), (2, 2), (0, 3)$. The last is vacuum, $E = 2$ cannot contain the three incoming legs, and $E = 4$ would be $3 \to 1$, excluded because the packet has $P^2 = 256/25$ while its sole massless output has square zero. Hence the source-connected column is exactly the $3 \to 3$ tree. The generic six-point Choi matrix preserves complement parity, so a positive even source has no negative-parity output. On a common compact regular acceptance the actual, unpartitioned tree column is

$$A_{\text{full},C} = 16\lambda^4 \sum_{B=0}^{9} K_{B,T} \otimes R_B, \qquad \|A_{\text{full},C}\|^2 \le 12960\lambda^8 \frac{T^2\mu(X)\mu(Y)}{d^2}.$$

Here every public channel has unit weight; no subordinate channel partition is inserted. Certificate
`REVERSE_PHYSICS_BT_COMPLETE_CONNECTED_ ORDER4_PACKET_COLUMN_V1`
therefore reduces connected outside leakage to positive-even three-body momentum outside the detector. This reduction also permits an exact removal of the compact momentum cutoff at fixed finite time. Put $P = (M, \mathbf{0})$, $M = 16/5$. Every mixed channel can be oriented as $q = P - p - k$, where the two spectator energies satisfy $0 \le E, K \le M/2$. Thus $q^0 = M - E - K \ge 0$, and $q = 0$ occurs exactly when $E = K = M/2$ and the spectator directions are antipodal. The transverse Jacobian of (q^0, q^z, q^x, q^y) with respect to two energies and two relative angular coordinates has determinant

$$-2M^2 = -\frac{512}{25}.$$

Using

$$d\Phi_3(P) = \frac{E\, dE\, d\Omega\, d\Omega_*}{512\pi^5}$$

and integrating the relative angle and both energy variables gives, exactly,

$$\int_{\Phi_3\times\Phi_3} \frac{d\Phi_3(x) d\Phi_3(y)}{D_B^2} = \frac{1}{3200\pi^6}$$

for each of the nine exchange channels; the intermediate dimensionless integral is $1/16$. The hard channel contributes $1/(6400\pi^6)$. Consequently the unpartitioned column exists globally as a Hilbert–Schmidt operator and obeys

$$\|A_{\text{full}}\|^2 \le \frac{1539}{400\pi^6}\lambda^8 T^2.$$

Certificate
`REVERSE_PHYSICS_BT_GLOBAL_CONNECTED_FINITE_ TIME_PACKET_COLUMN_V1`
therefore closes the connected order-λ^4 momentum domain, including $q_B = 0$ as an L^2 equivalence class. Disconnected spectator terms, the matching forward coefficient, all-time dynamics, and Eq. (19) remain open. There is, however, a nonempty detector class on which the disconnected gate vanishes exactly. At the certified rational incoming/outgoing centers, put all six external momenta in incoming convention. The minimum Euclidean squares of every subset momentum sum of sizes one, two, and three are

$$2, \qquad \frac{32}{625}, \qquad \frac{17794}{10625},$$

respectively. Hence sufficiently small compact packet neighborhoods avoid every component momentum delta. Exhaustive enumeration gives 202 disconnected set partitions in ten size profiles; every profile contains a component with at most three legs. Since distributional and external-mass derivatives preserve support, every disconnected contribution through order λ^4 pairs to zero with this fully rearranged detector. Thus

$$P_Y(U_T - I)P_X = \lambda^4 P_Y A_{\text{full}} P_X + O(\lambda^5),$$

and $P_Y P_X = 0$ removes the identity and forward coefficient from the first click term. For the dressed scalar source the complete leading physical coefficient is therefore

$$q^{(8)}_{Y\leftarrow X} = 16\lambda^8 \left\| \sum_{B=1}^{9} P_Y K_{B,T} P_X F \right\|^2 \le \frac{81}{200\pi^6}\lambda^8 T^2.$$

Certificate
`REVERSE_PHYSICS_BT_FULLY_REARRANGED_ PHYSICAL_PACKET_PROBABILITY_V1`
establishes a complete leading finite-time transition probability for this selected open detector class. Overlap detectors, higher orders, all-time scattering, and general Eq. (19) remain open. The first overlap stratum can also be closed locally. There is an exact hard nonforward rational configuration with one tagged identity spectator and no other vanishing component momentum sum of sizes one, two, or three. Of all 203 set partitions, the only supported disconnected partition is therefore the tagged two-leg identity times the active four-leg block. It is the unique order-λ^2 transition; the connected six-leg tree begins at order λ^4. The active four-point mass-jet vector is $r_4 = 2\sum_{i<j} e_{ij}$. Under the complement-pairing metric it lies in the positive eigenspace and obeys $r_4^\sharp r_4 = 24$, reproducing

$$\frac{d\sigma}{d\Omega} = \frac{3\lambda^4}{32\pi^2 s}.$$

Tensoring a normalized spectator packet therefore gives the complete leading tagged-detector coefficient

$$q_{\text{tag}} = \frac{3\lambda^4 \Delta\Omega}{32\pi^2 s\,\text{Area}} + O(\lambda^5).$$

At the certified point $s = 64/25$, its rational coefficient is $75/2048$. Certificate
`REVERSE_PHYSICS_BT_TAGGED_SPECTATOR_ PHYSICAL_PACKET_PROBABILITY_V1`
thus closes the first spectator-overlap stratum at leading order. A single finite-resolution detector crossing this stratum and the fully rearranged bulk, the identity/collinear strata, higher orders, all-time scattering, and Eq. (19) remain open. These are in fact the only local hard nonforward strata. An exact census of all 202 disconnected six-label partitions gives 162 containing a singleton. Of the remaining partitions, the fifteen $2+4$ cases split into the nine spectator cylinders $\Delta_{ia} = \{p_i = k_a\}$ and six impossible same-side pairs. The ten $3+3$ cases split into the all-incoming/all-outgoing total and nine mixed cases; a mixed conservation equation $p_i + p_j = k_a$ would imply $2p_i \cdot p_j = 0$ and is excluded by hard noncollinearity. Finally, the fifteen perfect matchings split into six forward permutations and nine matchings containing a same-side pair. Two distinct spectator equalities either identify same-side momenta or, by total conservation, force the third equality and hence a forward permutation. Thus the nine cylinders are pairwise disjoint on the hard nonforward domain, and no generic two-spectator stratum exists.

Certificate
`REVERSE_PHYSICS_BT_HARD_NONFORWARD_PHYSICAL_ STRATIFIED_ATLAS_V1`

therefore promotes the two calculations to a complete local atlas: the bulk starts with the connected $3 \to 3$ probability at order λ^8, while each spectator cylinder starts with the active $2 \to 2$ probability at order λ^4. This is distributionally stratum-local. For one unresolved finite-resolution record, the order-λ^2 spectator and order-λ^4 connected amplitudes can interfere at probability order λ^6. Their common positive external-jet carrier is now explicit. For tag pair $\{0,3\}$, the active four-point jet embeds with coefficient two on the six neutral complement pairs containing exactly one tag label and zero on the other four, preserving its norm 24. Pairing with $c = (J - I)\beta/4$ gives incidence weights five and six. At the certified tagged fixture the four weight-six channels obey $\delta = 0$, $D = 2$, and the exact tree cross kernel is

$$I^{(6)}_{\text{tree}} = 16\sqrt{2}\lambda^6 W(T), \quad W(T) = 12T + \frac{125}{256}\sin\frac{16T}{5} + \frac{125}{128}\sin\frac{8T}{5} + \frac{125}{8}\sin\frac{2(\sqrt{17}-3)T}{5}.$$

Moreover $W(T) \geq [221 - 50\sqrt{17}]T/8 > 0$ for $T > 0$ and $W(T)/T \to 12$. Certificate
`REVERSE_PHYSICS_BT_TAGGED_CONNECTED_ FINITE_TIME_INTERFERENCE_V1`
therefore proves that the two tree sectors do not decouple: four connected factorization channels resonate on the spectator cylinder. This is an external-mass/species cross kernel, not a normalized packet probability or the complete order-λ^6 coefficient. Common packet/coarea normalization, the active one-loop term, source dressing and survival/ collinear completion remain the next physical gate; all-time scattering and Eq. (19) remain open. The missing spectator normalization can be fixed for a declared finite box. The public cross commutator gives the positive ghost-even mode norm $N_s = 2E_sV$. Normalizing both external spectator legs cancels N_s in the disconnected identity tree but leaves $1/N_s$ on the connected tree. Restoring the scale *kappa*, with $E_s = 6kappa/5$ and $W_kappa(T) = kappa^{-2}w(kappaT)$, gives

$$q^{(6)}_{\text{cross}} = \frac{125\sqrt{2}\lambda^6 W_\kappa(T)\Delta\Omega}{12288\pi^2\kappa^3 \text{Area}\, V}.$$

Certificate
`REVERSE_PHYSICS_BT_TAGGED_CONNECTED_ FINITE_VOLUME_NORMALIZATION_V1`
therefore makes the tree cross contribution dimensionless on the declared cell. It vanishes as V^{-1} at fixed finite T, but at fixed V its ratio to the leading tagged probability grows asymptotically as $(10\sqrt{2}/3)\lambda^2 T/(\kappa^2 V)$. Fixed-time thermodynamic suppression is not all-time decoupling: the joint limit depends on $T/(\kappa^2 V)$. A compact-packet replacement and the loop, source and survival terms are still required for the complete order-λ^6 probability. The compact-packet replacement changes the interpretation of this limit. For normalized positive spectator packets f, g in the public measure $d\nu = d^3p/(2E_p)$, let $c_{gf} = \langle g, f\rangle_\nu$ and let $C_{gf}(T) = \langle g, \mathcal{W}_{\kappa,T} f\rangle_\nu$, where the active- contracted ten-channel kernel has incidence weights five and six. Then

$$\frac{q^{(6)}_{\text{cross}}[g,f]}{q^{(4)}_{\text{tag}}} = \frac{2\sqrt{2}\lambda^2}{3}\Re(\overline{c_{gf}}C_{gf}(T)).$$

On a compact hard tube with $D_A \geq d_0 > 0$ this kernel is bounded by $54T/d_0$ and is Hilbert–Schmidt. Certificate
`REVERSE_PHYSICS_BT_TAGGED_CONNECTED_ COMPACT_PACKET_INTERFERENCE_V1`
proves that the single-mode V^{-1} limit is not the continuum limit of a fixed packet. With N equal momentum cells of measure h, a uniform packet has matrix element $(h/N)\sum_{ij}\mathcal{W}_{ij}$; at fixed packet measure $Nh = \mu$ this tends to the finite double packet integral, whereas $N = 1$

recovers $\mathcal{W}/N_s$. Since the diagonal fixture is strictly positive, continuity gives nonzero positive compact packets for each fixed $T > 0$. The tree cross therefore survives continuum packet normalization, but source, loop and survival completion still decide the physical probability. The possible intervening odd probability order is fixed without computing an endpoint coefficient. Total Fock parity $\Pi_F = (-1)^N$ is distinct from both BT ghost parity and $SO^+(1,1)$ charge. The exact covariance $S_\lambda[\phi] = S_{-\lambda}[-\phi]$ and the corresponding sign covariance of the BT composite map make every order-n evolution and dressed-projector coefficient have parity $(-1)^n$. The leading three-particle source and its order-λ^2 tagged output are odd, while the complete order-λ^3 output, including first source and detector corrections, is even. Since the cross-Krein form is block diagonal in total particle-number parity,

$$q^{(5)}_{\text{tag}} = 2\Re\langle y_2, y_3\rangle_K = 0.$$

Certificate
`REVERSE_PHYSICS_BT_TAGGED_PACKET_ LAMBDA5_PARITY_SELECTION_V1`
therefore sharpens the tagged remainder to order λ^6 for the covariantly dressed packet family. It does not determine that even-order coefficient: the tree cross, active loop, second-order source/detector and survival terms remain to be assembled. On the selected covariantly pulled experiment the order-λ^6 ledger is in fact smaller. With fixed odd three-particle BT projectors, total Fock parity gives $P_{\text{out}}T_3P_{\text{in}} = 0$. Exhaustive tagged support then gives

$$T_4 = C_{4,\text{tree}} + I_s \otimes L_{4,\text{active loop}} + S_{2,s} \otimes A_{2,\text{active tree}},$$

and hence

$$q^{(6)}_{\text{tag}} = 2\Re\langle T_2, C_{4,\text{tree}}\rangle_K + 2\Re\langle T_2, I_s \otimes L_{4,\text{active loop}}\rangle_K + 2\Re\langle T_2, S_{2,s} \otimes A_{2,\text{active tree}}\rangle_K.$$

Certificate
`REVERSE_PHYSICS_BT_TAGGED_PACKET_ LAMBDA6_OBJECT_LEDGER_V1`
classifies these as the only three summands. The last term is the renormalized order-two spectator self-energy, including its counterterms, times the active tree. Nonforward support removes pure survival but not this dressed-spectator transition. Pulling source and effect together by the same coefficientwise two-sided R_t leaves the finite trace invariant, so separate source and detector dressing terms would double count a similarity transformation. The tree term is known; the renormalized active four-point one-loop and spectator two-point compact-packet kernels are missing. The hard logarithm constrains the former's scale derivative but supplies neither finite-time packet kernel. On the declared normal-ordered massless unit-residue auxiliary carrier the spectator term can be discharged. Certificate
`REVERSE_PHYSICS_BT_TAGGED_PACKET_NORMAL_ ORDERED_SPECTATOR_REDUCTION_V1`
uses the quartic auxiliary graph identities to show that the unique order-λ^2 two-point graph is a one-vertex, momentum-independent cross tadpole. Normal ordering removes it, while the massless and unit-residue conditions fix the complete two-point counterterm basis. Thus $S_{2,s} = 0$ as a finite-time packet operator in this scheme and only the active four-point one-loop cross remains missing. The scheme is declared, not claimed to be uniquely forced by the public Letter; an unmatched finite two-point convention must reinstate the spectator term. The remaining covariant active loop is also finite and explicit in the declared $\overline{\text{MS}}$ coupling scheme. Certificate
`REVERSE_PHYSICS_BT_AUXILIARY_ACTIVE_ ONE_LOOP_MSBAR_V1`

enumerates the six neutral auxiliary assignments. Their bubble weights are permutations of $(2,4,4)$, the positive tree norm is 24, and the tree–loop pairing is $40(B_s+B_t+B_u)$. Since the renormalized massless bubble is $B_X=L_X+2$, the complete real active density is

$$\frac{d\sigma^{(6)}_{\mathrm{active},\overline{\mathrm{MS}}}}{d\Omega}=\frac{5\lambda^6}{256\pi^4 s}(L_s+L_t+L_u+6).$$

Its logarithm reproduces the independent hard certificate. On compact hard coarea support it defines a bounded Hilbert–Schmidt covariant packet kernel. By itself this covariant expression supplies neither the finite-time transients nor affiliation with the same finite-duration BT Dyson evolution as the connected tree. That affiliation is now explicit. On the energy-diagonal sharp interval $[0,T]$, the ordered second-Dyson cross divided by the tree duration is

$$\int_0^T\left(1-\frac{\tau}{T}\right)\sin(\delta\tau)d\tau=\frac{1}{\delta}-\frac{\sin(\delta T)}{T\delta^2}.$$

It is the Hilbert transform of the unit-mass Fejer kernel $|F_T(\nu)|^2/(2\pi T)$. Certificate
`REVERSE_PHYSICS_BT_FINITE_TIME_ACTIVE_ LOOP_AFFILIATION_V1`
therefore evaluates the renormalized finite-time bubble as

$$B_T(P)=\log\frac{\mu^2}{|(P^0)^2-|\mathbf{P}|^2|}+2-C(T|P^0-|\mathbf{P}||)-C(T|P^0+|\mathbf{P}||),\quad C(z)=\frac{\sin z}{z}-\mathrm{Ci}(z).$$

In the total-three-particle centre frame of the tagged fixture, its active loop contribution is

$$\frac{125\lambda^6\Delta\Omega}{16384\pi^4\kappa^2\,\mathrm{Area}}\left[L_*+6-C(4\kappa T/5)-C(16\kappa T/5)-4C(4\sqrt{2}\kappa T/5)\right].$$

The unequal s-channel gaps record the boost of the active pair in this finite-time frame. The transient vanishes at large T, and the compact hard packet kernel is Hilbert–Schmidt. Combining it with the independently normalized connected-tree functional gives

$$q_{\mathrm{tag}}[f;T]=\frac{75\lambda^4\Delta\Omega}{2048\pi^2\kappa^2\,\mathrm{Area}}\left[1+\lambda^2R_6[f;T,\mu]\right]+O(\lambda^8),$$

$$R_6=\frac{2\sqrt{2}}{3}\Re C_{ff}(T)+\frac{5}{24\pi^2}\left[L_*+6-C(4\kappa T/5)-C(16\kappa T/5)-4C(4\sqrt{2}\kappa T/5)\right].$$

Certificate
`REVERSE_PHYSICS_BT_COMPLETE_TAGGED_Q6_ PHYSICAL_PROBABILITY_V1`
exhausts the order-λ^6 ledger in the declared normal-ordered MSbar scheme: $C_{ff}=\langle f,\mathcal{W}_{\kappa,T}f\rangle$ is the compact connected-tree functional, the bracket is the active loop, and the spectator self-energy term is zero. Coupling/Fock parity makes the next remainder $O(\lambda^8)$. The exact sign wall is packet-, duration-, scale-, and scheme-dependent. Since the leading coefficient is positive and R_6 is finite on compact hard support, the truncated probability is positive for sufficiently small coupling. This is a selected BT physical probability beyond leading order, not all-order positivity or general Eq. (19).

The selected tagged result also extends to every spectator-label channel on its permutation orbit. The group $S_3^{\mathrm{in}}\times S_3^{\mathrm{out}}$ acts transitively on the nine cylinders Δ_{ia}, and the stabilizer of Δ_{00} has order four. The exact ten-channel incidence split is recomputed for every tag:

six channels retain weight five and four retain weight six, reproducing the selected fixture. Introduce an explicit ten-valued classical record consisting of the nine cylinder labels and one fully rearranged bulk label. Its characteristic effects obey

$$\Pi_c\Pi_d = \delta_{cd}\Pi_c, \qquad \Pi_{\rm bulk} + \sum_{i,a} \Pi_{ia} = I_{10}.$$

For the invariant equal-block preparation $\rho = I_{10}/10$, each cylinder has weight $1/10$ and their union has weight $9/10$. Label covariance transports the complete selected q_4, q_6 coefficients to every cylinder, while the bulk begins at probability order λ^8. Hence

$$q_{\rm rec}[f;T] = \frac{9}{10} q_4 \left[1 + \lambda^2 R_6[f;T,\mu]\right] + O(\lambda^8).$$

Certificate
`REVERSE_PHYSICS_BT_NINE_CYLINDER_RECORDED_ Q6_INSTRUMENT_V1`
therefore closes all nine labelled outcomes through λ^6 on the selected permutation orbit. The factor $9/10$ is the declared classical preparation weight, not a new interaction coefficient. Because the stratum record is retained, amplitudes in different blocks are not coherently added. Arbitrary active energies and final-state shapes, a detector that erases the record, forward and collinear completion, all-time scattering, and Eq. (19) remain open.

The selected-angle restriction can be removed along an exact continuous fixed-energy family. With $c = \cos\theta \in (-1, 1)$ and $\kappa = 1$, keep $p_0 = k_0 = (6/5, 6/5, 0, 0)$ and take

$$p_{1,2} = (1, -3/5, \pm 4/5, 0), \qquad k_{1,2} = (1, -3/5, \pm 4c/5, \pm 4\sqrt{1-c^2}/5).$$

Then

$$s = \frac{64}{25}, \qquad t = -\frac{32(1-c)}{25}, \qquad u = -\frac{32(1+c)}{25}.$$

All ten channel functions are exact, and the same four weight-six channels remain resonant for every interior angle. Put

$$a_t = \frac{2}{5}(\sqrt{17-8c} - 3), \qquad a_u = \frac{2}{5}(\sqrt{17+8c} - 3).$$

The complete tree bracket is

$$\begin{aligned} W(c,T) = {} & 12T + \frac{125}{256}\sin\frac{16T}{5} + \frac{125}{128}\sin\frac{8T}{5} \\ & + \frac{10\sin(a_t T)}{-t} + \frac{10\sin(a_u T)}{-u}. \end{aligned}$$

The factorizations $-t = a_t D_t$, $-u = a_u D_u$, with $D_t, D_u > 12/5$, and $\sin x \geq -x$ for $x \geq 0$ give the uniform strict bound

$$\boxed{W(c,T) \geq \frac{13}{24} T > 0} \qquad (-1 < c < 1,\ T > 0).$$

Thus the resonant tree interference is not a ninety-degree artifact. The finite-time active loop on the same family is

$$\begin{aligned} B_*(c,T,\mu) = {} & \log\frac{15625(\mu/\kappa)^6}{65536(1-c^2)} + 6 - C(4\kappa T/5) - C(16\kappa T/5) \\ & - 2C\left(\frac{4\kappa T\sqrt{2(1-c)}}{5}\right) - 2C\left(\frac{4\kappa T\sqrt{2(1+c)}}{5}\right), \end{aligned}$$

so the fibrewise probability is

$$q_{\mathrm{tag}}(c;f,T)=q_4\{1+\lambda^2R_6(c;f,T,\mu)\}+O(\lambda^8), qquad R_6=\frac{2\sqrt{2}}{3}\Re C_{ff}(c,T)+\frac{5B_*}{24\pi^2}.$$

On every $|c|\le c_\star<1$ the loop gaps have a positive uniform lower bound. On a compact spectator packet tube with $D_A\ge d_0>0$, the tree kernel retains the exact bound $54T/d_0$; the special value $27T$ applies on the normalized angle fibre where $d_0=2$. These estimates give a finite uniform M_R and hence positivity of the truncated bracket whenever $\lambda^2M_R<1$. Certificate
`REVERSE_PHYSICS_BT_CONTINUOUS_ANGLE_Q6_ FAMILY_V1`
proves this fixed-energy continuous theorem and transports it to all nine spectator labels. This fibrewise theorem itself retains the angle as an orthogonal classical record. The endpoints $c=\pm1$, arbitrary energy/shape kinematics, real–virtual completion, all orders and Eq. (19) remain open.

For two distinct angle modes the record can nevertheless be erased by an explicit positive effect. On their record space let

$$P_+=\frac{1}{2}\begin{pmatrix}1&1\\1&1\end{pmatrix},\qquad P_-=I_2-P_+,\qquad E_\epsilon=P_++(1-\epsilon)P_-,\quad 0<\epsilon\le1.$$

The effect has off-diagonal entries $\epsilon/2$, spectrum $\{1,1-\epsilon\}$ and positive complement $I-E_\epsilon=\epsilon P_-$. At $\epsilon=1$ the click is the pure symmetric angle mode and contains no label identifying either angle. The fixed-s four-point density is angle independent, so after aligning the common tree phase its leading vector is $X_2=x_2(1,1)^T$ and

$$E_\epsilon X_2=X_2.$$

For the arbitrary complete order-λ^4 output X_4, this gives

$$\langle X_2,E_\epsilon X_4\rangle=\langle X_2,X_4\rangle.$$

Thus the complete coherent probability through the known order is

$$q_\epsilon(c_1,c_2;f,T)=2q_4\left\{1+\lambda^2\frac{R_6(c_1;f,T,\mu)+R_6(c_2;f,T,\mu)}{2}\right\}+O(\lambda^8),$$

independently of ϵ. Exact rational modes $c_1=0$ and $c_2=3/5$ have null conserved momenta on one common integer lattice after scaling by 25. Detector dependence first becomes possible in the known order-λ^4 amplitude norm:

$$\langle Y_4,E_\epsilon Y_4\rangle-\|Y_4\|^2=-\frac{\epsilon}{2}\|Y_4(c_1)-Y_4(c_2)\|^2.$$

Certificate
`REVERSE_PHYSICS_BT_TWO_ANGLE_COHERENT_ Q6_DETECTOR_V1`
therefore closes the two-mode operational coherence gate through λ^6 and localizes the first distinction at λ^8.

The displayed variance is in fact the complete *relative* q_8 coefficient. Exact total-Fock parity removes the odd selected blocks, so the exhaustive order-eight ledger is

$$q_8[I_2]=2\,\mathrm{Re}\langle X_2,X_6\rangle+\|X_4\|^2,\qquad q_8[E_\epsilon]=2\,\mathrm{Re}\langle X_2,E_\epsilon X_6\rangle+\langle X_4,E_\epsilon X_4\rangle.$$

Although X_6 is unknown, its cross cannot change the comparison:

$$\langle X_2, E_\epsilon X_6\rangle = \langle E_\epsilon X_2, X_6\rangle = \langle X_2, X_6\rangle.$$

Thus

$$\boxed{q_8[E_\epsilon] - q_8[I_2] = -\frac{\epsilon}{2}\|X_4(c_1) - X_4(c_2)\|^2 \le 0.}$$

Certificate
REVERSE_PHYSICS_BT_TWO_ANGLE_Q8_COHERENCE_ DIFFERENCE_V1
computes this full difference, not either absolute q_8 coefficient. The complete X_4 Gram data and X_2–X_6 interference, continuum-angle coherence, and selection by the public closed-system BT dynamics remain open. No endpoint, all-order or Eq. (19) completion follows.

There is nevertheless an absolute coefficient at the complementary dark port which does not require X_6. Set the effect equal to P_- itself. Since $P_-X_2 = 0$, its probability begins two orders later:

$$q_-(\lambda) = \lambda^8 Q_{8,-} + O(\lambda^{10}), \qquad Q_{8,-} = \langle X_4, P_-X_4\rangle = \frac{1}{2}\|X_4(c_2) - X_4(c_1)\|^2.$$

In particular $\langle X_2, P_-X_6\rangle = \langle P_-X_2, X_6\rangle = 0$. Let $\bar{q}_4 = q_4/\lambda^4 = \|x_2\|^2$ be the one-cell leading coefficient. The complete fibrewise relation

$$2\,\mathrm{Re}\langle x_2, X_4(c_i)\rangle = \bar{q}_4 R_6(c_i)$$

and Cauchy–Schwarz give

$$\frac{Q_{8,-}}{\bar{q}_4} \ge \frac{\{R_6(c_2) - R_6(c_1)\}^2}{8}.$$

For the common-lattice fixture $c_1 = 0, c_2 = 3/5$ at $\kappa T = 1$, exact alternating-series enclosures give

$$\Delta W > \frac{1}{16}, \qquad \Delta B > \frac{1}{220}.$$

With $N_s = 12\kappa V/5 > 0$, the complete contrast is

$$\Delta R_6 = \frac{2\sqrt{2}\,\Delta W}{3N_s} + \frac{5\Delta B}{24\pi^2} > \frac{49}{511104},$$

where the last strict bound drops the positive tree term and uses $\pi < 22/7$. Therefore

$$\boxed{\frac{Q_{8,-}}{\bar{q}_4} > \frac{2401}{2089818390528} > \frac{1}{10^9}.}$$

Certificate
REVERSE_PHYSICS_BT_DARK_PORT_ABSOLUTE_Q8_ LOWER_BOUND_V1
computes this strictly positive absolute dark-port coefficient on the declared finite-volume reduced-mode apparatus. The loop contrast is independent of the common scale and finite local scheme constant; its positive contribution makes the displayed lower bound uniform in finite positive N_s. This does not compute the recorded or symmetric bright-port absolute coefficients, thicken the modes into compact continuum packets, or control the $O(\lambda^{10})$ remainder at finite coupling.

The effect itself has a finite Hamiltonian realization. For the phased antisymmetric angle projector

$$P_-(\phi) = |-_\phi\rangle\langle -_\phi|, \qquad |-_\phi\rangle = \frac{|c_1\rangle - e^{i\phi}|c_2\rangle}{\sqrt{2}},$$

couple a degenerate pointer qubit by

$$H_{\text{int}} = gP_-(\phi) \otimes \sigma_y.$$

Since P_- is a projector, its duration-τ evolution is exactly

$$U_\tau = P_+(\phi) \otimes I_A + P_-(\phi) \otimes [\cos(g\tau)I_A - i\sin(g\tau)\sigma_y].$$

Preparing the pointer in $|0\rangle_A$ and reading it in the computational basis gives

$$K_{\text{click}} = P_+(\phi) + \cos(g\tau)P_-(\phi), \qquad K_{\text{no}} = \sin(g\tau)P_-(\phi),$$

so the measured effects are precisely

$$E_{\text{click}} = P_+(\phi) + [1-\epsilon]P_-(\phi), \qquad E_{\text{no}} = \epsilon P_-(\phi), \qquad \epsilon = \sin^2(g\tau).$$

The phase is selected by the physical shifter $S_\phi = \text{diag}(1, e^{i\phi})$, for which $P_-(\phi) = S_\phi P_-(0) S_\phi^\dagger$. If the leading BT phase is δ, calibration means $\phi = \delta$. A mismatch $\Delta = \delta - \phi$ gives the exact leading response

$$\langle X_2, E_{\text{click}} X_2\rangle = 2|x_2|^2[1 - \epsilon\sin^2(\Delta/2)].$$

Certificate
`REVERSE_PHYSICS_BT_TWO_ANGLE_FINITE_APPARATUS_ HAMILTONIAN_V1`
therefore replaces the abstract effect by a bounded finite detector interaction and selects both ϵ and the relative phase by apparatus parameters. The device is additional physics coupled to the certified BT output modes. Its coupling, duration and phase are not predicted by the public closed-system BT Hamiltonian, and no spacetime-local or continuum-angle detector has been constructed.

The finite effect does have an exact local quadratic vertex on the two rational modes. Their common total four-momentum is $P = (2, -6/5, 0, 0)\kappa$. On the vacuum-to-pair matrix elements the local densities

$$D_+ =:\phi^2:, \qquad D_- =:\phi^2: - \frac{625}{72\kappa^2} :(\partial_y\phi)^2:$$

have respective weights $(1, 1)$ and $(1, -1)$. Therefore

$$\alpha = \frac{1 - e^{-i\phi}}{2\sqrt{2}}, \qquad \beta = \frac{1 + e^{-i\phi}}{2\sqrt{2}}$$

synthesize the desired transition weights $(1, -e^{-i\phi})/\sqrt{2}$. Because $\langle e|\sigma_x|g\rangle = 1$ and $\langle e|\sigma_y|g\rangle = i$, two real Pauli quadratures give a Hermitian local interaction with this complex transition density. In the selected basis $|g, +_\phi\rangle, |g, -_\phi\rangle, |e, 0_{\text{field}}\rangle$, its normalized compression is

$$\frac{H_{\text{comp}}}{G} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

whose absorption/pass effects are exactly $\sin^2(G\tau)P_-(\phi)$ and $P_+(\phi)+\cos^2(G\tau)P_-(\phi)$.

This exact compression cannot be promoted to an invariant local detector sector. On the continuous fixed-energy family, compact smearing depends only on the common total momentum. Any quadratic density with finitely many derivatives restricts along an angular orbit $z = e^{i\theta}$ to a finite Laurent polynomial $p(z)$. Exact support at only two angles would make p vanish on their open complement; multiplying by a power of z gives an ordinary polynomial with infinitely many roots, hence the zero polynomial, contradicting its nonzero selected weights. Certificate
`REVERSE_PHYSICS_BT_TWO_ANGLE_LOCAL_DETECTOR_ COMPRESSION_V1`
therefore proves the selected local matrix elements and finite-derivative continuum selectivity no-go, not $\Pi e^{-iH\tau}\Pi = e^{-i\Pi H\Pi\tau}$. A finite angular packet calculation must therefore retain the complement before exponentiation. The interaction remains additional anisotropic scalar apparatus physics; it is neither selected by the public BT Hamiltonian nor a Lorentzian or gravitational detector.

At the selected fixed total momentum the packet calculation has an invariant full-sphere completion. In the centre-of-momentum frame,

$$\frac{P}{\kappa} = (2, -6/5, 0, 0), \qquad \frac{P^2}{\kappa^2} = \frac{64}{25},$$

and every outgoing massless pair is

$$k_1^* = \frac{4\kappa}{5}(1, n), \qquad k_2^* = \frac{4\kappa}{5}(1, -n), \qquad n \in S^2.$$

The lab boost has $\beta_x = -3/5$, $\gamma = 5/4$. Writing $n = (x, \sqrt{1-x^2}\cos\varphi, \sqrt{1-x^2}\sin\varphi)$ gives $k^0_{1,2}/\kappa = 1 \mp 3x/5$, so the earlier equal-lab-energy family is exactly the equator $x = 0$. The invariant two-body measure is a common P^2-dependent factor times $d\Omega = dx\, d\varphi$; its equatorial angular factor is $d\varphi$, not dc for $c = \cos\varphi$. Exchange $n \sim -n$ halves every norm and cancels from their ratios.

With $\zeta = e^{2i\varphi}$ and normalized projective Fejér kernels of order $N = 20$ centered at the two target pair lines, set

$$p(\varphi) = \frac{K_{20,0}(\varphi) - K_{20,1}(\varphi)}{1-\rho}, \qquad \rho = K_{20,0}(\varphi_1) < 10^{-3}.$$

Then $p(\pi/2) = 1$ and $p(\arccos(3/5)) = -1$ exactly, with Laurent support $-19, \ldots, 19$. Its invariant equatorial leakage into the complement of

$$B_1^\varphi = [\pi/4, 3\pi/8], \qquad B_0^\varphi = [3\pi/8, 5\pi/8]$$

is an exact element of $\mathbb{Q}(\pi, \sqrt{2})$ satisfying

$$\eta_\varphi = 0.00066128445650298648\ldots < \frac{331}{500000}.$$

The entire Laurent polynomial has one local homogeneous lift. For $m \geq 0$,

$$(1-x^2)^{19} e^{2im\varphi} = (n_y^2 + n_z^2)^{19-m}(n_y + in_z)^{2m},$$

and the negative modes use $n_y - in_z$. Every term has degree 38, so

$$\boxed{F(n) = (1 - x^2)^{19} p(\varphi)}$$

is the symbol of one finite constant-coefficient local quadratic density of maximum derivative order 38. It is antipodally even and descends to the unordered pair sphere. On the two azimuthal bins above and the common latitude band $|x| \le 1/2$, the squared latitude factor is $(1 - x^2)^{38}$. Its exact rational integral factorizes from the azimuthal integral, giving

$$1 - \eta_{S^2} = (1 - \eta_\varphi)(1 - \eta_x), \qquad \eta_x = 0.000002332953298014222291\ldots,$$

and the exact algebraic interval certificate proves

$$\boxed{0 < \eta_{S^2} = 0.000663615867055246985\ldots < \frac{83}{125000} < 10^{-3}.}$$

Strict rational square bounds for $\sqrt{2}$ and alternating Machin-series bounds for π decide the inequalities; the decimals are displays only.

Let B be the normalized selected bright packet, D its selected dark partner and R the normalized complement. The complete coupled direction is

$$v = \sqrt{1 - \eta_{S^2}}\, B + \sqrt{\eta_{S^2}}\, R.$$

Retaining R gives the exact resonant rotating-wave star Hamiltonian $H/G = |e\rangle\langle v| + |v\rangle\langle e|$ and hence

$$U_\tau = I + [\cos(G\tau) - 1](H/G)^2 - i \sin(G\tau)(H/G).$$

Its selected packet effect is

$$E_{\text{absorb}} = (1 - \eta_{S^2}) \sin^2(G\tau) P_B, \qquad E_{\text{pass}} = P_D + [1 - (1 - \eta_{S^2}) \sin^2(G\tau)] P_B.$$

At $G\tau = \pi/2$ the absorption efficiency is above 124917/125000 and the total field remainder is below 83/125000. At arbitrary time the outside-packet population, maximized over the selected detector/bright input plane, is $4\eta_{S^2}(1 - \eta_{S^2}) < 1/375$. Certificate `REVERSE_PHYSICS_BT_FIXED_P_TWO_SPHERE_PACKET_ DETECTOR_V1` therefore constructs a leakage-controlled local packet detector on the complete invariant angular shell at fixed P, without asserting exact angle support or deleting the residual. Finite invariant-mass and total-momentum thickness, number-scattering and counter-rotating terms, public-BT selection, transfer of the earlier equal-weight relative-q_8 formula to these unequal-norm packet modes, Eq. (19), gravity and Lorentzian causality remain open.

The fixed-P distribution can be thickened without replacing its local density. Work in the target centre-of-momentum apparatus frame,

$$P_0 = (M_0, 0, 0, 0), \qquad M_0 = \frac{8\kappa}{5},$$

and smear the same degree-38 pointwise quadratic density with a complex Gaussian modulation satisfying

$$|\widehat{h}(P)|^2 = \exp[-\|P - P_0\|_E^2/\sigma^2], \qquad \frac{\sigma}{M_0} = \frac{1}{50000}.$$

Two real detector quadratures realize this complex transition switching and its adjoint. The density remains spacetime local, but the Gaussian switching is Schwartz rather than compactly supported.

On the core

$$\|P - P_0\|_E \leq \frac{M_0}{10000} = 5\sigma,$$

rotationless centre-of-momentum coordinates map the angular direction by a spatial boost with singular values $1, 1, \gamma$. The exact bound

$$\gamma - 1 \leq \frac{\epsilon^2}{2(1-2\epsilon)}, \qquad \epsilon = \frac{1}{10000},$$

combines with the Fejér coefficient estimates

$$|F| \leq A_0 R^{38}, \qquad |\nabla F| \leq (38A_0 + A_1)R^{37},$$

where

$$A_0 = \frac{2}{1-\rho}, \qquad A_1 = \frac{2}{1-\rho}\frac{133}{10}.$$

Throughout the core, the resulting worst-case angular complement obeys

$$\eta_{\text{core}} < 0.000882012816932776710\ldots.$$

Outside the five-sigma ball, the squared degree-38 response is bounded by $A_0^2(1+sR)^{76}$ with $s = \sigma/M_0$ and $R = \|P - P_0\|_E/\sigma$. Since

$$(1+sR)^{76} \leq \exp[(76\epsilon/25)R^2], \qquad R \geq 5,$$

the complete four-dimensional radial tail is bounded by an elementary Gaussian integral. Finite rational lower sums for the required exponentials give

$$\eta_{P,\text{tail}} < 0.000000586640212958096882\ldots$$

and hence

$$\boxed{\eta_{\text{complete}} < 0.000882599457145734807\ldots < \frac{883}{10^6} < 10^{-3}.}$$

Thus the switched pair-annihilation response is nonzero and square integrable on a genuine open four-momentum set. Normalizing it as $v = w/\|w\|$ gives the leading positive effects

$$E_{\text{click}} = \zeta|v\rangle\langle v|, \qquad E_{\text{no}} = I - E_{\text{click}}, \qquad 0 \leq \zeta = g^2\|w\|^2 \leq 1.$$

The same switching exposes a useful kinematic separation. A number term transfers $q = k_{\text{in}} - k_{\text{out}}$ and therefore obeys $q^2 = -2k_{\text{in}} \cdot k_{\text{out}} \leq 0$. Completing the Euclidean square in the apparatus frame gives

$$\operatorname{dist}_E(P_0, \{q : q^2 \leq 0\})^2 = \frac{M_0^2}{2}.$$

The wrong-sign future-pair argument lies in the past causal cone, whose distance squared from P_0 is M_0^2. Their squared Fourier envelopes are therefore bounded pointwise by

$$e^{-1{,}250{,}000{,}000}, \qquad e^{-2{,}500{,}000{,}000},$$

respectively. Certificate
REVERSE_PHYSICS_BT_SCHWARTZ_FOUR_MOMENTUM_ PACKET_RESPONSE_V1
proves this normalizable leading response and spectral gap. It does not promote the pointwise number bound to an operator estimate: unrestricted mean momentum and the derivative weights still require a Schur or Hilbert–Schmidt argument. Nor does a spacetime switching define the exact time-independent star Hamiltonian used at fixed energy. Complete time ordering, recorded or bright-port absolute q_8, Eq. (19), gravity and Lorentzian causality remain open.

The operator-bounded successor first makes the off-shell realization explicit. Use the finite bidifferential density

$$\mathcal{D}_F(x) =: F\left(\frac{i\partial_1 - i\partial_2}{M_0}\right)\phi(x_1)\phi(x_2) : \Big|_{x_1=x_2=x}.$$

It is jet-local through order 38. Its pair symbol is $F((k_1-k_2)/M_0)$ with switching transfer k_1+k_2, whereas its number symbol is $F((k+k')/M_0)$ with transfer $k-k'$. On the declared one-particle apparatus band

$$K = \{k : M_0/4 \le |k| \le 3M_0/4\}, \qquad \mu(K) = \frac{\pi M_0^2}{2},$$

the number compression has kernel

$$N_K(k',k) = \widehat{h}(k-k')F((k+k')/M_0)$$

and is Hilbert–Schmidt. The cone gap and $|F((k+k')/M_0)| \le A_0(3/2)^{38}$ give

$$\frac{\|N_K\|^2}{\|w\|^2} < 2^{153-1,250,000,000} < 10^{-1,000,000}.$$

The complete wrong-sign pair vector obeys independently

$$\frac{\|w_-\|^2}{\|w\|^2} < 2^{47-2,499,999,924} < 10^{-1,000,000}.$$

Including both adjoints therefore gives, at first Dyson order on the declared domain,

$$\frac{\|E_{\text{undesired}}\|}{\|A_{\text{pair}}\|} < 10^{-400,000}, \qquad \frac{\|(A+E)^\dagger(A+E) - A^\dagger A\|}{\|A\|^2} < 10^{-399,999}.$$

These exact exponential comparisons use rational prefactor ledgers and $e > 2$, not floating-point exponentials. Certificate
REVERSE_PHYSICS_BT_COMPACT_ENERGY_QUADRATIC_ SECTOR_BOUND_V1
thus closes every quadratic frequency sector at first Dyson order on K. It does not control scattering from K to its complement, unrestricted energies, higher time ordering, recorded or bright-port absolute q_8, Eq. (19), gravity or Lorentzian causality.

The auxiliary preparation nevertheless has an exact scalar preimage on the finite covariant detector ideal. Put

$$P_u = |u_0\rangle\langle u_0|_K, \qquad \psi_{\phi,+} = R_t^\dagger u_0, \qquad P_{\phi,+} = R_t^\dagger P_u R_t.$$

The coefficientwise formal identity $R_t^\dagger R_t = R_t R_t^\dagger = 1$ gives

$$\langle \psi_{\phi,+}, \psi_{\phi,+} \rangle_K = 1, qquad P_{\phi,+}^2 = P_{\phi,+} = P_{\phi,+}^\dagger, qquad R_t P_{\phi,+} R_t^\dagger = P_u.$$

Pulling back the click and no-click effects by the same R_t preserves their finite trace, relative completeness and probabilities. Hence the displayed rate is also the leading rate for this explicitly dressed perfect-square scalar source.

The source cannot be confused with the characteristic projector in Eq. (18). On a three-particle string x, the charge is $q(x) = 2\,\mathrm{popcount}(x) - 3$, so zero charge is absent. The cross-Krein metric pairs V_q only with V_{-q} and vanishes on every $V_q \times V_q$. Any nonzero charge-invariant range therefore contains a null vector and cannot be positive. For u_0 the state charges are $\{-3, +3\}$ and the projector charges are $\{-6, 0, +6\}$; Eq. (16) transfers these to scalar Laurent support $\{Z^{-6}, 1, Z^6\}$. Both orbit branches are forced. Certificate
`REVERSE_PHYSICS_BT_SCALAR_ DRESSED_POSITIVE_SOURCE_ AFFILIATION_V1`
therefore establishes a leading physical-scalar probability jet for a coupling-dressed, shift-breaking source. It does not construct the standard $P_\chi^{(\phi)}$, repair its ghost-parity obstruction, prove general Eq. (19), or supply a convergent global R_t or all-time probability.

At the leading order needed here, the scalar state is no longer merely an inverse symbol. On $|0_{\phi;t}\rangle = R_t^\dagger |0_{\mathrm{BT}}\rangle$ the pulled creators are

$$d_\Upsilon^\dagger = \frac{Z^{-1} a_1^\dagger}{\sqrt{2E}}, \qquad d_\Omega^\dagger |0_{\phi;t}\rangle = \frac{Z(a_2^\dagger - 2iEt\, a_1^\dagger)}{4E^2\sqrt{2E}} |0_{\phi;t}\rangle.$$

For the three distinct detector modes,

$$\psi_{\phi,+}^{(0)} = \frac{1}{\sqrt{2}} \left[Z^{-3} \prod_j \frac{a_{1j}^\dagger}{\sqrt{2E_j}} + Z^3 \prod_j \frac{a_{2j}^\dagger - 2iE_j t\, a_{1j}^\dagger}{4E_j^2 \sqrt{2E_j}} \right] |0_{\phi;t}\rangle.$$

The two pure branches are null. Their cross pairing is one because each leg contributes $(2E)^3/[\sqrt{2E}\, 4E^2 \sqrt{2E}] = 1$; hence the displayed state has norm one. If $\psi_\phi = \psi_0 + \lambda\psi_1 + \cdots$ and $A = \lambda^4 A_4 + \lambda^5 A_5 + \cdots$, its leading probability is $\lambda^8 \langle A_4\psi_0, A_4\psi_0 \rangle_K$. Certificate
`REVERSE_PHYSICS_BT_SCALAR_ DRESSED_SOURCE_FREE_ NORMAL_FORM_V1`
therefore proves that unknown nonlinear source corrections cannot affect the certified leading rate. This is a finite-mode Gaussian-core state, not a vector in the infrared-obstructed ordinary massless Fock thermodynamic completion.

The classified endpoint freedom does not supply this complement by one fixed prescription on the minimal two-profile probe lift. Take $v = (1, z(1-z))$ and parameterize a reflection-even extension relative to the symmetric triple-plus reference by (c_0, c_1, c_2). The reference acts on $1, z(1-z), z^2(1-z)^2$ by $(0, 0, 3/2)$, while the three endpoint jets have action matrix

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & -4 \\ 0 & 0 & 4 \end{pmatrix}.$$

Consequently its induced Gram is

$$K(c) = \begin{pmatrix} 2c_0 & -2c_1 - 4c_2 \\ -2c_1 - 4c_2 & 3/2 + 4c_2 \end{pmatrix}.$$

The equation $K(c) = \left(\begin{smallmatrix} 0 & -\rho \\ -\rho & -2 \end{smallmatrix}\right)$ has the unique solution

$$c_0 = 0, \qquad c_1 = \frac{7}{4} + \frac{\rho}{2}, \qquad c_2 = -\frac{7}{8}.$$

Three exact physical fixtures have distinct ρ, so one ρ-independent coefficient triple cannot match even those three fibres. Changing the fixed affine reference changes the intercept but leaves $dc_1/d\rho = 1/2$. Certificate
`REVERSE_PHYSICS_BT_ENDPOINT_COMPLEMENT_ MATCHING_V1`
records the exact endpoint actions, affine matching law, physical fixtures, and subtraction witness. A spectator-dependent coefficient or a different dynamical profile remains possible; neither is derived from the public order-λ map. The statement is therefore a fixed-profile endpoint obstruction, not an all-local-counterterm no-go or a proof of Eq. (19). Resolution locality is not spacetime locality.

Remark 14.5 (Physical ultraviolet hard-scattering law). The Born rate, perfect-square beta function, and projected virtual hard log are

$$\frac{d\sigma_{\text{Born}}}{d\Omega} = \frac{3\lambda^4}{32\pi^2 s}, \qquad \beta_\lambda = -\frac{5\lambda^3}{16\pi^2}, \qquad \frac{d\sigma_{\text{virt,log}}}{d\Omega} = \frac{5\lambda^6}{256\pi^4 s}(L_s + L_t + L_u).$$

At fixed nonforward angle, all three channel invariants scale with s. The exact scale-derivative identity is therefore

$$-4\left(\frac{5}{16}\right)\left(\frac{3}{32}\right) + 2 \cdot 3\left(\frac{5}{256}\right) = 0.$$

Thus the explicitly computed hard logarithm supplies precisely the coefficient required by the independently computed running of the Born rate. With

$$D(s) = \lambda_0^{-2} + \frac{5}{16\pi^2}\log\frac{s}{s_0}, \qquad \lambda_0 = \lambda(\sqrt{s_0}),$$

the leading ultraviolet logarithms resum to

$$\frac{d\sigma^{\text{LL}}_{\text{hard}}}{d\Omega} = \frac{3}{32\pi^2 s D(s)^2} = \frac{24\pi^2}{25 s \log^2(s/\Lambda^2)}, \qquad \Lambda^2 = s_0 e^{-16\pi^2/(5\lambda_0^2)}.$$

For a fixed detector acceptance $\theta_0 \le \theta \le \pi - \theta_0$, $0 < \theta_0 < \pi/2$, this gives

$$\lim_{s\to\infty} s\log^2(s/\Lambda^2)\,\sigma^{\text{LL}}_{\text{window}} = \frac{96\pi^3}{25}\cos\theta_0.$$

The result is positive on the asymptotically-free UV branch, and its leading $1/\log^2 s$ coefficient is invariant under analytic coupling-scheme changes. It is the physical short-distance baseline for a nonforward acceptance, not an infrared-complete exclusive particle-number probability. Real–virtual endpoint cancellation, the Jordan/R_t dressing, incoming degenerate sectors, and next-to-leading-log finite terms remain unconstructed.

Where the double pole goes. One structural remark, since it bears on which route is the likelier to extend. The obstruction that withdraws the cutting rules at coincident poles is that the cut weight there is a derivative of a delta rather than a measure of definite sign. That is a property of the fourth-order *variable* and not of the theory: in the two-field frame of [4] the same dynamics is carried by two second-order fields whose propagator is off-diagonal, so every pole is simple and every cut an ordinary delta. The double pole is removed by the change of variables rather than by any additional hypothesis. That is a plausible reason why the Krein route reaches tree level at the locus where the positive route stops, and it suggests the positive-metric programme could be continued past its own Sec. VI in those variables rather than in the fourth-order field.

Open questions. Whether the two pointed sectors admit parity vacua $(|0_A\rangle \pm |0_B\rangle)/\sqrt{2}$ or are superselected requires finite-volume overlap and zero-mode analysis — in the hyperbolic polar coordinates $r = UV$, $\chi = \frac{1}{2}\log(U/V)$ (a coordinate presentation of the $O(1,1)$ model: boost $=$ χ-shift, exchange $= \chi \mapsto -\chi$, conserved current $j = V\partial U - U\partial V = 2r\,\partial\chi$) this becomes a boundary/self-adjoint-extension problem at the null locus $r = 0$, where the conserved charge produces a singular q^2/r barrier. The quantum implementation of the exchange on a completion, and any probability interpretation beyond [4], remain open; so does a normal-ordered operator Ward identity for the internal charge (the classical identity, with its exact regulator breaking $\epsilon u(1+u) - \mu^2 v$, is verified). The hierarchy conjecture's 5:1 prediction is confirmed at fourth order (Proposition 7.4), and its 5:3 and 7:1 instances are confirmed at sixth order (Proposition 7.5); higher coprime loci and the mechanism excluding even mode-2 transfers remain open. The ordinary scalar independent-mass route is obstructed: Remark 14.3 supplies its collinear logarithmic obstruction. Remark 14.4 shows that coherent projector transport passes the finite coefficient and charge preflights. A complete construction must derive the splitting generator from the broken-vacuum BT/PS asymptotic Hamiltonian, construct the continuum and incoming degenerate sectors, and only then evaluate the quotient trace. The gravitational extension is established at tree level in the sequel: the exact Einstein–Weyl cubic and quartic vertices, physical helicities, and Ward/BRST checks produce a second-order metric obstruction. That result does not import the scalar infrared proposition; any Bateman–Turok-type gravitational completion would need its own parent/daughter and asymptotic-state construction. No CPT claim is made: the exchange is a linear sector map, not a constructed antiunitary CPT operator.

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A The Weyl–Moyal recursion

Operators are represented by Weyl symbols on the two canonical pairs (x, p), (y, q); the star product is $f \star g = \sum_n \frac{1}{n!}(i/2)^n \Lambda^n(f, g)$ with $\Lambda = \overleftarrow{\partial}_x \overrightarrow{\partial}_p - \overleftarrow{\partial}_p \overrightarrow{\partial}_x + \overleftarrow{\partial}_y \overrightarrow{\partial}_q - \overleftarrow{\partial}_q \overrightarrow{\partial}_y$, a finite sum on polynomials. For quadratic h_0, $[h_0, \mathcal{W}[f]] = i\mathcal{W}[\{h_0, f\}]$ exactly; for cubic–cubic commutators the Λ^3 (constant) term contributes. Expanding $h_\zeta = e^{-X/2} \star (h_0 + \zeta v) \star e^{X/2}$, $X = \zeta R_1 + \zeta^2 R_2 + \zeta^3 R_3$, with $e^{-X/2} A e^{X/2} = A + \frac{1}{2}[A, X] + \frac{1}{8}[[A, X], X] + \frac{1}{48}[[[A, X], X], X] + \cdots$, the order-$\zeta^2$ and ζ^3 Hermiticity conditions are

$$[h_0, R_2] = (v_2^\dagger - v_2) + \tfrac{1}{2}[R_1, v + v^\dagger], \tag{23}$$

$$[h_0, R_3] = -(T_3 - T_3^\dagger), \tag{24}$$

$$T_3 = \tfrac{1}{8}\big([[h_0, R_1], R_2] + [[h_0, R_2], R_1]\big) + \tfrac{1}{48}[[[h_0, R_1], R_1], R_1] + \tfrac{1}{2}[v, R_2] + \tfrac{1}{8}[[v, R_1], R_1],$$

using $[h_0, R_1] = v^\dagger - v$ and the fact that $[v - v^\dagger, R_1]$ is Hermitian. Equations $[h_0, R] = S$ are solved monomial by monomial in the mode variables $a_1^m a_1^{\dagger n} a_2^r a_2^{\dagger s}$, where $[h_0, M] = \Omega_M M$ with $\Omega_M = (n - m)\omega_1 + (s - r)\omega_2$: $R_M = S_M/\Omega_M$ off the kernel; the kernel component is the obstruction.

B The 3:1 projection and gauge independence

At $\omega_1 = 3\omega_2$ the second-order source $S_2 = \frac{1}{2}[R_1, v + v^\dagger]$ is a quartic polynomial; its kernel monomials must satisfy $(n - m)3\omega_2 + (s - r)\omega_2 = 0$ with degree ≤ 4, i.e. $(m - n, r - s) = \pm(1, -3)$, realized only by $a_1 a_2^{\dagger 3}$ and $a_1^\dagger a_2^3$. Direct evaluation of the two coefficients gives $\mathfrak{o}_+^{(2)} = \frac{27\sqrt{3}}{320\omega_2^4}(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3)$. Gauge independence: a change $R_1 \to R_1 + G$ with $[G, h_0] = 0$ shifts S_2 by $\frac{1}{2}[G, v + v^\dagger]$; transfer bookkeeping shows that for every Hermitian kernel generator (N_1, N_2 polynomials and the resonant quartics $a_1 a_2^{\dagger 3} + a_1^\dagger a_2^3$, $i(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3)$) the shifted source has no kernel component: the transfers of $v + v^\dagger$, shifted by $\pm(1, -3)$ or $(0, 0)$, never land on $\pm(1, -3)$; machine verification confirms $\Pi_{\ker \mathrm{ad}_{h_0}}[G, v + v^\dagger] = 0$ term by term. Anti-Hermitian generators A_1 enter through $\frac{1}{2}[v - v^\dagger, A_1]$, and the same bookkeeping gives $\Pi_{\ker \mathrm{ad}_{h_0}}[v - v^\dagger, A_1] = 0$. Kernel additions to R_2 drop out of the second-order equation since $[h_0, R_2^{\ker}] = 0$.

C The effective shell matrix

Standard second-order degenerate perturbation theory on the shell $\mathcal{E}_{27} = \mathrm{span}\{|j, 27 - 3j\rangle\}$ gives $K_{ab}^{(2)} = \sum_{m \notin \mathcal{E}} v_{am} v_{mb}/(27 - E_m)$ (the first-order block vanishes: cubic v cannot realize the transfer $\pm(1, -3)$). Evaluating the sums yields the rational diagonal $D(n_1, n_2) = (633n_1^2 - 1836n_1 n_2 - 285n_1 + 3969n_2^2 + 3051n_2 + 818)/26880$ and the antisymmetric off-diagonal quoted in Theorem 6.2. For a tridiagonal matrix only the products $K_{j,j+1}K_{j+1,j} \in \mathbb{Q}$ enter the characteristic polynomial, computed exactly by the three-term recurrence $p_j = (D_j - \kappa)p_{j-1} - K_{j-1,j}K_{j,j-1}p_{j-2}$; the Sturm count of the resulting degree-10 rational polynomial is exactly 8.

D The transfer lattice

Transfer additivity: the grading by ad_{h_0} satisfies $[h_0, AB] = [h_0, A]B + A[h_0, B]$, so transfers add under operator products, hence under star products and commutators. Degree bound:

each commutator of polynomial factors lowers total degree by at least two (Λ^1), so order-n objects, built from n cubic factors and $n-1$ commutators (with $[h_0,\cdot]$ degree-preserving), have degree $\leq 3n-2(n-1)=n+2$; Moyal corrections lower degree by even amounts, fixing the parity $\equiv n \bmod 2$. A kernel monomial with transfer (d_1,d_2), $d_1 d_2 < 0$, requires degree $\geq |d_1|+|d_2|$ with matching per-mode parities, giving the candidate sets of Theorem 7.1.

E Field conventions, the source formula, and truncation completeness

Conventions. Finite volume; discrete momenta k (lattice units with $m_L = 4$ in Theorem 12.1); modes (b,k) with $[a_{b,k}, a^\dagger_{b',k'}] = \delta_{bb'}\delta_{kk'}$; unit-norm external states; $g=1$; deformation parameter ζ multiplying H_3, ζ^2 multiplying H_4. Positive-frame legs from the branch eigenvectors: u-legs $(a_+ + a_+^\dagger)/\sqrt{2\omega_+\delta}$ (healthy) and $(a_- - a_-^\dagger)/\sqrt{2\omega_-\delta}$ (ghost, quarter-turn); v-legs $\rho_\pm$ times u-legs, $\rho_\pm = \mp\delta/(2g)$. Interactions $H_3 = -g\int uv^2$, $H_4 = -\frac{g}{2}\int u^2v^2$ are pure potentials. The transported fourth-order field of Section 10 is fixed by matching the free two-point function $[e^{-i\omega_1 t}/2\omega_1 - e^{-i\omega_2 t}/2\omega_2]/\delta$ of [14]; the even-ghost rule follows because a monomial with g ghost legs, c_g of them creators, has v-coefficient $\propto i^{\,\#\mathrm{legs}}(-1)^{c_g}$ and $v^\dagger$-coefficient $\propto$ the conjugate with $c_g \to g-c_g$, whose difference vanishes for odd g.

Source formula. With $E_{\mathrm{in}} = E_{\mathrm{out}} = E$ and R_1 solved off shell, $\langle \mathrm{out}|[R_1,w]|\mathrm{in}\rangle = \sum_n [(v^\dagger - v)_{\mathrm{out},n} w_{n,\mathrm{in}} + w_{\mathrm{out},n}(v^\dagger - v)_{n,\mathrm{in}}]/(E-E_n)$ with $w = v + v^\dagger$; the algebraic identity $(v^\dagger - v)w + w(v^\dagger - v) = 2(v^\dagger v^\dagger - vv)$ reduces this to (15).

Truncation completeness (proof of Lemma 12.2). Second-order chains pass through 1-, 3-, or 5-particle intermediates. 1-particle: momentum $k_a + k_b$ fixed. 3-particle: the first vertex annihilates one external and creates a pair $(q,\, k_x - q)$; for the second vertex to reach the 2-particle out-state, the surviving created particle must be an out-particle, forcing $q \in \{k_c, k_d, k_x - k_c, k_x - k_d\}$; annihilating both created particles instead returns the in-momenta, giving zero overlap with a distinct out-state. 5-particle: the created triple (p,q,r), $p+q+r=0$, must contain both out-particles, fixing the third momentum. Hence all internal momenta lie in the s,t,u-generated set, and enlarging the momentum cutoff cannot change the element — as the exact agreement at $K_{\max} = 4, 6$ confirms.

F The doubled confluent parity

With change-of-basis matrix C whose columns are (c,d) in branch coordinates, $P_\delta^{(c,d)} = C^{-1}\,\mathrm{diag}(1,-1)C = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}$. For the doubled space take $C_A = C$ and the oppositely oriented C_B (columns $c_B = (b_+ - b_-)/2$, $d_B = (b_+ + b_-)/\delta$); the cross map $\kappa(b^A_\pm) = \pm b^B_\pm$ has matrix $T^{-1}\kappa_{\mathrm{branch}}T$ with $T = \mathrm{diag}(C_A, C_B)$, which evaluates to the δ-independent exchange $(c_A, d_A) \leftrightarrow (c_B, d_B)$, squaring to the identity.

G Verification checks and their classification

Check	Supports	Type
ID1–ID5	Theorem 3.1	exact symbolic
ID6–ID8	Theorem 4.1	exact symbolic
ID9a–b	Theorem 5.1	exact symbolic
ID10, O3e	Comp. Prop. 8.2	high-precision numeric
SR1–SR3	Theorem 7.1	exact combinatorial
O3a–O3d	Theorem 7.2	exact symbolic
PT1–PT2	Theorem 6.2 (structure)	exact symbolic
Sturm count	Theorem 6.2 (pair)	exact rational
PT3	Theorem 6.2 (truncated op.)	numeric
PT4	broken-PT tongues	numeric (effective matrix)
PS1–PS2	Proposition 9.1	exact symbolic
PS-A–PS-G	Theorem 10.1, Comp. Prop.	exact symbolic + numeric
PS-H	Jordan-chain lemma	exact symbolic
TF1–TF7	Theorem 11.1, Prop. 11.2	exact symbolic
SO1–SO7	Theorem 12.1 (numerics)	exact + numeric
HX1	Theorem 12.1 (exact value)	exact symbolic
HX2–HX3	Theorem 13.1	exact symbolic
PT proposition	Proposition 6.1	exact symbolic
$\omega_2^{-13/2}$ ratio	Theorem 7.2	exact symbolic
R_1 Laurent	Theorem 8.1	exact symbolic
DQ1–DQ2, DQ9	$O(1,1)$ polar form, current, Ward	exact symbolic
DQ3–DQ6	doubled pairing, graph, pseudo-unitarity	numeric
DQ7–DQ8	Comp. Prop. 12.4	exact symbolic
FO1–FO9	Comp. Prop. 7.4, Conj. 7.3	exact symbolic
ON1–ON4	Lemma 14.1, Comp. Prop. 14.2	exact symbolic + numeric
BT-RV	Remark 14.3	exact symbolic/rational
BT-AG	Remark 14.4	exact rational/algebraic
BT-DR	Remark 14.4	exact rational/semifinite trace
BT-AN	Remark 14.4	exact rational/Abel–Naimark
BT-PC	Remark 14.4	exact symbolic/rank–Jordan
BT-RJ	Remark 14.4	exact threshold/rigged–Jordan
BT-LC	Remark 14.4	exact local coherent/Poisson
BT-UV	Remark 14.5	exact rational/RG consistency

Scripts, in `symbolic/`, each printing `ALL PASS`:
`verify_interaction_deformation.py` `verify_interaction_order3.py`
`verify_pt_breaking.py` `verify_perfect_square.py` `verify_two_field.py`
`verify_sector_obstruction.py` `verify_hardening.py`
`verify_doubled_theory.py` `verify_51_order4.py`
`verify_obstruction_null.py`

The scalar real–virtual extension is reproduced independently from the repository root by

```
ulimit -v 500000; python3 \
  reverse_physics/bt_real_virtual_axis_gluing.py --check
ulimit -v 500000; python3 \
  reverse_physics/verify_bt_real_virtual_axis_gluing.py
ulimit -v 500000; python3 -m unittest -v \
  reverse_physics.tests.test_bt_real_virtual_axis_gluing
```

The corrected finite-projector and asymptotic-generator preflight is reproduced by

```
ulimit -v 500000; python3 \
  reverse_physics/bt_asymptotic_generator_preflight.py --check
ulimit -v 500000; python3 \
  reverse_physics/verify_bt_asymptotic_generator_preflight.py
ulimit -v 500000; python3 -m unittest -v \
  reverse_physics.tests.test_bt_asymptotic_generator_preflight
```

The dynamically derived order-λ R_t kernel, Appendix C label consistency test, secular cancellation, and endpoint Gram are reproduced by

```
ulimit -v 500000; python3 \
  reverse_physics/bt_rt_jordan_kernel.py --check
ulimit -v 500000; python3 \
  reverse_physics/verify_bt_rt_jordan_kernel.py
ulimit -v 500000; python3 -m unittest -v \
  reverse_physics.tests.test_bt_rt_jordan_kernel
```

The physical ultraviolet hard/window law is reproduced by

```
ulimit -v 500000; python3 \
  reverse_physics/bt_uv_hard_scattering_law.py --check
ulimit -v 500000; python3 \
  reverse_physics/verify_bt_uv_hard_scattering_law.py
ulimit -v 500000; python3 -m unittest -v \
  reverse_physics.tests.test_bt_uv_hard_scattering_law
```